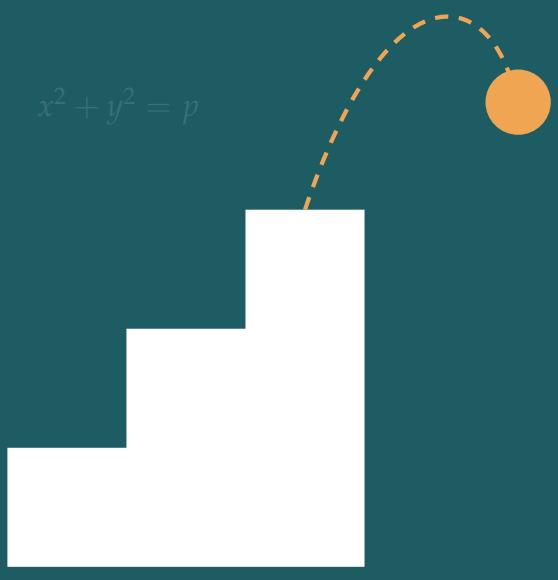
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2006 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2006 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2006 Worked Solutions

Worked solutions to all 37 questions of the 2006 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2006, Question 1

Pure • Difference of Two Squares

> **STEP I 2006, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/06-s1-q1

The key idea

If $(n+m)^2 - 33\,127 = k^2$, then $33\,127 = (n+m)^2 - k^2$ is a difference of two squares, and so it factorises as $(n+m-k)(n+m+k)$. This is Fermat's old trick for factorising: start just above $\sqrt{N}$ and step upwards until $a^2 - N$ is a square. Running the argument backwards, every factorisation of $33\,127$ gives a value of m, so counting the values of m comes down to counting factorisations.

Solution

Finding n

Since $180^2 = 32\,400$, the square root of $33\,127$ is a little above 180. We find

$$182^2 = 33\,124 < 33\,127 < 33\,489 = 183^2,$$

so $n = 182$. Note that $33\,127 = 182^2 + 3$, so it is not itself a perfect square.

A small value of m

We try $m = 1, 2, \ldots$ in turn:

$$183^2 - 33\,127 = 33\,489 - 33\,127 = 362, \quad \text{not a square } (19^2 = 361),$$
$$184^2 - 33\,127 = 33\,856 - 33\,127 = 729 = 27^2.$$

So $m = 2$ works, with $(n+m)^2 - 33\,127 = 27^2$.

Factorising $33\,127$

Rearranging and using the difference of two squares,

$$33\,127 = 184^2 - 27^2 = (184 - 27)(184 + 27) = 157 \times 211,$$

so we may take $p = 157$ and $q = 211$, both greater than 1.

Exactly two values of m

Both factors are prime. For 157: since $13^2 = 169 > 157$, we only need to test the primes $2, 3, 5, 7, 11$, and none divides 157. For 211: since $15^2 = 225 > 211$, we test $2, 3, 5, 7, 11, 13$, and none divides 211. So $33\,127 = 157 \times 211$ is the prime factorisation, and the only ways to write 33 127 as a product uv of positive integers with $u \le v$ are

$$1 \times 33\,127 \qquad \text{and} \qquad 157 \times 211.$$

Now suppose $(n+m)^2 - 33\,127 = k^2$ with $k \ge 0$. We take $a = n + m > 0$, which is the situation the question has in mind (we start at n and move upwards). Then

$$(a-k)(a+k) = 33\,127.$$

Here $a + k > 0$, so $a - k > 0$ as well, and $a - k \le a + k$. So $(a-k,\, a+k)$ must be one of the two pairs (u, v) above, and then

$$a = \frac{u+v}{2}, \qquad k = \frac{v-u}{2}.$$

These are integers because u and v are both odd.

> **Ask yourself** Why does each factorisation give exactly one m, and why can no other m sneak in?

The map runs both ways. Given a and k, the pair $(a-k, a+k)$ is a factorisation; given a factorisation (u, v), the formulas above recover a and k uniquely. So the values of a (and hence $m = a - 182$) correspond one-to-one with the factor pairs, and there are exactly two factor pairs.

- $(u, v) = (157, 211)$ gives $a = 184$, $k = 27$, so $m = 184 - 182 = 2$, the value we found.
- $(u, v) = (1, 33\,127)$ gives $a = 16\,564$, $k = 16\,563$, so

$$m = 16\,564 - 182 = 16\,382.$$

Hence there are exactly two values of m, namely $m = 2$ and $m = 16\,382$. As a check of the second, $16\,564^2 - 16\,563^2 = 16\,564 + 16\,563 = 33\,127$.

(If we also allowed $n + m$ to be negative, then a could be -184 or $-16\,564$, giving the mirror-image values $m = -366$ and $m = -16\,746$. The question clearly intends $n + m$ to be positive.)

Checks

A direct computer search over every m from 0 up to 16 564 finds that $(182+m)^2 - 33\,127$ is a perfect square only for $m = 2$ and $m = 16\,382$, and that the divisors of 33 127 are exactly $1, 157, 211, 33\,127$. By hand, $157 \times 211 = 157 \times 200 + 157 \times 11 = 31\,400 + 1\,727 = 33\,127$.

Common slips

- Forgetting the trivial factorisation $1 \times 33\,127$, and so finding only one value of m.
- Claiming that 157 and 211 are prime without checking: the argument that there are *exactly* two values depends on it.
- Not checking that u and v have the same parity. Here both are odd, so a and k are integers, but for an even number that is not a multiple of 4 no factorisation would work.

Beyond STEP

This is Fermat's factorisation method, which works quickly when N has two factors close to $\sqrt{N}$, as here. The same idea, of finding $a^2 \equiv k^2 \pmod{N}$ rather than $a^2 - N = k^2$ exactly, is the basis of the quadratic sieve, one of the fastest general-purpose factoring algorithms. The link between factorisations and representations as a difference of squares is a nice first taste of the elementary number theory in Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP I 2006, Question 2

Pure • Areas and Optimisation

STEP I 2006, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q2

The key idea

Fix the tether point by a single number, its distance x from one corner, and follow the rope as it wraps around the barn. The grazing region then splits into a semicircle and four quarter-circles whose radii are simple linear functions of x. So A is a quadratic in x on a closed interval, and both parts of the question come down to finding its largest and smallest values.

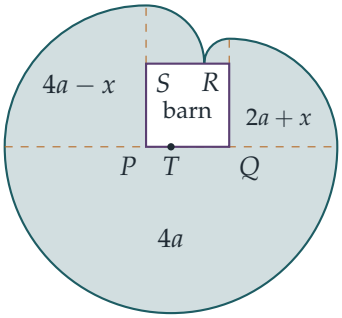
Solution

Setting up

Label the barn $PQRS$, with the goat tethered at T on side PQ, where $PT = x$ and $TQ = 2a - x$ for $0 \leq x \leq 2a$. By symmetry it doesn't matter which side the goat is on.

The goat can reach a point when the shortest path from T to it that avoids the barn is at most $4a$. Taut, that path is straight except where it bends around the corners of the barn. Going round the barn from T:

- **In front of PQ.** Every point on the far side of the line PQ can be seen directly from T, so the goat grazes a semicircle of radius $4a$.
- **Beside PS and QR.** Once the rope bends round P, $4a - x$ is left, and it sweeps a quarter-circle centred at P alongside side PS. Likewise, $4a - (2a - x) = 2a + x$ is left at Q, giving a quarter-circle centred at Q alongside QR.
- **Behind the barn.** Since $x \leq 2a$, we have $4a - x \geq 2a$, so the rope reaches S with $(4a - x) - 2a = 2a - x$ to spare, and that sweeps a quarter-circle centred at S behind side SR. In the same way, the rope reaches R with $(2a + x) - 2a = x$ left over, giving a quarter-circle of radius x centred at R.



Ask yourself Could the two pieces behind the barn overlap, so that we count some grass twice?

The quarter-circles centred at S and R lie on the same side of the line SR. Their radii, $2a - x$ and x, add up to $2a = SR$, so the two circles just touch at the point of SR that is $2a - x$ from S. So they don't overlap, and between them the goat reaches the whole of side SR. No other pieces overlap: the semicircle lies on one side of the line PQ and the four quarter-circles lie on the other, each in its own region beside or behind the barn. And a point behind the barn is reachable exactly when it lies within the remaining rope of S or of R, so nothing is missed.

The area

Adding the semicircle and the four quarter-circles,

$$\begin{aligned} A &= \tfrac{1}{2}\pi(4a)^2 + \tfrac{1}{4}\pi\left[(4a-x)^2 + (2a+x)^2 + (2a-x)^2 + x^2\right] \\ &= 8\pi a^2 + \tfrac{1}{4}\pi\left(4x^2 - 8ax + 24a^2\right) \\ &= \pi\left(x^2 - 2ax + 14a^2\right) \\ &= \pi\left[(x-a)^2 + 13a^2\right]. \end{aligned}$$

This formula holds for every x in $[0, 2a]$. At the ends ($x = 0$ or $x = 2a$, goat tied at a corner) one quarter-circle shrinks to a point, and the semicircle and the quarter-circle of radius $4a$ join up into three-quarters of a circle, which is what we would expect.

The maximum and the minimum

On $0 \le x \le 2a$ we have $-a \le x - a \le a$, so $0 \le (x-a)^2 \le a^2$. Hence

$$13\pi a^2 \le A \le 14\pi a^2,$$

which proves $A \le 14\pi a^2$. Equality on the right holds exactly when $x = 0$ or $x = 2a$, that is, when the goat is tied at a corner. The left-hand bound is attained when $x = a$, so the minimum value of A is

$$A_{\min} = 13\pi a^2,$$

reached when the goat is tethered at the midpoint of a side.

Checks

- At a corner ($x = 0$), we get three-quarters of a circle of radius $4a$ ($12\pi a^2$), a quarter-circle of radius $2a$ round the far corner of the adjacent side (πa^2), and a quarter-circle of radius $2a$ behind the barn (πa^2). That makes $14\pi a^2$, as the formula says.
- We estimated the grazed area on a fine grid with $a = 1$, testing each point by its true shortest distance round the barn (a straight line, or one bent at the corners). We got 40.845 at the midpoint against $13\pi \approx 40.841$, and 43.988 at a corner against $14\pi \approx 43.982$, with similar agreement at other tether points.

Common slips

- Forgetting the pieces behind the barn: the rope is long enough to go round two corners on each side.
- Worrying that the two pieces behind the barn overlap, or forcing an overlap into the calculation. Their radii always add up to exactly $2a$, so the circles only touch.

Beyond STEP

The taut rope follows a *geodesic*, the shortest path in a region with an obstacle. Here it is made of straight segments that bend only at the corners of the barn, so the set the goat can reach is a "ball" in a non-Euclidean way of measuring distance. Shortest paths around obstacles lead to the calculus of variations and to motion-planning problems in computer science. Siklos, *Advanced Problems in Mathematics*, has more problems of this kind, where a careful case analysis turns a geometry question into simple algebra.

WORKED SOLUTION

STEP I 2006, Question 3

Pure • Roots of Polynomials

STEP I 2006, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q3

The key idea

Think about the graph, not the formula. A curve that starts below the axis and must end above it has to cross, and the value at $x = 0$ (the constant term) tells you which side of the axis the curve is on at the origin. For the cubic, the turning points do the work: the curve crosses the axis three times exactly when its local maximum is above the axis and its local minimum is below it.

Solution

Part (i)

Let $f(x) = x^2 + bx + c$. The graph is a parabola opening upwards, so $f(x) \to +\infty$ as $x \to \pm\infty$. If $c < 0$ then $f(0) = c < 0$: the curve is below the axis at $x = 0$.

Since f is continuous, is positive for large negative x and is negative at 0, it crosses the axis somewhere in $(-\infty, 0)$. In the same way it crosses again in $(0, \infty)$. These two roots lie on opposite sides of 0, so they are different. A quadratic has at most two roots, so the equation has exactly two distinct real roots, one negative and one positive. Hence $c < 0$ is a *sufficient* condition.

It is *not necessary*. For example, $x^2 - 3x + 2 = (x - 1)(x - 2)$ has the distinct real roots 1 and 2, but $c = 2 > 0$. (The condition that is both necessary and sufficient is $b^2 > 4c$; when $c < 0$ this holds automatically, because $b^2 \geq 0 > 4c$.)

Part (ii)

We claim that the equation has two distinct positive roots if and only if

$$b^2 > 4c, \qquad b < 0, \qquad c > 0.$$

Necessary. Suppose the roots are $\alpha \neq \beta$ with $\alpha, \beta > 0$. Then $x^2 + bx + c = (x - \alpha)(x - \beta)$,

so

$$c = \alpha\beta > 0, \qquad b = -(\alpha+\beta) < 0, \qquad b^2 - 4c = (\alpha+\beta)^2 - 4\alpha\beta = (\alpha-\beta)^2 > 0.$$

Sufficient. If $b^2 > 4c$ the roots α, β are real and distinct. Their product is $c > 0$, so they have the same sign. Their sum is $-b > 0$, so that common sign is positive.

Ask yourself Why aren't $b < 0$ and $c > 0$ enough on their own?

Take $x^2 - x + 1$: here $b = -1 < 0$ and $c = 1 > 0$, but $b^2 - 4c = -3 < 0$, so there are no real roots at all. In graph terms, $c > 0$ puts the curve above the axis at $x = 0$ and $b < 0$ puts the vertex at $x = -\frac{b}{2} > 0$. We also need the vertex to lie below the axis: its height is $c - \frac{b^2}{4}$, which is negative exactly when $b^2 > 4c$. Each of the three conditions is needed.

Part (iii)

Let $g(x) = x^3 + px + q$, so $g'(x) = 3x^2 + p$. A real cubic has three roots counted with multiplicity, and its non-real roots come in complex conjugate pairs. Note also that $g(0) = q < 0$ in both cases.

The case $p > 0$, $q < 0$. Here $g'(x) = 3x^2 + p > 0$ for every x, so g is strictly increasing and has exactly one real root. Since $g(0) = q < 0$ and g is increasing, the root is positive. The remaining two roots are a pair of non-real complex conjugates.

So there is one real root, which is positive and simple, and two non-real complex conjugate roots.

The case $p < 0$, $q < 0$. Write $p = -3a^2$ with $a > 0$, that is, $a = \sqrt{-p/3}$. Then $g'(x) = 3(x^2 - a^2)$, so g is increasing on $(-\infty, -a]$, decreasing on $[-a, a]$ and increasing on $[a, \infty)$. There is a local maximum at $x = -a$ and a local minimum at $x = a$, with

$$g(-a) = -a^3 + 3a^3 + q = 2a^3 + q, \qquad g(a) = a^3 - 3a^3 + q = q - 2a^3.$$

Since $q < 0$, the local minimum $g(a)$ is negative. On $[0, a]$, g decreases from $g(0) = q < 0$, so it has no roots there. On $[a, \infty)$ it increases from $g(a) < 0$ to $+\infty$, so it has exactly one root greater than a, which is positive. Any other real root must lie in $(-\infty, -a]$ or $(-a, 0)$, so it is negative, and whether such roots exist depends only on the sign of the local maximum $g(-a)$.

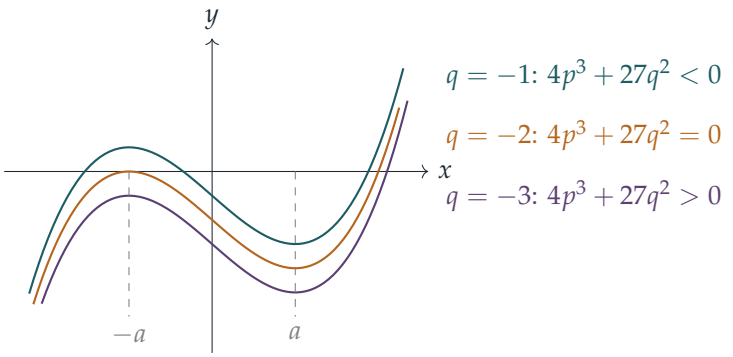
Ask yourself How does the sign of $g(-a)$ connect to $4p^3 + 27q^2$?

Multiply the two turning values:

$$g(-a)\,g(a) = (q + 2a^3)(q - 2a^3) = q^2 - 4a^6 = q^2 - 4\Big(-\frac{p}{3}\Big)^3 = \frac{4p^3 + 27q^2}{27}.$$

Since $g(a) < 0$, the sign of $g(-a)$ is the *opposite* of the sign of $4p^3 + 27q^2$. There are three cases.

- $4p^3 + 27q^2 > 0$. Then $g(-a) < 0$, so $g < 0$ on all of $(-\infty, a]$ (it increases up to the negative value $g(-a)$, then decreases). The only real root is the positive one. As before, there is one positive real root and two non-real complex conjugate roots.
- $4p^3 + 27q^2 = 0$. Then $g(-a) = 0$ and $g'(-a) = 0$, so $x = -a$ is a repeated root. Since the sum of the roots is 0 (there is no x^2 term), the third root is $2a$. The roots are $-a$, $-a$, $2a$: a negative double root and a positive simple root, all real.
- $4p^3 + 27q^2 < 0$. Then $g(-a) > 0$. Since $g(x) \to -\infty$ as $x \to -\infty$, there is a root in $(-\infty, -a)$. Since $g(-a) > 0 > g(0)$, there is another in $(-a, 0)$. Together with the root greater than a, this gives three distinct real roots: two negative and one positive.



The sketch shows $y = x^3 - 3x + q$ (so $p = -3$, $a = 1$) for $q = -1, -2, -3$: raising the local maximum through the axis turns one real root into three.

In every case with $q < 0$ there is exactly one positive root; this is consistent with the product of the roots being $-q > 0$.

Checks

- With $p = -3$, $q = -1$ we get $4p^3 + 27q^2 = -108 + 27 = -81 < 0$, and numerically the roots are about -1.532, -0.347 and 1.879: two negative, one positive, as predicted.
- With $p = -3$, $q = -2$: $4p^3 + 27q^2 = 0$ and $x^3 - 3x - 2 = (x+1)^2(x-2)$, giving roots $-1, -1, 2$.
- A numerical root-finder applied to thousands of random (p, q) in each case, and a test of random (b, c) against the conditions in (ii), agreed with every claim above.

Common slips

- In (ii), giving only $b < 0$ and $c > 0$ (which ensure that *if* the roots are real they are positive) and forgetting $b^2 > 4c$.
- Proving only necessity, or only sufficiency, in (ii). "Necessary and sufficient" needs both directions.
- In (iii), forgetting the boundary case $4p^3 + 27q^2 = 0$, where there is a repeated root.

Beyond STEP

The quantity $-(4p^3 + 27q^2)$ is the *discriminant* of the cubic $x^3 + px + q$, the exact analogue of $b^2 - 4ac$. It equals the product of the squared differences of the roots, so it vanishes exactly when there is a repeated root and is positive exactly when all three roots are real and distinct. Discriminants, and the way roots relate to coefficients, lead into Galois theory and the question of which equations can be solved by radicals. Allenby's *Rings, Fields and Groups* is a good place to start.

WORKED SOLUTION

STEP I 2006, Question 4

Pure • Regular Polygon Areas

STEP I 2006, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

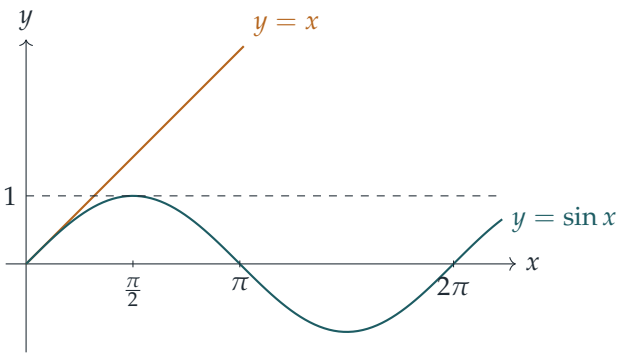
step.maths.org/questions/06-s1-q4

The key idea

The line $y = x$ is the tangent to $y = \sin x$ at the origin, and the sine curve bends away below it. That one picture gives both the inequality and the approximation. For the polygon, split it into n congruent isosceles triangles meeting at the centre; every quantity then becomes trigonometry of the half-angle π/n, and the inequality $\sin x < x$ turns out to be exactly what fixes the sign of the derivative.

Solution

The two graphs



Both curves pass through the origin with gradient 1 there, since $\frac{\mathrm{d}}{\mathrm{d}x}\sin x = \cos x = 1$ at $x = 0$. So $y = x$ is the tangent to the sine curve at O.

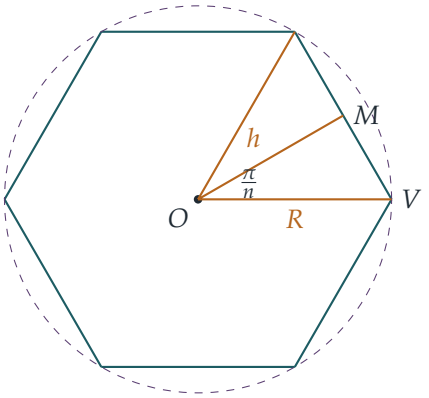
(i) For $0 < x \leq \frac{\pi}{2}$, the sine curve is concave ($\frac{\mathrm{d}^2}{\mathrm{d}x^2}\sin x = -\sin x < 0$), so it lies below its tangent at O, giving $\sin x < x$. (Equivalently: $g(x) = x - \sin x$ has $g(0) = 0$ and $g'(x) = 1 - \cos x > 0$ for $0 < x < 2\pi$, so g is strictly increasing there.) For $x > \frac{\pi}{2}$ we have $\sin x \leq 1 < \frac{\pi}{2} < x$. So $x > \sin x$ for every $x > 0$.

(ii) Because the line is the tangent at O, the two graphs are almost indistinguishable

close to the origin: for small x the height $\sin x$ is very nearly the height x, so $\dfrac{\sin x}{x} \approx 1$. (A sharper statement: since $\sin x \geq x - \frac{x^3}{6}$ for $x > 0$, we have $1 - \frac{x^2}{6} \leq \frac{\sin x}{x} < 1$.)

The area of the polygon

Join the centre O of the polygon to its vertices. This cuts the polygon into n congruent isosceles triangles, each with base equal to a side $s = P/n$ and apex angle $2\pi/n$ at O. Drop the perpendicular from O to the midpoint M of a side; it bisects the apex angle.



In the right-angled triangle OMV (with V a vertex), the angle at O is π/n and $MV = \frac{1}{2}s$. So the distance $h = OM$ from the centre to a side is

$$h = \frac{s/2}{\tan(\pi/n)} = \frac{P}{2n\tan(\pi/n)}.$$

Each of the n triangles has area $\frac{1}{2}sh$, so the polygon has area

$$A = n \cdot \tfrac{1}{2} \cdot \frac{P}{n} \cdot \frac{P}{2n\tan(\pi/n)} = \frac{P^2}{4n\tan(\pi/n)},$$

as required.

The area increases with n

With P fixed, write $A = \dfrac{P^2}{4f(n)}$ where $f(n) = n\tan\dfrac{\pi}{n}$. Since $f(n) > 0$ for $n \geq 3$, A increases exactly when f decreases. By the product and chain rules,

$$f'(n) = \tan\frac{\pi}{n} + n\sec^2\frac{\pi}{n} \cdot \left(-\frac{\pi}{n^2}\right) = \tan\theta - \theta\sec^2\theta, \qquad \text{where } \theta = \frac{\pi}{n}.$$

? **Ask yourself** How do we decide the sign of $\tan\theta - \theta\sec^2\theta$?

Take out the positive factor $\sec^2\theta$ and use $\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta$:

$$f'(n) = \sec^2\theta\left(\sin\theta\cos\theta - \theta\right) = \tfrac{1}{2}\sec^2\theta\left(\sin 2\theta - 2\theta\right).$$

For $n \geq 3$ we have $0 < \theta \leq \frac{\pi}{3}$, so $2\theta > 0$ and part (i) gives $\sin 2\theta < 2\theta$. Hence $f'(n) < 0$, and

$$\frac{\mathrm{d}A}{\mathrm{d}n} = -\frac{P^2 f'(n)}{4f(n)^2} > 0.$$

So, with the perimeter fixed, the area increases as n increases.

Comparison with the circumcircle

The smallest circle containing the polygon is the circumcircle, centred at O through all the vertices. To see that no smaller circle will do, let the vertices be $V_1, \ldots, V_n$ with $\overrightarrow{OV_k} = \mathbf{v}_k$, $|\mathbf{v}_k| = R$. By symmetry $\sum \mathbf{v}_k = \mathbf{0}$, so for any point C with $\overrightarrow{OC} = \mathbf{c}$,

$$\frac{1}{n}\sum_{k=1}^{n}|\mathbf{v}_k - \mathbf{c}|^2 = R^2 - \frac{2}{n}\,\mathbf{c}\cdot\sum_{k=1}^{n}\mathbf{v}_k + |\mathbf{c}|^2 = R^2 + |\mathbf{c}|^2 \geq R^2.$$

So some vertex is at distance at least R from C, and a circle centred at C containing the polygon has radius at least R. The circumradius is $R = OV$, and from the same right-angled triangle,

$$\tfrac{1}{2}s = R\sin\theta \quad\Longrightarrow\quad R = \frac{P}{2n\sin\theta}.$$

So the ratio of the areas is

$$\frac{A}{\pi R^2} = \frac{P^2}{4n\tan\theta}\cdot\frac{4n^2\sin^2\theta}{\pi P^2} = \frac{n}{\pi}\cdot\frac{\sin^2\theta}{\tan\theta} = \frac{\sin\theta\cos\theta}{\theta} = \frac{\sin 2\theta}{2\theta},$$

using $n/\pi = 1/\theta$. For large n, $2\theta = 2\pi/n$ is small, so by part (ii) the ratio is approximately 1.

Checks

For a square ($n = 4$, $\theta = \pi/4$) the formula gives $A = P^2/16$, which is right since each side is $P/4$; the area ratio is $\sin(\pi/2)/(\pi/2) = 2/\pi \approx 0.637$, matching a square of side $\sqrt{2}$ in a unit circle. For a hexagon the ratio is $3\sqrt{3}/(2\pi) \approx 0.827$. We checked the area formula against the shoelace formula for $n = 3, \ldots, 39$, checked $f'(n) < 0$ against a numerical derivative, and found the ratio to be 0.99934 at $n = 100$ and 0.9999934 at $n = 1000$.

Common slips

- Using the full apex angle $2\pi/n$ in the right-angled triangle instead of the half-angle π/n.
- Forgetting the chain-rule factor $-\pi/n^2$ when differentiating $\tan(\pi/n)$.
- Claiming $\tan\theta < \theta\sec^2\theta$ "because $\tan\theta \approx \theta$": an approximation can't settle a sign. The exact inequality $\sin 2\theta < 2\theta$ does.

Beyond STEP

The limit $\sin x / x \to 1$ is the fact that makes the derivative of $\sin x$ equal to $\cos x$, and a first analysis course proves it carefully from a definition of sin rather than from a picture. The monotonicity result is a discrete glimpse of the isoperimetric inequality: among all closed curves of a given length, the circle encloses the most area, and the polygon areas increase towards the circle's value $P^2/(4\pi)$. For a rigorous treatment of limits and derivatives, try Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 2006, Question 5

Pure • Integration by Substitution

STEP I 2006, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q5

The key idea

Both integrands contain an awkward square root, and the cleanest way to kill a square root of a linear expression is to call the whole root a new variable. Once the root has gone, what is left is a rational function, which partial fractions can always integrate. In part (ii) the exponentials are best removed first, so that the square root becomes the root of something linear, and then the same trick applies.

Solution

Part (i)

Since $x > 4$, we have $2x + 1 > 9$, so we may take $u = \sqrt{2x+1} > 3$. From $u^2 = 2x + 1$,

$$2u\,\mathrm{d}u = 2\,\mathrm{d}x, \qquad \text{so} \qquad \mathrm{d}x = u\,\mathrm{d}u,$$

and

$$x - 4 = \frac{u^2 - 1}{2} - 4 = \frac{u^2 - 9}{2}.$$

Substituting,

$$\int \frac{3}{(x-4)\sqrt{2x+1}}\,\mathrm{d}x = \int \frac{3}{\frac{1}{2}(u^2-9)\,u}\,u\,\mathrm{d}u = \int \frac{6}{u^2-9}\,\mathrm{d}u.$$

The factor u from $\mathrm{d}x$ cancels the square root exactly; that is why this substitution works. Now $u^2 - 9 = (u-3)(u+3)$, and

$$\frac{6}{(u-3)(u+3)} = \frac{1}{u-3} - \frac{1}{u+3},$$

as you can check by recombining: the numerator is $(u+3) - (u-3) = 6$. Hence

$$\int \frac{6}{u^2-9}\,\mathrm{d}u = \ln(u-3) - \ln(u+3) + K = \ln\left(\frac{u-3}{u+3}\right) + K,$$

where no modulus signs are needed because $u > 3$ makes both $u-3$ and $u+3$ positive. Putting $u = \sqrt{2x+1}$ back gives

$$\int \frac{3}{(x-4)\sqrt{2x+1}}\,\mathrm{d}x = \ln\left(\frac{\sqrt{2x+1}-3}{\sqrt{2x+1}+3}\right) + K,$$

as required.

Part (ii)

Let I be the integral. The obvious substitution, copying part (i), is to name the square root:

$$w = \sqrt{\mathrm{e}^x + 1}, \qquad w^2 = \mathrm{e}^x + 1.$$

Then $2w\,\mathrm{d}w = \mathrm{e}^x\,\mathrm{d}x$, so $\mathrm{d}x = \dfrac{2w}{\mathrm{e}^x}\,\mathrm{d}w = \dfrac{2w}{w^2-1}\,\mathrm{d}w$. The limits become

$$x = \ln 3 :\ w = \sqrt{3+1} = 2, \qquad x = \ln 8 :\ w = \sqrt{8+1} = 3.$$

Therefore

$$I = \int_2^3 \frac{2}{(w^2-1)\,w} \cdot \frac{2w}{w^2-1}\,\mathrm{d}w = \int_2^3 \frac{4}{(w^2-1)^2}\,\mathrm{d}w.$$

Ask yourself How do we split $\dfrac{4}{(w^2-1)^2}$ into partial fractions without a long calculation?

Start from the simple split $\dfrac{2}{w^2-1} = \dfrac{1}{w-1} - \dfrac{1}{w+1}$ and square it:

$$\begin{aligned}\frac{4}{(w^2-1)^2} &= \frac{1}{(w-1)^2} - \frac{2}{(w-1)(w+1)} + \frac{1}{(w+1)^2} \\ &= \frac{1}{(w-1)^2} + \frac{1}{(w+1)^2} - \frac{1}{w-1} + \frac{1}{w+1},\end{aligned}$$

using the simple split once more on the middle term. Each piece now integrates at sight:

$$\int \frac{4}{(w^2-1)^2}\,\mathrm{d}w = -\frac{1}{w-1} - \frac{1}{w+1} - \ln\left(\frac{w-1}{w+1}\right) + C,$$

with no modulus signs needed since $w \geq 2$ on the range of integration. Evaluating,

$$\begin{aligned}\text{at } w = 3 :&\quad -\tfrac{1}{2} - \tfrac{1}{4} - \ln\tfrac{2}{4} = -\tfrac{3}{4} + \ln 2, \\ \text{at } w = 2 :&\quad -1 - \tfrac{1}{3} - \ln\tfrac{1}{3} = -\tfrac{4}{3} + \ln 3.\end{aligned}$$

So

$$I = \left(-\tfrac{3}{4} + \ln 2\right) - \left(-\tfrac{4}{3} + \ln 3\right) = \frac{7}{12} + \ln\frac{2}{3},$$

which is the result we had to show.

You could equally do this in two steps, first $v = \mathrm{e}^x$ (giving $\int_3^8 \frac{2}{v^2\sqrt{v+1}}\,\mathrm{d}v$) and then $w^2 = v+1$; the single substitution above does both at once.

Checks

- Differentiate the answer to (i): with $u = \sqrt{2x+1}$, $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u}$, and $\frac{\mathrm{d}}{\mathrm{d}u}\ln\frac{u-3}{u+3} = \frac{6}{u^2-9}$, so the derivative is $\frac{6}{u(u^2-9)} = \frac{3}{(x-4)\sqrt{2x+1}}$.
- Numerically, $\frac{7}{12} + \ln\frac{2}{3} \approx 0.58333 - 0.40547 = 0.17787$, and Simpson's rule applied to the original integrand on $[\ln 3, \ln 8]$ gives 0.177868, agreeing to six figures.
- A sanity check on sign and size: the integrand in (ii) is positive and falls from about 0.33 to about 0.08 over an interval of length $\ln\frac{8}{3} \approx 0.98$, so an answer a little under 0.2 is just what we expect.

Common slips

- Forgetting to change the limits in (ii) when changing variable, or substituting x-limits into a w-expression.
- Writing $\frac{4}{(w^2-1)^2}$ as only $\frac{A}{(w-1)^2} + \frac{B}{(w+1)^2}$: the repeated factors need the linear terms $\frac{1}{w-1}$ and $\frac{1}{w+1}$ as well.
- Getting $\mathrm{d}x = \mathrm{d}u$ instead of $\mathrm{d}x = u\,\mathrm{d}u$ in (i), which leaves a stray square root.

Beyond STEP

The reason both parts end in logarithms and rational terms is a general theorem: every rational function has an elementary antiderivative built from rational functions, logarithms and inverse tangents, via partial fractions; and any integrand that is rational in x and $\sqrt{ax+b}$ becomes rational after the substitution $u^2 = ax + b$. Extending this to roots of quadratics leads to Euler's substitutions, and asking which integrals *cannot* be done in elementary terms leads to Liouville's theory of integration in finite terms. For a careful treatment of integration and why it works, read Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 2006, Question 6

Pure • Pell Equation Transformations

STEP I 2006, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q6

The key idea

Each part asks whether a linear change of coordinates preserves a quadratic expression. The direct route is to substitute, expand and compare coefficients of a^2, ab and b^2. Behind the scenes there is a neater picture: $x^2 - 2y^2$ is the "size" of the number $x + y\sqrt{2}$, and multiplying by $3 + 2\sqrt{2}$ (whose size is $9 - 8 = 1$) leaves that size unchanged. Seeing this tells you exactly how to build the map in (iii).

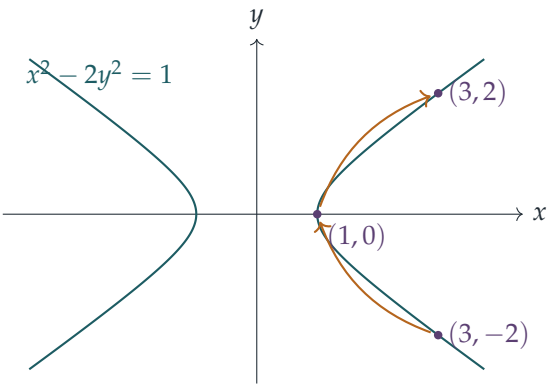
Solution

Part (i)

Suppose $a^2 - 2b^2 = 1$. Then

$$\begin{aligned}(3a + 4b)^2 - 2(2a + 3b)^2 &= (9a^2 + 24ab + 16b^2) - 2(4a^2 + 12ab + 9b^2) \\ &= a^2 - 2b^2 \\ &= 1,\end{aligned}$$

so $(3a + 4b,\ 2a + 3b)$ lies on $x^2 - 2y^2 = 1$, as required. In fact the first two lines are an identity, true for every a and b: the map preserves $x^2 - 2y^2$ everywhere, not just on the curve.



The map slides points along the right-hand branch: $(3,-2) \mapsto (1,0) \mapsto (3,2) \mapsto (17,12) \mapsto \cdots$.

Part (ii)

Suppose $Ma^2 - Nb^2 = 1$. Expanding,

$$M(5a+6b)^2 - N(4a+5b)^2 = (25M-16N)a^2 + (60M-40N)ab + (36M-25N)b^2.$$

The image point lies on the curve exactly when this equals $1 = Ma^2 - Nb^2$. Subtracting $Ma^2 - Nb^2$, the condition becomes

$$\begin{aligned} 0 &= (24M-16N)a^2 + (60M-40N)ab + (36M-24N)b^2 \\ &= 4(3M-2N)(2a^2+5ab+3b^2) \\ &= 4(3M-2N)(2a+3b)(a+b). \end{aligned}$$

Ask yourself Do we really need $3M = 2N$, or could the other factors be zero instead?

The condition has to hold for *every* point on the curve. When $M, N > 0$ the curve $Mx^2 - Ny^2 = 1$ is a hyperbola with infinitely many points. The lines $2x + 3y = 0$ and $x + y = 0$ each meet it in at most two points, so most points on the curve have $(2a+3b)(a+b) \neq 0$. Hence we need $3M - 2N = 0$. Conversely, if $3M = 2N$ the condition holds at every point.

So $3M = 2N$. Then $2 \mid 3M$, so M is even. The smallest choice is $M = 2$, which gives $N = 3$. As a final check,

$$2(5a+6b)^2 - 3(4a+5b)^2 = (50-48)a^2 + (120-120)ab + (72-75)b^2 = 2a^2 - 3b^2.$$

The smallest positive integers are $M = 2$ and $N = 3$.

Part (iii)

We want positive integers with

$$(Pa+Qb)^2 - 3(Ra+Sb)^2 = a^2 - 3b^2 \quad \text{for all } a, b,$$

which certainly makes the image of every point of the curve lie on it. Comparing coefficients:

$$P^2 - 3R^2 = 1, \qquad PQ = 3RS, \qquad Q^2 - 3S^2 = -3.$$

The first equation asks for a positive integer point (P, R) on $x^2 - 3y^2 = 1$. The smallest is $(2,1)$, so try $P = 2$, $R = 1$. The second equation then gives $Q = \frac{3}{2}S$, and the third becomes $\frac{9}{4}S^2 - 3S^2 = -3$, that is, $S^2 = 4$. So $S = 2$ and $Q = 3$:

$$P = 2, \quad Q = 3, \quad R = 1, \quad S = 2.$$

Check: $(2a + 3b)^2 - 3(a + 2b)^2 = 4a^2 + 12ab + 9b^2 - 3a^2 - 12ab - 12b^2 = a^2 - 3b^2 = 1$. So $(2a + 3b,\ a + 2b)$ lies on $x^2 - 3y^2 = 1$.

Ask yourself Where does a map like this come from?

Write $(a + b\sqrt{3})(2 + \sqrt{3}) = (2a + 3b) + (a + 2b)\sqrt{3}$. The norm $x^2 - 3y^2$ of $x + y\sqrt{3}$ is multiplicative, and $2 + \sqrt{3}$ has norm $4 - 3 = 1$, so multiplying by it preserves the norm. Part (i) works the same way with $3 + 2\sqrt{2}$. For (ii), $(a\sqrt{2} + b\sqrt{3})(5 + 2\sqrt{6}) = (5a + 6b)\sqrt{2} + (4a + 5b)\sqrt{3}$, which is why the form $2x^2 - 3y^2$ appears there. Other answers are valid in (iii): multiplying by $(2 + \sqrt{3})^2 = 7 + 4\sqrt{3}$ gives $(7a + 12b,\ 4a + 7b)$.

Checks

- Start from $(1,0)$ on $x^2 - 3y^2 = 1$. Repeating the map in (iii) gives $(2,1)$, $(7,4)$, $(26,15)$, $(97,56)$, and each satisfies $x^2 - 3y^2 = 1$ (for example $26^2 - 3 \cdot 15^2 = 676 - 675 = 1$).
- For (ii), $(1/\sqrt{2}, 0)$ is on $2x^2 - 3y^2 = 1$ and maps to $(5/\sqrt{2},\ 4/\sqrt{2})$. Then $2 \cdot \frac{25}{2} - 3 \cdot \frac{16}{2} = 25 - 24 = 1$.
- A brute-force search over $1 \le M, N < 30$, testing actual points of each hyperbola, confirms that $(2,3)$ is the smallest pair that works.

Common slips

- In (ii), setting only the ab coefficient to zero without checking that the a^2 and b^2 coefficients then come out as M and $-N$. They do here, but that has to be shown.
- Saying "the identity must hold for all a, b" without justification, when the question only gives points on the curve. You need an argument, such as the factorisation above, that rules out the other factors vanishing.

Beyond STEP

These are Pell equations $x^2 - dy^2 = 1$. Their integer solutions are generated by powers of a fundamental unit $x_1 + y_1\sqrt{d}$ in the ring $\mathbb{Z}[\sqrt{d}]$, and the maps here are multiplication

by such units. In matrix language, each map is a 2×2 integer matrix that preserves a quadratic form, an element of an "integral orthogonal group". Davenport's *The Higher Arithmetic* has a lovely chapter-length treatment of Pell's equation via continued fractions.

WORKED SOLUTION

STEP I 2006, Question 7

Pure • Integrals with Modulus Signs

STEP I 2006, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q7

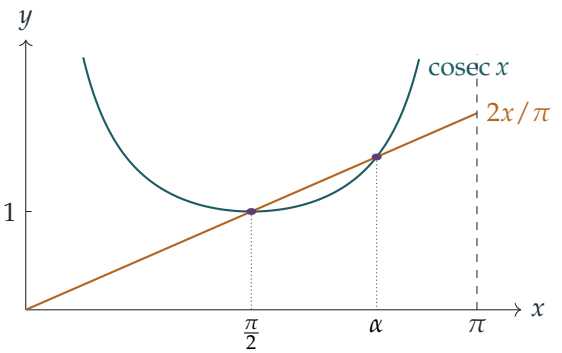
The key idea

Dividing $x \sin x = \pi/2$ by $\frac{\pi}{2} \sin x$ (which is positive on $(0, \pi)$) turns it into $\operatorname{cosec} x = 2x/\pi$, so the roots are where a straight line meets a U-shaped curve, and a line can cross a convex curve at most twice. For the integrals, the whole game with a modulus is to find where the expression inside changes sign, split the range there, and integrate each piece with the correct sign.

Solution

Part (i): the sketch and the roots

On $0 < x < \pi$, $\operatorname{cosec} x$ tends to $+\infty$ at both ends, is symmetric about $x = \pi/2$ and has its minimum value 1 there. The line $y = 2x/\pi$ runs from $(0, 0)$ to $(\pi, 2)$ and passes through $(\pi/2, 1)$, the bottom of the cosec curve.



Since $\sin x > 0$ on $(0, \pi)$,

$$x \sin x = \frac{\pi}{2} \iff \frac{1}{\sin x} = \frac{2x}{\pi},$$

so the roots are the x-coordinates of the intersections in the sketch. One is visible immediately: $x = \pi/2$, since $\frac{\pi}{2}\sin\frac{\pi}{2} = \frac{\pi}{2}$.

To be sure there are exactly two, let $h(x) = \operatorname{cosec} x - 2x/\pi$. Then

$$h''(x) = \operatorname{cosec} x\,(\operatorname{cosec}^2 x + \cot^2 x) > 0 \quad \text{on } (0,\pi),$$

so h is strictly convex, and a strictly convex function has at most two zeros. (If it had three, $a < b < c$, then b would lie strictly between a and c while $h(b)$ lay on, rather than strictly below, the chord joining $(a,0)$ and $(c,0)$.) Now $h'(\pi/2) = -\cot\frac{\pi}{2}\operatorname{cosec}\frac{\pi}{2} - \frac{2}{\pi} = -\frac{2}{\pi} < 0$, so h is negative just to the right of $\pi/2$, while $h(x) \to +\infty$ as $x \to \pi^-$. By continuity h has a zero α with $\pi/2 < \alpha < \pi$. So there are exactly two roots, $\pi/2$ and α, and α is the larger.

Part (i): the integral

From the sketch (and the convexity argument), the cosec curve lies below the line between the two roots and above it beyond α. Multiplying $\operatorname{cosec} x \lessgtr 2x/\pi$ by the positive quantity $\frac{\pi}{2}\sin x$, this says

$$x\sin x - \frac{\pi}{2} \geq 0 \text{ on } \left[\tfrac{\pi}{2},\alpha\right], \qquad x\sin x - \frac{\pi}{2} \leq 0 \text{ on } [\alpha,\pi].$$

Ask yourself How do we deal with the modulus sign?

We split the integral at α, where the sign changes, and on the second piece we integrate the negative of the expression.

Integrating by parts with $u = x$, $\mathrm{d}v = \sin x\,\mathrm{d}x$,

$$\int x\sin x\,\mathrm{d}x = -x\cos x + \int \cos x\,\mathrm{d}x = \sin x - x\cos x.$$

So an antiderivative of $x\sin x - \pi/2$ is

$$F(x) = \sin x - x\cos x - \frac{\pi x}{2},$$

and the integral we want is

$$I = \big[F(x)\big]_{\pi/2}^{\alpha} - \big[F(x)\big]_{\alpha}^{\pi} = 2F(\alpha) - F\left(\tfrac{\pi}{2}\right) - F(\pi).$$

The three values are

$$F(\alpha) = \sin\alpha - \alpha\cos\alpha - \frac{\alpha\pi}{2},$$

$$F\left(\tfrac{\pi}{2}\right) = 1 - 0 - \frac{\pi^2}{4},$$

$$F(\pi) = 0 - \pi(-1) - \frac{\pi^2}{2} = \pi - \frac{\pi^2}{2}.$$

Hence

$$I = 2\sin\alpha - 2\alpha\cos\alpha - \alpha\pi - 1 + \frac{\pi^2}{4} - \pi + \frac{\pi^2}{2}$$
$$= 2\sin\alpha + \frac{3\pi^2}{4} - \alpha\pi - \pi - 2\alpha\cos\alpha - 1,$$

which is the required result.

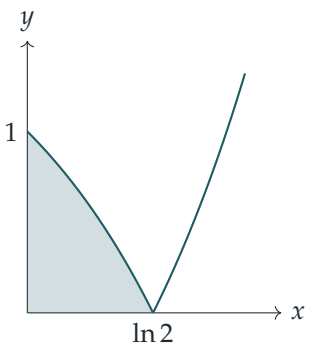
Part (ii)

Ask yourself Which of the two modulus signs actually does anything here?

On the positive x-axis, $x \geq 0$, so $e^x \geq 1$ and the inner modulus is harmless: $|e^x - 1| = e^x - 1$. The curve is therefore

$$y = |e^x - 2| \qquad (x \geq 0).$$

This equals $2 - e^x$ for $0 \leq x \leq \ln 2$, falling from $y = 1$ at $x = 0$ to $y = 0$ at $x = \ln 2$, and equals $e^x - 2$ for $x \geq \ln 2$, rising without bound.



So the curve touches the x-axis only at $x = \ln 2$, and the only region enclosed by the positive x-axis, the y-axis and the curve is the shaded one over $0 \leq x \leq \ln 2$ (beyond $\ln 2$ the curve climbs away and nothing closes up). Its area is

$$\int_0^{\ln 2} (2 - e^x)\,dx = \Big[2x - e^x\Big]_0^{\ln 2} = (2\ln 2 - 2) - (0 - 1)$$
$$= 2\ln 2 - 1 = \ln 4 - 1,$$

as required.

Checks

Bisection gives $\alpha \approx 2.4433$, and then the formula in (i) evaluates to about 0.61343; Simpson's rule applied directly to $|x\sin x - \pi/2|$ on $[\pi/2, \pi]$ gives the same value to

14 decimal places. A grid search on $(0, \pi)$ finds exactly two sign changes of $x \sin x - \pi/2$, near 1.5708 and 2.4433. In (ii), $\ln 4 - 1 \approx 0.38629$, again matching a numerical integration. As a sanity check on the sign in (i), the answer must be positive, and it is.

Common slips

- Integrating $x \sin x - \pi/2$ straight across $[\pi/2, \pi]$ without splitting at α: this gives a signed area (here negative), not the integral of the modulus.
- Missing the root at $x = \pi/2$ because the line passes through the minimum of the cosec curve, and the picture can look like a tangency. It isn't one: the curve has gradient 0 there and the line has gradient $2/\pi$, so they cross.
- In (ii), keeping both modulus signs and splitting into too many cases. Settle the sign of $e^x - 1$ first, using $x \geq 0$.

Beyond STEP

The root α has no closed form, and yet we could still compute the integral exactly in terms of it: a common situation in analysis, where a quantity is pinned down by an equation rather than a formula. The fact that a strictly convex function meets a line at most twice is one of many neat consequences of convexity, which also underlies Jensen's inequality and much of optimisation. Bartle and Sherbert's *Introduction to Real Analysis* treats continuity, the intermediate value theorem and convex functions carefully.

WORKED SOLUTION

STEP I 2006, Question 8

Pure • Vectors in Three Dimensions

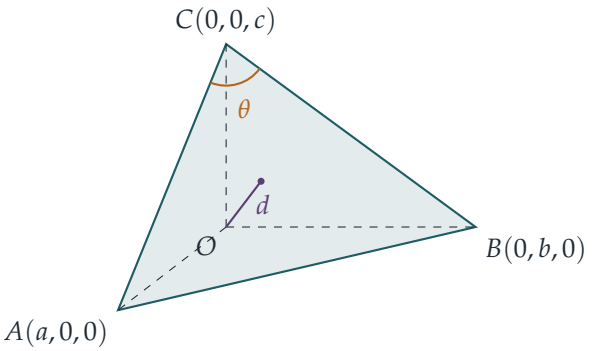
STEP I 2006, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q8

The key idea

The tetrahedron $OABC$ has three mutually perpendicular edges at O, so its volume is easy if we use OAB as the base. The same solid can also be viewed with ABC as the base, and then the height is exactly the distance d we want. Working out the volume in two ways gives an equation for d, provided we know the area of ABC. That area comes from the angle at C, which the scalar product gives us.



Solution

Part (i)

Take triangle OAB, which lies in the plane $z = 0$, as the base. Since OA and OB lie along the x- and y-axes, the angle AOB is a right angle, so

$$\text{area of } OAB = \tfrac{1}{2}ab.$$

The height is the distance of C from the plane $z = 0$, which is c, because OC lies along the z-axis and so is perpendicular to that plane. Hence

$$V = \tfrac{1}{3} \times \tfrac{1}{2}ab \times c = \frac{abc}{6}.$$

Part (ii)

The angle ACB is the angle between the vectors $\overrightarrow{CA}$ and $\overrightarrow{CB}$:

$$\overrightarrow{CA} = \begin{pmatrix} a \\ 0 \\ -c \end{pmatrix}, \qquad \overrightarrow{CB} = \begin{pmatrix} 0 \\ b \\ -c \end{pmatrix}.$$

Their scalar product is $0 + 0 + c^2 = c^2$, and their lengths are $\sqrt{a^2 + c^2}$ and $\sqrt{b^2 + c^2}$. So

$$\cos\theta = \frac{\overrightarrow{CA} \cdot \overrightarrow{CB}}{|\overrightarrow{CA}|\,|\overrightarrow{CB}|} = \frac{c^2}{\sqrt{(a^2 + c^2)(b^2 + c^2)}},$$

which is the required result.

For the area we need $\sin\theta$. Since $0 < \theta < \pi$, $\sin\theta > 0$, so

$$\begin{aligned} \sin\theta = \sqrt{1 - \cos^2\theta} &= \sqrt{\frac{(a^2 + c^2)(b^2 + c^2) - c^4}{(a^2 + c^2)(b^2 + c^2)}} \\ &= \frac{\sqrt{a^2b^2 + b^2c^2 + c^2a^2}}{\sqrt{(a^2 + c^2)(b^2 + c^2)}}, \end{aligned}$$

because the c^4 terms cancel when we expand $(a^2 + c^2)(b^2 + c^2)$.

Ask yourself Which formula for the area of a triangle fits the information we now have?

We know two sides, CA and CB, and the angle θ between them, so we use $\frac{1}{2}\,CA \cdot CB \sin\theta$. The awkward square roots cancel:

$$\text{area of } ABC = \tfrac{1}{2}\sqrt{(a^2 + c^2)(b^2 + c^2)} \cdot \frac{\sqrt{a^2b^2 + b^2c^2 + c^2a^2}}{\sqrt{(a^2 + c^2)(b^2 + c^2)}} = \tfrac{1}{2}\sqrt{a^2b^2 + b^2c^2 + c^2a^2}.$$

Pleasingly, the answer is symmetric in a, b and c, as it must be: the triangle ABC doesn't care which vertex we called C.

The distance d

Now regard ABC as the base of the same tetrahedron. Its height is the perpendicular distance from the opposite vertex O to the plane of ABC, which is d. Since the volume doesn't depend on which face we call the base, part (i) and part (ii) give

$$\frac{abc}{6} = \tfrac{1}{3} \times \tfrac{1}{2}\sqrt{a^2b^2 + b^2c^2 + c^2a^2} \times d, \qquad \text{so} \qquad d = \frac{abc}{\sqrt{a^2b^2 + b^2c^2 + c^2a^2}}.$$

Squaring and inverting (everything is positive),

$$\frac{1}{d^2} = \frac{a^2b^2 + b^2c^2 + c^2a^2}{a^2b^2c^2} = \frac{1}{c^2} + \frac{1}{a^2} + \frac{1}{b^2},$$

dividing each term of the numerator by $a^2b^2c^2$. This is the required result.

Checks

- Take $a = b = c = 1$. Then ABC is equilateral with side $\sqrt{2}$, of area $\frac{\sqrt{3}}{4} \cdot 2 = \frac{\sqrt{3}}{2}$, which matches $\frac{1}{2}\sqrt{3}$. The plane is $x + y + z = 1$, at distance $1/\sqrt{3}$ from O, and indeed $1/d^2 = 3$.
- With $a = 1$, $b = c = 2$ the formulae give volume $\frac{2}{3}$, area $\sqrt{6} \approx 2.449$ and $d = 2/\sqrt{6} \approx 0.8165$, so $1/d^2 = \frac{3}{2} = 1 + \frac{1}{4} + \frac{1}{4}$.
- Numerically, for 10 000 random (a, b, c), the volume (by the scalar triple product), the area (by the cross product) and d (by locating the foot of the perpendicular directly) all agreed with the formulae to within 10^{-12}.

Common slips

- Using $\overrightarrow{AC}$ with $\overrightarrow{CB}$: mixing directions flips the sign of the scalar product and gives the supplement of θ.
- Forgetting that the height for base ABC is the perpendicular from O to the plane, not the length of any edge.

Beyond STEP

The plane through A, B and C is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, with normal $(\frac{1}{a}, \frac{1}{b}, \frac{1}{c})$, and the general point-to-plane distance formula gives d in one line. The area result is a three-dimensional Pythagoras theorem (de Gua's theorem): the squared area of ABC equals the sum of the squared areas of the three right-angled faces at O, since $\frac{1}{4}(a^2b^2 + b^2c^2 + c^2a^2)$ is exactly that sum. Both ideas come from the cross product and determinants, which you meet early in a first linear algebra course; Strang's *Introduction to Linear Algebra* is a good place to see them.

WORKED SOLUTION

STEP I 2006, Question 9

Mechanics • Connected Particles

STEP I 2006, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q9

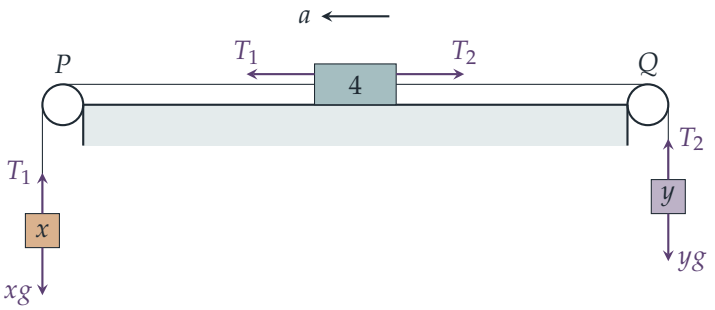
The key idea

The motion splits into two stages, each with constant acceleration. While both strings are attached, the whole system of three bodies moves as one, driven by the difference of the two hanging weights. Once the string is cut, only the lighter particle pulls on the block, now against the motion, so the block decelerates uniformly. The speed at the moment of cutting is the link between the two stages, and the constraint $x + y = 6$ lets us write everything in terms of y.

Solution

Model

We treat the block and the two hanging masses as particles, the strings as light and inextensible, and the table and pulleys as smooth. Since $x > y$, the block moves towards P, the particle of mass x descends and the particle of mass y rises. Let T_1 and T_2 be the tensions in the strings over P and Q.



Stage 1: both strings attached

Let the common acceleration be a_1 (block towards P, particle x down, particle y up). Newton's second law for each body gives

$$\begin{aligned} xg - T_1 &= xa_1, \\ T_1 - T_2 &= 4a_1, \\ T_2 - yg &= ya_1. \end{aligned}$$

Adding eliminates the tensions:

$$(x-y)g = (x+y+4)a_1 = 10a_1, \qquad \text{so} \qquad a_1 = \frac{(x-y)g}{10} = \frac{(6-2y)g}{10},$$

using $x = 6 - y$. This is positive because $y < x$ means $y < 3$.

Starting from rest with constant acceleration, the block covers distance d in time t_1 and reaches speed V, where

$$d = \tfrac{1}{2}a_1 t_1^2, \qquad V^2 = 2a_1 d.$$

Hence

$$t_1 = \sqrt{\frac{2d}{a_1}} = \sqrt{\frac{20d}{(6-2y)g}} = \sqrt{\frac{d}{5g}}\,\frac{10}{\sqrt{6-2y}}, \qquad V = \sqrt{\frac{(6-2y)gd}{5}}.$$

Stage 2: after the cut

Now $T_1 = 0$. The block is still moving towards P with speed V, and the particle of mass y is still rising with speed V. Only the string over Q acts on the block, and it pulls back towards Q. Let the deceleration be a_2. Then

$$\begin{aligned} T_2 &= 4a_2 \quad \text{(block)}, \\ yg - T_2 &= ya_2 \quad \text{(particle } y\text{, decelerating while rising)}, \end{aligned}$$

so

$$a_2 = \frac{yg}{4+y}.$$

Ask yourself How do we know the string over Q stays taut after the cut?

If the string went slack, the particle would decelerate at g while the block would keep its speed. With the string taut the tension is $T_2 = 4a_2 = 4yg/(4+y) > 0$, which is consistent, and the common deceleration a_2 is less than g. So the string really does stay taut, and the block and particle move together until they stop.

The block comes to rest after a further time

$$\begin{aligned} t_2 = \frac{V}{a_2} &= \sqrt{\frac{(6-2y)gd}{5}} \cdot \frac{4+y}{yg} \\ &= \sqrt{\frac{d}{5g}}\,\sqrt{6-2y}\,\left(1 + \frac{4}{y}\right), \end{aligned}$$

where we wrote $\sqrt{(6-2y)gd/5}\,/\,g = \sqrt{d/(5g)}\sqrt{6-2y}$ and $(4+y)/y = 1 + 4/y$.

Total time

The block first returns to rest at the end of stage 2, so the total time is

$$t_1 + t_2 = \sqrt{\frac{d}{5g}}\left[\frac{10}{\sqrt{6-2y}} + \left(1+\frac{4}{y}\right)\sqrt{6-2y}\right] = \sqrt{\frac{d}{5g}}\,\mathrm{f}(y),$$

as required.

The stationary value of f

The physical range is $0 < y < 3$. Write $u = 6 - 2y$, so $\mathrm{d}u/\mathrm{d}y = -2$ and

$$\mathrm{f}(y) = 10u^{-1/2} + \left(1+\frac{4}{y}\right)u^{1/2}.$$

Differentiating, with the product rule on the second term,

$$\begin{aligned}\mathrm{f}'(y) &= 10\cdot\left(-\tfrac{1}{2}\right)u^{-3/2}(-2) - \frac{4}{y^2}u^{1/2} + \left(1+\frac{4}{y}\right)\tfrac{1}{2}u^{-1/2}(-2)\\ &= \frac{10}{u^{3/2}} - \frac{4u^{1/2}}{y^2} - \frac{y+4}{y\,u^{1/2}}.\end{aligned}$$

Multiplying by $y^2u^{3/2}$, which is positive for $0 < y < 3$, the condition $\mathrm{f}'(y) = 0$ becomes

$$10y^2 - 4u^2 - (y^2+4y)u = 0.$$

Substituting $u = 6 - 2y$:

$$\begin{aligned}10y^2 - 4(36 - 24y + 4y^2) - (6y^2 - 2y^3 + 24y - 8y^2) &= 0\\ 2y^3 - 4y^2 + 72y - 144 &= 0\\ 2(y-2)(y^2+36) &= 0.\end{aligned}$$

Since $y^2 + 36 > 0$, the only solution is

$$y = 2,$$

which lies in $0 < y < 3$ (and gives $x = 4$).

Because $\mathrm{f}(y) \to \infty$ as $y \to 0^+$ and as $y \to 3^-$, this single stationary point is the minimum: $\mathrm{f}(2) = 10/\sqrt{2} + 3\sqrt{2} = 8\sqrt{2}$, so the shortest possible time to rest is $8\sqrt{2d/(5g)}$.

Checks

- Stepping the motion numerically (time step 2×10^{-6} s) for random $y \in (0,3)$ and d reproduced $\sqrt{d/(5g)}\,\mathrm{f}(y)$ to five significant figures; for example $y = 1.671$, $d = 0.581$ gives 1.27096 s both ways with $g = 9.8$.
- A numerical derivative of f changes sign exactly once on $(0,3)$, at $y = 2$, and $\mathrm{f}(2) = 11.3137\ldots = 8\sqrt{2}$.

- Limiting cases make sense: as $y \to 3$ the masses nearly balance, $a_1 \to 0$ and stage 1 takes forever; as $y \to 0$ nothing slows the block down after the cut, so stage 2 takes forever.

Common slips

- Using mass $4 + x + y$ in stage 2: once the string is cut, the particle of mass x is no longer part of the system.
- Forgetting that the particle of mass y is still moving upwards at speed V after the cut, and so treating stage 2 as starting from rest.
- Dropping $\mathrm{d}u/\mathrm{d}y = -2$ when differentiating $\sqrt{6 - 2y}$.

Beyond STEP

Splitting a motion into stages joined by continuity of velocity is the basic idea behind piecewise-defined dynamics, which reappears in impulsive forces and in control problems where a force is switched on or off. Asking which choice of y minimises the time is a first taste of optimisation in mechanics, which grows into the calculus of variations (the brachistochrone is the classic example). Kleppner and Kolenkow, *An Introduction to Mechanics*, treats connected bodies and constraint forces carefully.

WORKED SOLUTION

STEP I 2006, Question 10

Mechanics • Projectile Motion

STEP I 2006, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q10

The key idea

The trajectory is a parabola and the roof is a straight line, so "just touches" means the line is a *tangent*: substituting one into the other gives a quadratic in x with a repeated root. The sign question is settled by asking where that repeated root lies. For the final part, the particles can only meet where P meets the roof, so Q must arrive at that point at exactly the moment P does; comparing horizontal positions at that time gives the result.

Solution

The trajectory

Model P as a particle moving freely under gravity, with no air resistance. Its initial velocity components are $V\cos 45^\circ = V/\sqrt{2}$ both horizontally and vertically, so at time t

$$x = \frac{Vt}{\sqrt{2}}, \qquad y = \frac{Vt}{\sqrt{2}} - \tfrac{1}{2}gt^2.$$

Eliminating $t = \sqrt{2}\,x/V$ gives $\frac{1}{2}gt^2 = gx^2/V^2$, so the trajectory is

$$y = x - \frac{gx^2}{V^2}.$$

Touching the roof

P meets the roof where $x - gx^2/V^2 = x\tan\alpha + b$, that is, where

$$\frac{g}{V^2}\,x^2 + (\tan\alpha - 1)\,x + b = 0. \qquad (*)$$

Touching (rather than crossing) means $(*)$ has a repeated root, so its discriminant is zero:

$$(\tan\alpha - 1)^2 = \frac{4gb}{V^2} \quad\Longrightarrow\quad V(\tan\alpha - 1) = \pm 2\sqrt{bg}.$$

Ask yourself Both signs satisfy the discriminant condition. What tells them apart?

The repeated root of $(*)$ is

$$x_0 = -\frac{(\tan\alpha - 1)V^2}{2g} = \frac{(1 - \tan\alpha)V^2}{2g}.$$

The particle moves in the direction of increasing x (since $x = Vt/\sqrt{2}$), so it can only touch the roof where $x_0 > 0$. That forces $\tan\alpha - 1 < 0$, and as $V > 0$ the left-hand side $V(\tan\alpha - 1)$ is negative. Hence

$$V(-1 + \tan\alpha) = -2\sqrt{bg},$$

as required. (With the positive root, the tangent point would be at $x_0 < 0$, on the backward continuation of the parabola, which P never visits.) Geometrically, the roof must be less steep than the initial direction of motion, $\alpha < 45°$; otherwise the roof stays above the line $y = x$, and the parabola lies below that line for $x > 0$.

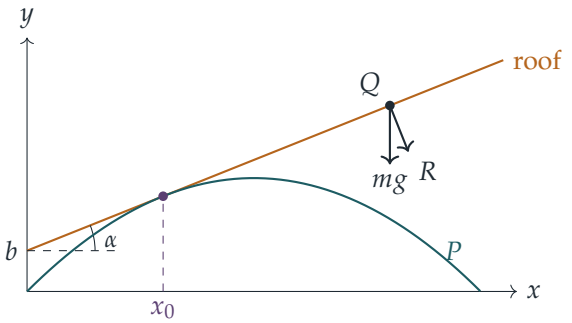
The time of touching

Since $x = Vt/\sqrt{2}$, the touching time is

$$T = \frac{\sqrt{2}\,x_0}{V} = \frac{\sqrt{2}\,(1 - \tan\alpha)V}{2g} = \frac{V(1 - \tan\alpha)}{\sqrt{2}\,g}.$$

The particle on the rail

Model Q as a particle on a smooth rail along the straight line $y = x\tan\alpha + b$; it is projected from $(0, b)$ along the rail in the direction of increasing x (it has to be, to reach P, whose x is always positive). The forces on Q are its weight mg and the normal reaction R of the rail, perpendicular to the rail. The figure shows these forces, together with P's path touching the roof at $x = x_0$.



There is no friction, so resolving along the rail, up the slope,

$$m\ddot{s} = -mg\sin\alpha,$$

where s is the distance of Q along the rail from $(0, b)$. With $s = 0$ and $\dot{s} = U$ at $t = 0$,

$$s = Ut - \tfrac{1}{2}gt^2 \sin\alpha, \qquad x_Q = s\cos\alpha.$$

P is never on the roof except at the touching point, and Q is always on the roof, so the particles can touch only if Q is at that point at time T. Both particles are then on the roof line, so it is enough that their x-coordinates agree at time T:

$$\left(UT - \tfrac{1}{2}gT^2 \sin\alpha\right)\cos\alpha = \frac{VT}{\sqrt{2}}.$$

Dividing by $T > 0$ and substituting $gT = V(1 - \tan\alpha)/\sqrt{2}$:

$$U\cos\alpha = \frac{V}{\sqrt{2}} + \tfrac{1}{2}\sin\alpha\cos\alpha \cdot \frac{V(1-\tan\alpha)}{\sqrt{2}}.$$

Multiplying by $2\sqrt{2}$, and using $\sin\alpha\cos\alpha\tan\alpha = \sin^2\alpha$,

$$\begin{aligned} 2\sqrt{2}\,U\cos\alpha &= 2V + V\sin\alpha\cos\alpha\,(1-\tan\alpha) \\ &= V\left(2 + \sin\alpha\cos\alpha - \sin^2\alpha\right), \end{aligned}$$

which is the required result.

Checks

- Units: $V(1-\tan\alpha)/g$ is a speed divided by an acceleration, so a time; and $\sqrt{bg}$ is a speed, as is $V(\tan\alpha - 1)$.
- Limiting case $\alpha \to 0$ (a flat roof): $b = V^2/(4g)$ is the familiar maximum height $(V\sin 45^\circ)^2/(2g)$, reached at $T = V/(\sqrt{2}g)$; and Q feels no force along the rail, so it must match P's horizontal speed, $U = V/\sqrt{2}$. The formula gives exactly this.
- Numerically, with $g = 9.8$, $V = 10$ and $\alpha = 0.3$ rad, the touching point is at $x_0 \approx 3.524$. Stepping Q along the rail with the derived U for random V and $\alpha < \pi/4$ puts it at P's position at time T to within 10^{-6}, and P stays below the roof at all other times.

Common slips

- Giving the acceleration of Q as $-g$ or $-g\cos\alpha$: only the component of gravity along the rail, $g\sin\alpha$, acts along it.
- Treating U as horizontal speed: U is the speed along the rail, so Q's horizontal displacement is $s\cos\alpha$.
- Justifying the negative root by saying "speeds are positive" only. It is the sign of x_0 (equivalently $\tan\alpha < 1$) that decides it.

Beyond STEP

The tangency condition is a special case of a general idea: the boundary of the region a family of curves can reach is their *envelope*, and finding it amounts to asking when a quadratic has a repeated root. For projectiles launched at a fixed speed in all directions, the envelope is the "parabola of safety". The motion of Q is the simplest constrained motion; first-year courses handle constraints systematically with Lagrangian mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers both projectiles and constrained motion carefully.

WORKED SOLUTION

STEP I 2006, Question 11

Mechanics • Elastic Collisions

> **STEP I 2006, Question 11**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/06-s1-q11

The key idea

We can't follow the collisions one by one: we don't even know the masses of $A_1, \ldots, A_{n-2}$. Instead we look only at the *final* state. Two facts constrain it. Momentum and kinetic energy are the same at the end as at the start. And once the collisions are over, no particle can be catching the one in front of it, so the final velocities increase (weakly) along the trough. Each part is then a short argument about which final states these two facts allow.

Solution

Setting up

We model the particles as points moving in a straight line without friction, with every collision perfectly elastic. Take the direction of A_0's initial motion as positive, and let v_k be the final velocity of A_k ($k = 0, 1, \ldots, n$). Write m_k for the mass of A_k, so $m_0 = m_{n-1} = m$ and $m_n = \lambda m$. Conservation gives

$$\sum_{k=0}^{n} m_k v_k = mu, \tag{M}$$

$$\sum_{k=0}^{n} m_k v_k^2 = mu^2. \tag{E}$$

The particles can't pass one another in the trough, so they stay in the order $A_0, A_1, \ldots, A_n$. If at the end $v_k > v_{k+1}$ for some k, then A_k would catch up with A_{k+1} and there would be another collision. So after all collisions

$$v_0 \leq v_1 \leq \cdots \leq v_n. \tag{O}$$

In particular, a particle moving forwards must have no stationary particle in front of it, and a particle moving backwards must have no stationary particle behind it.

Part (i)

Suppose only A_j moves at the end, with velocity v. Then (M) and (E) give $m_j v = mu$ and $m_j v^2 = mu^2$. Dividing, $v = u$, and then $m_j = m$.

Since $v = u > 0$, the ordering (O) needs every particle in front of A_j to have velocity at least u. They are all at rest, so there are none, and $A_j = A_n$. But A_n has mass $\lambda m \neq m$ because $\lambda > 1$. This is a contradiction, so exactly one particle can't be moving at the end.

Part (ii)

Suppose only A_{n-1} and A_n move, with velocities p and q. Dividing (M) and (E) by m,

$$p + \lambda q = u, \qquad p^2 + \lambda q^2 = u^2.$$

Substituting $p = u - \lambda q$ into the second equation gives $-2\lambda uq + \lambda^2 q^2 + \lambda q^2 = 0$, that is

$$\lambda q\big((1+\lambda)q - 2u\big) = 0.$$

We need $q \neq 0$ because A_n is moving, so

$$q = \frac{2u}{1+\lambda}, \qquad p = u - \frac{2\lambda u}{1+\lambda} = -\frac{(\lambda-1)u}{1+\lambda} < 0.$$

So A_{n-1} would be moving backwards. But $n \geq 2$, so there is a particle A_{n-2} behind it, and by assumption that particle is at rest. Then $v_{n-2} = 0 > p = v_{n-1}$, which contradicts (O): A_{n-1} would run into A_{n-2}. So this final state is impossible too.

Part (iii)

The main tool here is the following fact.

Lemma. At the end of the motion, some particle is moving backwards.

Proof. Suppose instead that $v_k \geq 0$ for every k. By (O), v_n is the largest velocity. Each term of (E) satisfies $m_k v_k^2 \leq v_n\, m_k v_k$, because $0 \leq v_k \leq v_n$ and $m_k > 0$. Adding these and using (M),

$$mu^2 = \sum m_k v_k^2 \leq v_n \sum m_k v_k = v_n\, mu, \qquad \text{so} \qquad v_n \geq u.$$

On the other hand, every term of (M) is non-negative, so the single term $\lambda m v_n$ is at most the total:

$$\lambda m v_n \leq mu, \qquad \text{so} \qquad v_n \leq \frac{u}{\lambda} < u,$$

using $\lambda > 1$ and $u > 0$. These two bounds contradict each other. □

Ask yourself The masses of $A_1, \ldots, A_{n-2}$ are never given. Why doesn't that matter?

The lemma only uses (M), (E), (O) and the mass of A_n, so it holds whatever the other masses are. That is why the question can be answered at all.

Now suppose only A_{n-2}, A_{n-1} and A_n move at the end.

If $n \geq 3$, there is a stationary particle A_{n-3} behind A_{n-2}. By (O), $v_{n-2} \geq v_{n-3} = 0$, and then $v_{n-1}, v_n \geq v_{n-2} \geq 0$. Every particle has velocity ≥ 0, which the lemma forbids.

If $n = 2$, the three particles are A_0, A_1, A_2, which is all of them, and the motion is completely determined. When A_0 hits A_1 (equal masses), the velocities are exchanged: A_0 stops and A_1 moves on with speed u. Then A_1 hits A_2, and by the calculation of part (ii) (with velocities u and 0 before the collision) it rebounds with velocity $-\frac{(\lambda-1)u}{\lambda+1}$ while A_2 moves off with $\frac{2u}{\lambda+1}$. Finally A_1 hits the stationary A_0, and again the equal masses exchange velocities. After that, A_0 moves backwards, A_1 is at rest and A_2 moves forwards, so (O) holds and there are no more collisions. A_1 ends at rest, so again it is impossible for all three to be moving.

Part (iv)

Suppose exactly two particles move at the end. By (M) the total momentum is $mu > 0$, so at least one of them moves forwards. By the lemma at least one moves backwards. So one, say A_i, has $v_i < 0$, and the other, A_j, has $v_j > 0$.

Ask yourself Which two particles can these be?

By (O), every particle behind A_i has velocity $\leq v_i < 0$, so it is moving. Only A_j is also moving, and it moves forwards, so nothing is behind A_i: $A_i = A_0$. Similarly, every particle in front of A_j has velocity $\geq v_j > 0$, so there are none and $A_j = A_n$.

So A_0 (mass m) and A_n (mass λm) are moving, with velocities $a < 0$ and $b > 0$. Equations (M) and (E) are

$$a + \lambda b = u, \qquad a^2 + \lambda b^2 = u^2,$$

exactly the equations of part (ii). Since $b \neq 0$,

$$b = \frac{2u}{\lambda+1}, \qquad a = -\frac{(\lambda-1)u}{\lambda+1}.$$

The two moving particles are therefore A_0, moving backwards with speed $\dfrac{(\lambda-1)u}{\lambda+1}$, and A_n, moving forwards with speed $\dfrac{2u}{\lambda+1}$.

This really can happen: the case $n = 2$ in part (iii) ends in exactly this state.

Checks

- With $\lambda = 3$ and $u = 4$, the speeds are 2 and 2. Then momentum is $m(-2) + 3m(2) = 4m$ and energy is $\frac{1}{2}m(4) + \frac{1}{2}(3m)(4) = 8m = \frac{1}{2}m(16)$, as it should be.
- As $\lambda \to 1$, A_0's final speed tends to 0 and A_n's to u: with equal masses everywhere, the velocity is just passed along the line, as in Newton's cradle.

- We simulated the full sequence of collisions for 60,000 random set-ups (n from 2 to 6, random masses for $A_1, \ldots, A_{n-2}$, random λ and spacings). A_0 always ended moving backwards, the final states in (i)–(iii) never occurred, and every time exactly two particles were moving they were A_0 and A_n with the speeds above.

Common slips

- Trying to track every collision. The intermediate masses are unknown, so only the final state can be analysed.
- Forgetting the ordering condition (O). Conservation alone allows the state in part (ii); it's the ordering that rules it out.
- Assuming A_{n-2} is not A_0. When $n = 2$ it is, and part (iii) needs a separate (short) argument.

Beyond STEP

The lemma is an inequality between two conserved quantities: with all velocities non-negative, the energy can't exceed (largest velocity) $\times$ (momentum). Arguments like this, which use conserved quantities and an ordering to rule out final states without solving the dynamics, are typical of the study of many-particle systems. A good next step is Kleppner and Kolenkow, *An Introduction to Mechanics*, which treats collisions, the centre-of-mass frame and conservation laws carefully.

WORKED SOLUTION

STEP I 2006, Question 12

Probability and Statistics • Conditional Probability

STEP I 2006, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q12

The key idea

A road is open only if *every* section on it is open, and the towns are cut off only if *every* road is blocked. So we work road by road, using independence, and always through complements. The second part is a conditional probability in which the choice of road matters: a one-section road is more likely to be open than a two-section road, so finding your road open tells you something about which road you chose. We handle that with the definition $\mathrm{P}(B \mid A) = \mathrm{P}(A \cap B)/\mathrm{P}(A)$, splitting by the road chosen.

Solution

Probability of being cut off

Call the two-section roads R_1 and R_2 and the one-section road R_3. Let $q = 1 - p$ be the probability that a single section is open.

A two-section road is open exactly when both of its sections are open, which by independence has probability $q^2 = (1-p)^2$. So

$$\mathrm{P}(R_1 \text{ blocked}) = \mathrm{P}(R_2 \text{ blocked}) = 1 - (1-p)^2 = p(2-p),$$
$$\mathrm{P}(R_3 \text{ blocked}) = p.$$

The three roads use five different sections, so the events "R_1 blocked", "R_2 blocked" and "R_3 blocked" depend on disjoint sets of independent sections and are therefore independent. Oxtown is cut off from Camville exactly when all three roads are blocked, so

$$\mathrm{P}(\text{cut off}) = p(2-p) \cdot p(2-p) \cdot p = p^3(2-p)^2,$$

as required.

The conditional probability

Let C_k be the event that I choose road R_k, so $\mathrm{P}(C_k) = \frac{1}{3}$ for each k, and my choice is independent of the flooding. Let O be the event that my road is open, and B the event that the other two roads are both blocked. We want $\mathrm{P}(B \mid O) = \mathrm{P}(B \cap O)/\mathrm{P}(O)$.

> **Ask yourself** Isn't the answer just P(the other two roads blocked), by independence?

It would be if we knew which road we had chosen. But "the other two roads" are a different pair depending on our choice, and the observation that our road is open makes the short road R_3 a more likely choice than before. So we must split by the road chosen and weight correctly, which is exactly what the formula below does.

The denominator. Using the law of total probability over the choice of road,

$$\begin{aligned}\mathrm{P}(O) &= \tfrac{1}{3}(1-p)^2 + \tfrac{1}{3}(1-p)^2 + \tfrac{1}{3}(1-p) \\ &= \tfrac{1}{3}(1-p)\big[2(1-p)+1\big] = \tfrac{1}{3}(1-p)(3-2p).\end{aligned}$$

The numerator. If I chose R_1, I need R_1 open and R_2, R_3 blocked; by independence of the roads this has probability $(1-p)^2 \cdot p(2-p) \cdot p$. Choosing R_2 gives the same. If I chose R_3, I need R_3 open and both two-section roads blocked: $(1-p) \cdot p^2(2-p)^2$. Hence

$$\begin{aligned}\mathrm{P}(B \cap O) &= \tfrac{1}{3}\Big[2p^2(2-p)(1-p)^2 + p^2(2-p)^2(1-p)\Big] \\ &= \tfrac{1}{3}\,p^2(1-p)(2-p)\big[2(1-p)+(2-p)\big] \\ &= \tfrac{1}{3}\,p^2(1-p)(2-p)(4-3p).\end{aligned}$$

The answer. Dividing, and cancelling the factor $\frac{1}{3}(1-p)$, which is non-zero for $p < 1$,

$$\mathrm{P}(B \mid O) = \frac{p^2(2-p)(4-3p)}{3-2p}.$$

Each of the factors p, $2-p$, $4-3p$ and $3-2p$ is linear, so this is fully factorised.

The limit $p \to 1$

As $p \to 1$ every factor tends to 1, so

$$\mathrm{P}(B \mid O) \to \frac{1 \cdot 1 \cdot 1}{1} = 1.$$

This makes sense. When flooding is almost certain, finding my own road open is a lucky fluke, and the other roads, which flood independently of mine, almost certainly did not share my luck: the knowledge that my road is open says nothing about theirs being open. So I should be nearly certain that any other choice would have failed. (At $p = 1$ itself the conditioning event has probability zero and the conditional probability is undefined; only the limit is meaningful.)

Checks

- At $p = 0$ nothing is ever blocked, and the formula gives 0, as it should.
- At $p = \frac{1}{2}$ the formula gives $\dfrac{\frac{1}{4} \cdot \frac{3}{2} \cdot \frac{5}{2}}{2} = \frac{15}{32}$. Enumerating all 2^5 patterns of blocked sections and all three road choices exactly gives the same value, and likewise for several other values of p, such as $p = \frac{2}{3}$, giving $\frac{32}{45}$.
- The answer is always between 0 and 1 for $0 \leq p < 1$: for example at $p = 0.99$ it is about 0.9996.

Common slips

- Taking a two-section road to be blocked with probability p^2 (both sections blocked) rather than $1 - (1 - p)^2$ (at least one blocked).
- Forgetting to condition: the probability of the other two roads being blocked is not the same as its conditional probability given that my randomly chosen road is open.
- Weighting the three choices equally *after* the observation; it is the prior probabilities $\frac{1}{3}$ that are equal, not the posterior ones.

Beyond STEP

Roads in parallel, each made of sections in series, is the simplest example of a *reliability network*; the probability that a network connects two points is studied in general in percolation theory, where the sharp change in behaviour as p varies is a central theme. The second part is Bayes' theorem in disguise: the observation "my road is open" updates the probabilities of which road was chosen. Grimmett and Welsh, *Probability: An Introduction*, covers conditional probability and independence in exactly this spirit.

WORKED SOLUTION

STEP I 2006, Question 13

Probability and Statistics • Poisson Mixtures

STEP I 2006, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q13

The key idea

Diamonds scattered independently at a constant average rate are a Poisson process, so the number of diamonds in m kilograms of chocolate is Poisson with mean m. The amount of chocolate is itself random, so we condition on it: work out the answer for a fixed amount, then average over the die rolls using the law of total probability (and its cousin for expectations). Each part then reduces to summing a series, finite in (i) and geometric in (ii).

Solution

Let D be the number of diamonds I find. Since the diamonds are hidden independently at an average rate of one per kilogram, the number in a bar of mass m kg has the Poisson distribution with mean m. In particular, given a mass of m kg,

$$\mathrm{P}(D = 0) = \mathrm{e}^{-m}, \qquad \mathrm{E}(D) = m.$$

Throughout, $100k$ grams is $0.1k$ kilograms.

Part (i)

Let N be the score on the die, so $\mathrm{P}(N = n) = \frac{1}{6}$ for $n = 1, \ldots, 6$. Given $N = n$, I buy $0.1n$ kg, so $D \mid N = n$ is Poisson with mean $0.1n$. By the law of total probability,

$$\begin{aligned}\mathrm{P}(D = 0) &= \sum_{n=1}^{6} \mathrm{P}(N = n)\,\mathrm{P}(D = 0 \mid N = n) = \frac{1}{6}\sum_{n=1}^{6} \mathrm{e}^{-0.1n} \\ &= \frac{1}{6}\left(\mathrm{e}^{-0.1} + \mathrm{e}^{-0.2} + \cdots + \mathrm{e}^{-0.6}\right).\end{aligned}$$

This is a geometric series with first term $\mathrm{e}^{-0.1}$, common ratio $\mathrm{e}^{-0.1}$ and six terms, so

$$\mathrm{P}(D = 0) = \frac{\mathrm{e}^{-0.1}}{6}\left(\frac{1 - \mathrm{e}^{-0.6}}{1 - \mathrm{e}^{-0.1}}\right),$$

as required.

For the expectation we condition in the same way:

$$\mathrm{E}(D) = \sum_{n=1}^{6} \mathrm{P}(N=n)\,\mathrm{E}(D \mid N=n) = \frac{1}{6}\sum_{n=1}^{6} 0.1n = \frac{0.1\times 21}{6} = 0.35.$$

In other words, $\mathrm{E}(D) = 0.1\,\mathrm{E}(N) = 0.1\times 3.5$, which is what we had to show.

Part (ii)

Now let T be the number of the throw on which the first 6 appears. The throws are independent, and $T=t$ means $t-1$ non-sixes followed by a six, so

$$\mathrm{P}(T=t) = \left(\tfrac{5}{6}\right)^{t-1}\tfrac{1}{6}, \qquad t = 1,2,3,\ldots$$

Given $T=t$, I buy $0.1t$ kg, so $\mathrm{P}(D=0 \mid T=t) = \mathrm{e}^{-0.1t}$. Hence

$$\mathrm{P}(D=0) = \sum_{t=1}^{\infty}\left(\tfrac{5}{6}\right)^{t-1}\tfrac{1}{6}\,\mathrm{e}^{-0.1t} = \frac{\mathrm{e}^{-0.1}}{6}\sum_{t=1}^{\infty}\left(\tfrac{5}{6}\mathrm{e}^{-0.1}\right)^{t-1}.$$

Ask yourself Why may we sum this to infinity?

It is a geometric series with ratio $r = \frac{5}{6}\mathrm{e}^{-0.1}$, and $0<r<1$, so it converges with sum $1/(1-r)$. Therefore

$$\mathrm{P}(D=0) = \frac{\mathrm{e}^{-0.1}}{6}\cdot\frac{1}{1-\frac{5}{6}\mathrm{e}^{-0.1}} = \frac{\mathrm{e}^{-0.1}}{6-5\mathrm{e}^{-0.1}},$$

multiplying top and bottom by 6. This is the required result.

For the expected number of diamonds,

$$\mathrm{E}(D) = \sum_{t=1}^{\infty}\mathrm{P}(T=t)\,\mathrm{E}(D\mid T=t) = \sum_{t=1}^{\infty}\left(\tfrac{5}{6}\right)^{t-1}\tfrac{1}{6}\times 0.1t = \frac{0.1}{6}\sum_{t=1}^{\infty} t\left(\tfrac{5}{6}\right)^{t-1}.$$

This is where the hint comes in. For $|x|<1$ the binomial expansion gives

$$(1-x)^{-2} = 1+2x+3x^2+4x^3+\cdots = \sum_{t=1}^{\infty} t x^{t-1}.$$

With $x=\frac{5}{6}$ the sum is $\left(1-\frac{5}{6}\right)^{-2} = 36$, so

$$\mathrm{E}(D) = \frac{0.1}{6}\times 36 = 0.6.$$

Again this is $0.1\,\mathrm{E}(T)$, where $\mathrm{E}(T)=6$ is the familiar mean number of throws needed to get a 6.

Checks

- Numerically, the probability in (i) is about 0.7150 and in (ii) about 0.6131. A simulation of 2×10^5 shoppers, with Poisson diamond counts, gave 0.7151 and 0.6136, and mean numbers of diamonds 0.350 and 0.599.
- The probability in (ii) is smaller than in (i). That makes sense: on average I buy 600 g rather than 350 g.
- Both expectations are 0.1 times the mean of the random multiplier. Expected diamonds equal expected kilograms, since the expected number of diamonds given the mass is linear in the mass.

Common slips

- Working out $\mathrm{e}^{-0.35}$ (or $\mathrm{e}^{-0.6}$) for the probability of no diamonds. That uses only the average mass, but e^{-m} is not linear in m, so the averaging must come after taking e^{-m}, not before. Expectations are linear, which is why the shortcut does work for $\mathrm{E}(D)$.
- Forgetting to convert grams to kilograms, so getting a mean of $100n$ rather than $0.1n$.
- Writing $\mathrm{P}(T = t) = (\frac{5}{6})^t \frac{1}{6}$, which counts one non-six too many.

Beyond STEP

Here the parameter of a Poisson distribution is itself random, which makes D a *mixture* distribution. The tools used, the law of total probability and the tower law $\mathrm{E}(D) = \mathrm{E}\big(\mathrm{E}(D \mid T)\big)$, are central in first-year probability. Probability generating functions handle such problems neatly: $\mathrm{P}(D = 0)$ is the generating function of T evaluated at $\mathrm{e}^{-0.1}$. Fixing a random number of trials and then a random count in each one leads on to compound distributions and branching processes. Grimmett and Welsh, *Probability: An Introduction*, develops all of these.

WORKED SOLUTION

STEP I 2006, Question 14

Probability and Statistics • Sampling Without Replacement

STEP I 2006, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q14

The key idea

Taking the red sweet on the rth attempt means $r-1$ blues followed by the red. With replacement every draw looks the same, so the probability is a simple product of powers; we then maximise over n by treating n as a continuous variable and differentiating the logarithm. Without replacement the cleanest view is to imagine drawing *all* the sweets: the red one is then equally likely to be in any position.

Solution

Throughout, $n \geq 0$ is an integer, and R is the number of the attempt on which the red sweet is first taken.

Part (i)

Because the sweet is returned and the bag shaken, each draw is independent of the others, and on every draw the red sweet is taken with probability $\frac{1}{n+1}$ and a blue one with probability $\frac{n}{n+1}$. The event $R = r$ means the first $r-1$ draws are blue and the rth is red, so by independence

$$P(R=r) = \left(\frac{n}{n+1}\right)^{r-1} \frac{1}{n+1} = \frac{n^{r-1}}{(n+1)^r}, \qquad r = 1, 2, 3, \ldots$$

(This is the geometric distribution with success probability $\frac{1}{n+1}$.)

Now fix r and think of this as a function of n:

$$f(n) = \frac{n^{r-1}}{(n+1)^r}.$$

If $r = 1$ then $f(n) = \frac{1}{n+1}$, which is largest at $n = 0$ (with no blue sweets, the first sweet is certainly red). So suppose $r \geq 2$. Then $f(0) = 0$ and $f(n) > 0$ for $n > 0$, so we may

take logarithms on $n > 0$, treating n as a real variable:

$$\ln f(n) = (r-1)\ln n - r\ln(n+1), \qquad \frac{\mathrm{d}}{\mathrm{d}n}\ln f(n) = \frac{r-1}{n} - \frac{r}{n+1} = \frac{(r-1)-n}{n(n+1)}.$$

Ask yourself n has to be a whole number. Is it legitimate to differentiate?

Yes, because we use the derivative only to find where f increases and decreases. For $n > 0$ the denominator $n(n+1)$ is positive, so the derivative is positive for $0 < n < r-1$ and negative for $n > r-1$. Hence f is strictly increasing on $[0, r-1]$ and strictly decreasing on $[r-1, \infty)$. Since $r-1$ is itself an integer, it beats every other integer, and there is no need to compare neighbouring integers.

So in every case (including $r = 1$) the probability is maximised by

$$n = r - 1,$$

with maximum value $\dfrac{(r-1)^{r-1}}{r^r}$ (read as 1 when $r = 1$). For example, with $r = 3$ the best bag has $n = 2$ blue sweets and gives probability $\frac{4}{27}$.

Part (ii)

Now the draws are not independent, but there are two easy routes.

Direct product. For $R = r$ we need blue on each of the first $r-1$ draws and then red. This is only possible if there are enough blues, $r - 1 \le n$. Given that the earlier draws were all blue, the kth draw is made from a bag of $n + 2 - k$ sweets, one of them red. So

$$P(R = r) = \frac{n}{n+1} \cdot \frac{n-1}{n} \cdot \frac{n-2}{n-1} \cdots \frac{n-r+2}{n-r+3} \cdot \frac{1}{n-r+2}.$$

Each numerator cancels the next denominator, and the product telescopes to

$$P(R = r) = \begin{cases} \dfrac{1}{n+1}, & 1 \le r \le n+1, \\ 0, & r > n+1. \end{cases}$$

Symmetry. Stopping at the red sweet makes no difference to which position it occupies. So imagine carrying on until the bag is empty. All $(n+1)!$ orders of the (distinguishable) sweets are equally likely, and so the red sweet is equally likely to be in any of the $n+1$ positions. That gives the same answer.

For the maximisation, fix r. If $n < r-1$ the probability is 0. For $n \ge r-1$ it is $\frac{1}{n+1}$, which strictly decreases as n increases. So the probability is maximised by the smallest admissible bag,

$$n = r - 1,$$

and the maximum probability is $\frac{1}{r}$. (With $n = r-1$ the red sweet is equally likely to be any of the r sweets, and drawing it last is one of the r equally likely outcomes.)

Checks

- In (i), $\sum_{r\geq 1} P(R=r) = \frac{1}{n+1}\cdot\dfrac{1}{1-n/(n+1)} = 1$, and in (ii) the $n+1$ equal values $\frac{1}{n+1}$ sum to 1.
- With $r=3$ in (i): $n=1,2,3$ give $\frac{1}{8}$, $\frac{4}{27}\approx 0.148$ and $\frac{9}{64}\approx 0.141$, so $n=2$ wins.
- We checked (ii) by listing every ordering of the sweets for $n\leq 6$. A simulation of (i) with $n=3$, $r=2$ gave 0.188, against the exact $\frac{3}{16}=0.1875$.

Common slips

- Writing $\left(\frac{n}{n+1}\right)^r \frac{1}{n+1}$ in (i): only $r-1$ blues come before the red.
- In (ii), forgetting that the answer is 0 when $r > n+1$. This is exactly what stops "make n as small as possible" from running away to $n=0$.

Beyond STEP

Both maximisations are really *maximum likelihood estimation*. If you see that the red sweet first appears on attempt r but don't know n, the value of n that makes your observation most probable is $n=r-1$ in both experiments. This is a cornerstone idea of first-year statistics. The symmetry argument in (ii) is an example of *exchangeability*, which makes many sampling-without-replacement problems easy. Both ideas are developed in Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP II 2006, Question 1

Pure • Periodic Recurrence Relations

STEP II 2006, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q1

The key idea

Because each term determines the next, the sequence repeats with period p as soon as $u_{p+1} = u_1 = 2$. So every part of (i) comes down to computing a few terms as functions of k and solving one polynomial equation. The equation for period 4 must contain the answers for periods 1 and 2 as factors (a sequence of period 1 or 2 also satisfies $u_5 = u_1$), and knowing this in advance makes the factorisation easy. Part (ii) is a one-line induction followed by the usual "let $n \to \infty$ in the recurrence" argument.

Solution

Part (i)

Write $f(u) = k - \dfrac{36}{u}$, so that $u_{n+1} = f(u_n)$. If $u_{p+1} = u_1$, then applying f repeatedly gives $u_{p+2} = u_2$, $u_{p+3} = u_3$, and so on, so $u_{n+p} = u_n$ for every n. Conversely, period p certainly needs $u_{p+1} = u_1 = 2$. So the sequence has period p *exactly* when $u_{p+1} = 2$ (with all the terms defined, that is, never 0) and no smaller period works.

(a) The sequence is constant if and only if $u_2 = 2$, that is, $k - 18 = 2$. So $k = 20$.

(b) We have $u_2 = k - 18$ and

$$u_3 = k - \frac{36}{k-18} \qquad (k \neq 18).$$

Setting $u_3 = 2$ gives $(k-2)(k-18) = 36$, so $k^2 - 20k = 0$, so $k = 0$ or $k = 20$. The value $k = 20$ gives a constant sequence, which has period 1, not 2. For $k = 0$ the sequence is $2, -18, 2, -18, \ldots$, which does have period 2. So $k = 0$.

(c) We need two more terms. With $k \neq 18$,

$$u_3 = \frac{k(k-18) - 36}{k-18} = \frac{k^2 - 18k - 36}{k-18},$$

and then, provided $k^2 - 18k - 36 \neq 0$,

$$u_4 = k - \frac{36(k-18)}{k^2-18k-36} = \frac{k^3 - 18k^2 - 72k + 648}{k^2 - 18k - 36}.$$

The condition $u_5 = 2$ says $k - 36/u_4 = 2$, that is, $u_4 = \dfrac{36}{k-2}$ (we need $k \neq 2$ here; $k = 2$ would give $36/u_4 = 0$, which is impossible). Cross-multiplying,

$$(k-2)(k^3 - 18k^2 - 72k + 648) = 36(k^2 - 18k - 36).$$

Expanding and collecting everything on the left,

$$k^4 - 20k^3 - 36k^2 + 792k - 1296 = 36k^2 - 648k - 1296,$$
$$k^4 - 20k^3 - 72k^2 + 1440k = 0.$$

Ask yourself How can we factorise a quartic like this without a lot of trial and error?

We know two roots before we start. Any sequence of period 1 or 2 also satisfies $u_5 = u_1$, so $k = 20$ and $k = 0$ must be roots. Taking out k and then $k - 20$,

$$k^4 - 20k^3 - 72k^2 + 1440k = k(k-20)(k^2 - 72).$$

So $k \in \{0, 20, 6\sqrt{2}, -6\sqrt{2}\}$. The values 0 and 20 give periods 2 and 1, so the new values are $k = \pm 6\sqrt{2}$.

Finally we check that the terms are all defined for $k = \pm 6\sqrt{2}$, using $k^2 = 72$:

- $u_2 = k - 18 \neq 0$;
- the numerator of u_3 is $k^2 - 18k - 36 = 36 - 18k$, which is non-zero since $k \neq 2$;
- the numerator of u_4 is $k(k^2 - 72) - 18k^2 + 648 = -1296 + 648 = -648 \neq 0$;
- $u_5 = 2$ by construction, and $k \neq 2$, $k \neq 18$ as required in the working.

Since $k \neq 0, 20$, the period is not 1 or 2, and a sequence with $u_5 = u_1$ whose least period is not 1 or 2 has least period exactly 4 (the least period must divide 4). So the sequence has period 4 exactly when $k = \pm 6\sqrt{2}$.

Part (ii)

Take $k = 37$. We prove $u_n \geq 2$ by induction. It holds for $n = 1$ since $u_1 = 2$. If $u_n \geq 2$, then $u_n > 0$, so $0 < \dfrac{36}{u_n} \leq 18$, and therefore

$$u_{n+1} = 37 - \frac{36}{u_n} \geq 37 - 18 = 19 \geq 2.$$

So $u_n \geq 2$ for all n, which is what we had to show. (In fact $u_n \geq 19$ for all $n \geq 2$.)

Ask yourself The equation for the limit has two roots. Which is the right one, and why can we let $n \to \infty$ in the recurrence?

Since $u_n \to \ell$ and every $u_n \geq 2$, we have $\ell \geq 2$; in particular $\ell \neq 0$, so $36/u_n \to 36/\ell$. Letting $n \to \infty$ on both sides of $u_{n+1} = 37 - 36/u_n$ (the left side also tends to ℓ) gives

$$\ell = 37 - \frac{36}{\ell} \quad \Longrightarrow \quad \ell^2 - 37\ell + 36 = 0 \quad \Longrightarrow \quad (\ell - 1)(\ell - 36) = 0.$$

The root $\ell = 1$ is ruled out by $\ell \geq 2$, so

$$\ell = 36.$$

Checks

- For $k = 6\sqrt{2} \approx 8.485$ the terms are $2, -9.515, 12.269, 5.551, 2, \ldots$, and for $k = -6\sqrt{2}$ they are $2, -26.485, -7.126, -3.433, 2, \ldots$. Both repeat after four steps.
- The factor $k^2 - 72$ has a tidy meaning: the map $u \mapsto (ku - 36)/u$ corresponds to the matrix $\begin{pmatrix} k & -36 \\ 1 & 0 \end{pmatrix}$, with trace k and determinant 36. Its eigenvalues are $6e^{\pm i\theta}$ with $k = 12\cos\theta$, and the fourth power is a multiple of the identity when $\theta = \pi/4$ or $3\pi/4$. This gives $k = \pm 6\sqrt{2}$, and $\theta = \pi/2$ gives $k = 0$ for period 2.
- With $k = 37$: $u_2 = 19$, $u_3 \approx 35.105$, $u_4 \approx 35.975$, $u_5 \approx 35.9993$, and the terms are clearly approaching 36, not 1.

Common slips

- Giving $k = 0, 20$ for (b), or all four roots for (c). The question asks for period 2 and period 4, and a constant sequence does not have period 2 in that sense.
- Taking $\ell = 1$ as well as $\ell = 36$. The bound $u_n \geq 2$ from the first half of (ii) is there to rule it out.
- Letting $n \to \infty$ without noting $\ell \neq 0$, which is needed for $36/u_n \to 36/\ell$.

Beyond STEP

The map $u \mapsto (ku - 36)/u$ is a Möbius transformation, and composing such maps corresponds to multiplying 2×2 matrices. That is why periodicity comes down to the eigenvalues having ratio a root of unity. For $k = 37$ the two fixed points 1 and 36 behave differently: 36 attracts nearby terms and 1 repels them, which you can see from $|f'(\ell)| = 36/\ell^2$. This is the start of the theory of fixed-point iteration and convergence of sequences, covered rigorously in Abbott, *Understanding Analysis*. Needham, *Visual Complex Analysis*, has a beautiful chapter on Möbius transformations and their classification.

WORKED SOLUTION

STEP II 2006, Question 3

Pure • Surds and Approximation

STEP II 2006, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q3

The key idea

The number $5+\sqrt{24}$ has a "conjugate" $5-\sqrt{24}$, and the two multiply to 1. Adding equal powers of a number and its conjugate makes all the surds cancel, so the sum is an integer. The conjugate is small, so its powers are tiny, and the big power must sit just below an integer. Part (ii) is the same argument with 5 replaced by N.

Solution

Part (i)

Let $a=5+\sqrt{24}$ and $b=5-\sqrt{24}$. Then

$$a+b=10, \qquad ab=25-24=1,$$

so $b=1/a$. The expression we want is therefore a^4+b^4. Using $a^2+b^2=(a+b)^2-2ab$ twice:

$$\begin{aligned} a^2+b^2 &= 10^2-2=98,\\ a^4+b^4 &= (a^2+b^2)^2-2(ab)^2=98^2-2=9602. \end{aligned}$$

So $a^4+\dfrac{1}{a^4}=9602$, an integer.

For the inequalities, recall that $\dfrac{1}{a}=b=5-\sqrt{24}$.

- $b>0.1 \iff \sqrt{24}<4.9 \iff 24<24.01$, which is true. (Squaring is reversible here because both sides are positive.)
- $b<\frac{2}{19} \iff \sqrt{24}>5-\frac{2}{19}=\frac{93}{19} \iff 24\cdot 19^2>93^2 \iff 8664>8649$, which is true.
- $\frac{2}{19}<0.11 \iff 2<2.09$, which is true.

So $0.1 < \dfrac{1}{5+\sqrt{24}} < \dfrac{2}{19} < 0.11$.

Ask yourself How do these bounds pin down a^4 to four decimal places?

From the first result, $a^4 = 9602 - b^4$. Raising the positive inequalities $0.1 < b < 0.11$ to the fourth power,

$$0.0001 < b^4 < 0.11^4 = 0.00014641,$$

and therefore

$$9601.99985359 < a^4 < 9601.9999.$$

Every number in this interval lies between 9601.99985 and 9601.9999, so it rounds to

$$\left(5+\sqrt{24}\right)^4 = 9601.9999 \quad \text{(to four decimal places)}.$$

(The sharper bound $b < \frac{2}{19}$ gives $b^4 < \frac{16}{130321} \approx 0.000123$, which is not needed here but shows where $\frac{2}{19}$ comes from: see the Checks.)

Part (ii)

Let $A = N + \sqrt{N^2-1}$ and $B = N - \sqrt{N^2-1}$, both real since $N \geq 2$. As before,

$$A + B = 2N, \qquad AB = N^2 - (N^2-1) = 1.$$

Step 1: $A^k + B^k$ is an integer. Expand both powers by the binomial theorem. Terms with an even power of $\sqrt{N^2-1}$ are integers and appear identically in both; terms with an odd power appear with opposite signs and cancel. So $S_k = A^k + B^k$ is an integer for every positive integer k. (Alternatively, A and B are the roots of $x^2 - 2Nx + 1 = 0$, so $S_k = 2NS_{k-1} - S_{k-2}$ with $S_0 = 2$, $S_1 = 2N$, and induction gives integers.)

Step 2: B is small. We claim $0 < B < \dfrac{1}{2N - \frac{1}{2}}$. Since $B = 1/A$ and $A > 0$, this is equivalent to

$$A > 2N - \tfrac{1}{2} \iff \sqrt{N^2-1} > N - \tfrac{1}{2}.$$

Both sides are positive (as $N \geq 2$), so we may square:

$$N^2 - 1 > N^2 - N + \tfrac{1}{4} \iff N > \tfrac{5}{4},$$

which holds because $N \geq 2$. So $0 < B < \left(2N - \frac{1}{2}\right)^{-1}$, and raising to the kth power (everything positive),

$$0 < B^k < \left(2N - \tfrac{1}{2}\right)^{-k}.$$

Step 3: conclude. We have $A^k = S_k - B^k$, so A^k lies at distance exactly B^k from the integer S_k. The integer nearest to A^k is at least as close as S_k (in fact it is S_k, since $B^k < \frac{1}{2}$). Hence

$$\left|A^k - (\text{nearest integer})\right| \leq B^k < \left(2N - \tfrac{1}{2}\right)^{-k},$$

which is the required result.

Checks

- Numerically, $(5+\sqrt{24})^4 \approx 9601.999896$, inside our interval and rounding to 9601.9999.
- With $N=5$, part (ii) gives $(2N-\frac{1}{2})^{-1} = \frac{1}{9.5} = \frac{2}{19}$: exactly the bound from part (i). So part (i) is the case $N=5$, $k=4$ in disguise.
- For $N=2$, $k=1$: $2+\sqrt{3} \approx 3.732$ is about 0.268 from 4, and $(3.5)^{-1} \approx 0.286$, so the bound holds. A computer checked N up to 1000 and k up to 24.

Common slips

- Squaring $\sqrt{N^2-1} > N-\frac{1}{2}$ without noting both sides are positive.
- Writing $a^4 \approx 9602.0000$: the value is *below* 9602 by between 0.0001 and 0.00015, and that is enough to change the fourth decimal place.
- Forgetting that the case $N=1$ is excluded for a reason: then $B=1$ and the argument collapses.

Beyond STEP

Numbers like $5+\sqrt{24}$ are units in the ring $\mathbb{Z}[\sqrt{6}]$: they have inverses $5-\sqrt{24}$ of the same form, and their powers give all the solutions of the Pell equation $x^2-24y^2=1$. Powers of such units get ever closer to integers, and this is one route to the theory of Pisot numbers, whose powers have the same near-integer behaviour. Davenport's *The Higher Arithmetic* treats Pell's equation and continued fractions beautifully.

WORKED SOLUTION

STEP II 2006, Question 4

Pure • Integration by Substitution

STEP II 2006, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q4

The key idea

The substitution $x = \pi - t$ leaves $\sin x$ unchanged but turns the factor x into $\pi - t$, so the integral reappears on the right-hand side and can be solved for, like an equation. This removes the awkward factor x entirely. After that, every integral in the question is a function of $\sin x$ times $\sin x$ (or $\cos x$), and a substitution $u = \cos x$ or $u = \sin x$ finishes the job. The real work is noticing how to bring parts (ii) and (iii) into the form the identity covers.

Solution

The identity

Let $I = \int_0^\pi x\, f(\sin x)\, \mathrm{d}x$. Put $x = \pi - t$, so $\mathrm{d}x = -\mathrm{d}t$; when $x = 0$, $t = \pi$, and when $x = \pi$, $t = 0$. Since $\sin(\pi - t) = \sin t$,

$$\begin{aligned} I &= \int_\pi^0 (\pi - t)\, f(\sin t)\, (-\mathrm{d}t) = \int_0^\pi (\pi - t)\, f(\sin t)\, \mathrm{d}t \\ &= \pi \int_0^\pi f(\sin t)\, \mathrm{d}t - \int_0^\pi t\, f(\sin t)\, \mathrm{d}t = \pi \int_0^\pi f(\sin x)\, \mathrm{d}x - I, \end{aligned}$$

because the name of the integration variable doesn't matter. Hence $2I = \pi \int_0^\pi f(\sin x)\, \mathrm{d}x$, which is the result we needed:

$$\int_0^\pi x\, f(\sin x)\, \mathrm{d}x = \tfrac{1}{2}\pi \int_0^\pi f(\sin x)\, \mathrm{d}x.$$

Part (i)

Here $f(\sin x) = \dfrac{\sin x}{3 + \sin^2 x}$, so the identity gives

$$\int_0^\pi \frac{x \sin x}{3 + \sin^2 x}\, \mathrm{d}x = \tfrac{1}{2}\pi J, \qquad J = \int_0^\pi \frac{\sin x}{3 + \sin^2 x}\, \mathrm{d}x.$$

Write $3+\sin^2 x = 4-\cos^2 x$ and substitute $u=\cos x$, $\mathrm{d}u = -\sin x\,\mathrm{d}x$, with u running from 1 down to -1:

$$J = \int_{-1}^{1} \frac{\mathrm{d}u}{4-u^2}.$$

By partial fractions, $\dfrac{1}{4-u^2} = \dfrac{1}{4}\left(\dfrac{1}{2-u}+\dfrac{1}{2+u}\right)$, so

$$J = \frac{1}{4}\left[\ln\frac{2+u}{2-u}\right]_{-1}^{1} = \frac{1}{4}(\ln 3 - \ln \tfrac{1}{3}) = \tfrac{1}{2}\ln 3.$$

(Both $2\pm u$ are positive on $[-1,1]$, so no modulus signs are needed.) Therefore

$$\int_0^{\pi} \frac{x\sin x}{3+\sin^2 x}\,\mathrm{d}x = \frac{\pi}{4}\ln 3.$$

Part (ii)

Ask yourself The identity only covers $[0,\pi]$. How do we deal with $[\pi,2\pi]$?

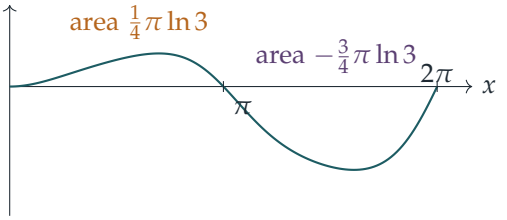
Split the range at π and shift the second piece back to $[0,\pi]$ with $x=\pi+t$. Then $\sin x = -\sin t$ and $\sin^2 x = \sin^2 t$, so

$$\begin{aligned}\int_{\pi}^{2\pi} \frac{x\sin x}{3+\sin^2 x}\,\mathrm{d}x &= -\int_0^{\pi} \frac{(\pi+t)\sin t}{3+\sin^2 t}\,\mathrm{d}t = -\pi J - \int_0^{\pi} \frac{t\sin t}{3+\sin^2 t}\,\mathrm{d}t \\ &= -\tfrac{1}{2}\pi\ln 3 - \tfrac{1}{4}\pi\ln 3 = -\tfrac{3}{4}\pi\ln 3,\end{aligned}$$

using $J=\frac{1}{2}\ln 3$ and part (i). Adding the integral over $[0,\pi]$ from part (i),

$$\int_0^{2\pi} \frac{x\sin x}{3+\sin^2 x}\,\mathrm{d}x = \frac{\pi}{4}\ln 3 - \frac{3\pi}{4}\ln 3 = -\frac{\pi}{2}\ln 3.$$

The sketch shows why the answer is negative: the integrand has the same shape on each half, but on $[\pi,2\pi]$ it is weighted by larger values of x.



Part (iii)

Since $\sin 2x = 2\sin x\cos x$ and $\sin x \geq 0$ on $[0,\pi]$,

$$|\sin 2x| = 2\sin x\,|\cos x| = 2\sin x\sqrt{1-\sin^2 x},$$

which is a function of $\sin x$ alone.

Ask yourself Why is it important that $|\cos x|$, not $\cos x$, appears here?

On $[0, \pi]$, $\cos x$ is *not* a function of $\sin x$: $\cos(\pi - x) = -\cos x$ while $\sin(\pi - x) = \sin x$. But $|\cos x| = \sqrt{1 - \sin^2 x}$ is, so the modulus is exactly what lets us use the identity:

$$\int_0^{\pi} \frac{x|\sin 2x|}{3 + \sin^2 x}\,\mathrm{d}x = \tfrac{1}{2}\pi \int_0^{\pi} \frac{|\sin 2x|}{3 + \sin^2 x}\,\mathrm{d}x.$$

The new integrand is unchanged by $x \mapsto \pi - x$, so the integral over $[0, \pi]$ is twice that over $[0, \frac{1}{2}\pi]$, where $\cos x \geq 0$ and the modulus can be dropped:

$$\int_0^{\pi} \frac{|\sin 2x|}{3 + \sin^2 x}\,\mathrm{d}x = 2\int_0^{\pi/2} \frac{2\sin x \cos x}{3 + \sin^2 x}\,\mathrm{d}x = 2\Big[\ln(3 + \sin^2 x)\Big]_0^{\pi/2} = 2\ln\tfrac{4}{3},$$

since the numerator is the derivative of the denominator. Therefore

$$\int_0^{\pi} \frac{x|\sin 2x|}{3 + \sin^2 x}\,\mathrm{d}x = \tfrac{1}{2}\pi \cdot 2\ln\tfrac{4}{3} = \pi \ln \frac{4}{3}.$$

Checks

- Simpson's rule gives 0.86285 for (i), -1.72570 for (ii) and 0.90378 for (iii), matching $\frac{1}{4}\pi\ln 3$, $-\frac{1}{2}\pi\ln 3$ and $\pi\ln\frac{4}{3}$.
- Answer (ii) is exactly -2 times answer (i). Splitting at π explains this: the two halves are $\frac{1}{4}\pi\ln 3$ and $-\frac{3}{4}\pi\ln 3$.
- A quick test of the identity: with $f \equiv 1$ both sides equal $\frac{1}{2}\pi^2$.

Common slips

- Using the analogous "identity" on $[0, 2\pi]$, which would give $\pi\int_0^{2\pi} f(\sin x)\,\mathrm{d}x = 0$. The proof needs $\sin(2\pi - t) = \sin t$, which is false.
- In (iii), writing $|\sin 2x| = 2\sin x\cos x$ throughout $[0, \pi]$, which gives 0 for the inner integral.

Beyond STEP

The trick of making an integral reappear and solving for it, and the use of symmetries like $x \mapsto \pi - x$, are standard tools in analysis and in evaluating definite integrals that have no elementary antiderivative. They lead to Fourier series, where integrals of x times trigonometric functions over a period are the basic objects. For more on this style of argument, see Siklos, *Advanced Problems in Mathematics*, and for the rigorous theory of the integral, Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2006, Question 5

Pure • Floor Function Areas

STEP II 2006, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q5

The key idea

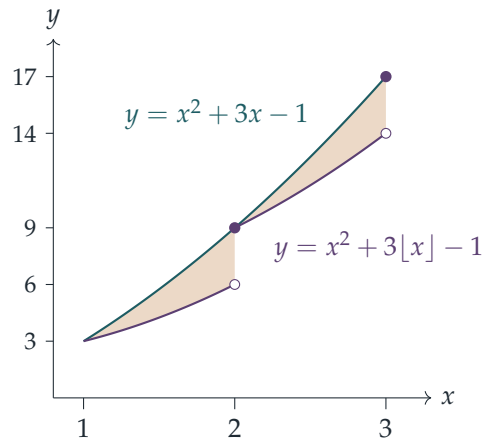
Between two consecutive integers $k \leq x < k+1$ the floor function is constant, equal to k. So on each such unit interval the jagged curve is an ordinary, smooth curve, and the area splits into a sum of strips, one per unit interval. Work out the area of the strip over $[k, k+1]$ as a function of k, then add up the strips for $k = 1, 2, \ldots, n-1$.

Solution

Throughout, write $f(x) = x^2 + 3x - 1$. On $1 \leq x \leq 3$ it rises from $f(1) = 3$ through $f(2) = 9$ to $f(3) = 17$, and it is increasing for all $x \geq 1$ since $f'(x) = 2x + 3 > 0$ there.

Part (i)

The second curve is $y = x^2 + 3\lfloor x \rfloor - 1$. On $[1, 2)$ we have $\lfloor x \rfloor = 1$, so it is $y = x^2 + 2$, rising from 3 towards 6; on $[2, 3)$ we have $\lfloor x \rfloor = 2$, so it is $y = x^2 + 5$, rising from 9 towards 14; and at $x = 3$ it equals 17. Each piece is the parabola $y = f(x)$ shifted down, and the two curves meet at every integer.



The vertical gap between the curves is

$$(x^2 + 3x - 1) - (x^2 + 3\lfloor x \rfloor - 1) = 3(x - \lfloor x \rfloor) \geq 0,$$

so the first curve is never below the second. On the interval $[k, k+1]$, with k an integer, the gap is $3(x-k)$ except at the single point $x = k+1$, which does not affect an integral. So the strip over $[k, k+1]$ has area

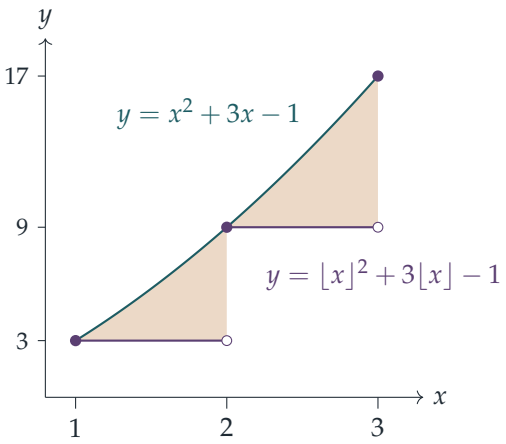
$$\int_k^{k+1} 3(x-k)\,\mathrm{d}x = \left[\tfrac{3}{2}(x-k)^2\right]_k^{k+1} = \tfrac{3}{2},$$

the same for every k. The interval $1 \leq x \leq n$ is made up of the $n-1$ unit intervals $[1,2], [2,3], \ldots, [n-1, n]$, so the area between the curves is

$$\tfrac{3}{2}(n-1).$$

Part (ii)

Now the second curve is $y = f(\lfloor x \rfloor)$, a step function: $y = f(1) = 3$ on $[1,2)$, $y = f(2) = 9$ on $[2,3)$, and $y = f(3) = 17$ at $x = 3$. Each step starts on the parabola at an integer and runs horizontally to the right.



Ask yourself Which curve is on top, and does it stay on top?

Since f is increasing for $x \geq 1$ and $\lfloor x \rfloor \leq x$, we have $f(\lfloor x \rfloor) \leq f(x)$ throughout $1 \leq x \leq n$. So the parabola is always on top and the area is simply the integral of $f(x) - f(\lfloor x \rfloor)$, with no sign changes to worry about.

On $[k, k+1]$, where $k \geq 1$ is an integer, the step sits at height $f(k)$ (again ignoring the single point $x = k+1$). Substituting $x = k + t$ with $0 \leq t \leq 1$,

$$\begin{aligned} f(k+t) - f(k) &= (k+t)^2 + 3(k+t) - 1 - (k^2 + 3k - 1) \\ &= 2kt + t^2 + 3t, \end{aligned}$$

so the strip over $[k, k+1]$ has area

$$\int_0^1 (2kt + t^2 + 3t)\,\mathrm{d}t = k + \tfrac{1}{3} + \tfrac{3}{2} = k + \tfrac{11}{6}.$$

Unlike part (i), the strips now grow with k, because the parabola gets steeper. Adding the strips for $k = 1, 2, \ldots, n-1$ and using $\sum_{k=1}^{n-1} k = \frac{1}{2}n(n-1)$,

$$\begin{aligned} \text{Area} &= \sum_{k=1}^{n-1} \left(k + \tfrac{11}{6}\right) = \tfrac{1}{2}n(n-1) + \tfrac{11}{6}(n-1) \\ &= \tfrac{1}{6}(n-1)(3n+11), \end{aligned}$$

which is the required result.

Checks

- With $n = 1$ there is no region at all, and both answers give 0.
- With $n = 2$, part (ii) gives $\frac{1}{6} \cdot 1 \cdot 17 = \frac{17}{6}$. Directly, $\int_1^2 (x^2 + 3x - 4)\,\mathrm{d}x = \frac{7}{3} + \frac{9}{2} - 4 = \frac{17}{6}$.
- A midpoint-rule integration of the gaps for $n = 2, \ldots, 7$ matched $\frac{3}{2}(n-1)$ and $\frac{1}{6}(n-1)(3n+11)$ to four decimal places; for example, $n = 4$ gives 11.5 in part (ii).

Common slips

- Counting n strips instead of $n-1$: the interval $[1, n]$ contains only $n-1$ unit intervals.
- In part (ii), using $\int_0^1 2kt\,\mathrm{d}t = 2k$ instead of k.
- Integrating $\lfloor x \rfloor$ as if it were x (or ignoring it) over a whole interval, rather than splitting at the integers first.

Beyond STEP

Part (ii) compares the area under a curve with the area of a "staircase" of rectangles under it, which is exactly a lower Riemann sum with step width 1. In a first analysis course, the integral is *defined* by squeezing such lower and upper sums together as the steps get narrower, and functions with jumps, like $\lfloor x \rfloor$, are still integrable because their jumps are isolated points. Abbott's *Understanding Analysis* gives a very readable account of the Riemann integral.

WORKED SOLUTION

STEP II 2006, Question 7

Pure • Tangents to an Ellipse

STEP II 2006, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q7

The key idea

The tangent comes quickest from the "replace x^2 by xx_0 and y^2 by yy_0" form, or equally from implicit differentiation. For the construction, just work out the coordinates of Q and the equation of PQ, then compare with the tangent at a general point. The words $\tan(\alpha/2) = k$ are a strong hint: they tell you to write $\cot\alpha$ and $\operatorname{cosec}\alpha$ in terms of $t = \tan(\alpha/2)$.

Solution

The tangent

Differentiating $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ implicitly with respect to x gives

$$\frac{2x}{a^2} + \frac{2y}{b^2}\frac{\mathrm{d}y}{\mathrm{d}x} = 0, \qquad \text{so} \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{b^2x}{a^2y}.$$

At $(a\cos\alpha, b\sin\alpha)$, with $\sin\alpha \neq 0$ (otherwise the tangent is vertical and the stated equation makes no sense), the gradient is

$$-\frac{b^2 a\cos\alpha}{a^2 b\sin\alpha} = -\frac{b\cot\alpha}{a}.$$

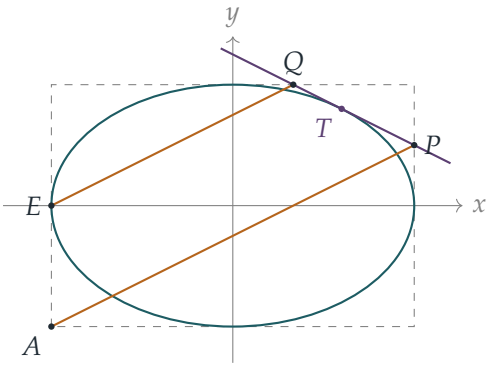
So the tangent is

$$\begin{aligned} y - b\sin\alpha &= -\frac{b\cot\alpha}{a}(x - a\cos\alpha) \\ \implies \quad y &= -\frac{b\cot\alpha}{a}x + b\frac{\cos^2\alpha}{\sin\alpha} + b\sin\alpha = -\frac{b\cot\alpha}{a}x + \frac{b(\cos^2\alpha + \sin^2\alpha)}{\sin\alpha}, \end{aligned}$$

that is, $y = -\dfrac{b\cot\alpha}{a}x + b\operatorname{cosec}\alpha$, as required.

The construction

Here is the picture with $a = 3$, $b = 2$, $k = \frac{1}{2}$. The ellipse sits inside the rectangle with corners $(\pm a, \pm b)$; A is the bottom-left corner, E the midpoint of the left side, P is on the right side and Q on the top side.



The line AP joins $(-a, -b)$ to (a, kb), so its gradient is

$$\frac{kb - (-b)}{a - (-a)} = \frac{(1+k)b}{2a}.$$

The parallel line through $E(-a, 0)$ is $y = \dfrac{(1+k)b}{2a}(x + a)$. Putting $y = b$ gives $x + a = \dfrac{2a}{1+k}$, so

$$Q = \left(\frac{a(1-k)}{1+k},\ b\right).$$

Now the gradient of PQ is

$$\frac{b - kb}{\dfrac{a(1-k)}{1+k} - a} = \frac{b(1-k)}{\dfrac{-2ak}{1+k}} = -\frac{b(1-k^2)}{2ak},$$

which is fine because $k \neq 0$. Using the point $P(a, kb)$, the line PQ is

$$y = kb - \frac{b(1-k^2)}{2ak}(x - a) = -\frac{b(1-k^2)}{2ak}\,x + \frac{2k^2 b + b(1-k^2)}{2k},$$

so

$$PQ: \qquad y = -\frac{b}{a} \cdot \frac{1-k^2}{2k}\,x + b \cdot \frac{1+k^2}{2k}.$$

Ask yourself How do we recognise this as the tangent at some point of the ellipse?

Compare it with the tangent at $(a\cos\alpha, b\sin\alpha)$ found above. We need an α with $\cot\alpha = \dfrac{1-k^2}{2k}$ and $\operatorname{cosec}\alpha = \dfrac{1+k^2}{2k}$. These are exactly the t-formulae: with $t = \tan(\alpha/2)$,

$$\tan\alpha = \frac{2t}{1-t^2}, \qquad \sin\alpha = \frac{2t}{1+t^2}, \qquad \text{so} \qquad \cot\alpha = \frac{1-t^2}{2t}, \qquad \operatorname{cosec}\alpha = \frac{1+t^2}{2t}.$$

Since $0 < k < 1$, we can choose $\alpha = 2\arctan k$, which lies in $(0, \frac{\pi}{2})$, so $\sin\alpha \neq 0$ and $\tan(\alpha/2) = k$. With $t = k$ the two expressions match both the gradient and the intercept of PQ. So PQ is the line $y = -\frac{b\cot\alpha}{a}x + b\operatorname{cosec}\alpha$, which is the tangent to the ellipse at $(a\cos\alpha, b\sin\alpha)$ where $\tan(\alpha/2) = k$. This is what we had to show.

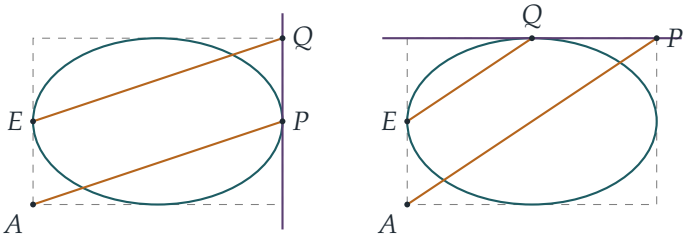
In coordinates, the point of contact is $T = \left(\dfrac{a(1-k^2)}{1+k^2}, \dfrac{2kb}{1+k^2}\right)$.

The cases $k = 0$ and $k = 1$

The algebra above divided by k, and the tangent formula needs $\sin\alpha \neq 0$, so we redo the construction directly.

$k = 0$. Now $P = (a, 0)$, and AP has gradient $\frac{b}{2a}$. The parallel line through E is $y = \frac{b}{2a}(x + a)$, which meets $y = b$ at $x = a$, so $Q = (a, b)$. Then PQ is the vertical line $x = a$. That is the tangent to the ellipse at $(a, 0)$, the point with $\alpha = 0$, and $\tan(0/2) = 0 = k$. So the result *does* still hold, even though the tangent can't be written in the form $y = mx + c$.

$k = 1$. Now $P = (a, b)$, and AP is the diagonal of the rectangle, with gradient $\frac{b}{a}$. The parallel line through E is $y = \frac{b}{a}(x + a)$, which meets $y = b$ at $x = 0$, so $Q = (0, b)$. Then PQ is the horizontal line $y = b$, the tangent at the top of the ellipse $(0, b)$. This is the point with $\alpha = \frac{\pi}{2}$, and $\tan(\pi/4) = 1 = k$. So the result holds here too (and the formula works: $\cot\frac{\pi}{2} = 0$ and $\operatorname{cosec}\frac{\pi}{2} = 1$ give $y = b$).



Checks

- Take $a = 3$, $b = 2$, $k = \frac{1}{2}$. Then $Q = (1, 2)$ and PQ is $y = -\frac{1}{2}x + \frac{5}{2}$. Also $\tan\alpha = \frac{4}{3}$, so $\cos\alpha = \frac{3}{5}$, $\sin\alpha = \frac{4}{5}$ and $T = (\frac{9}{5}, \frac{8}{5})$. This lies on PQ ($-\frac{9}{10} + \frac{5}{2} = \frac{8}{5}$), and the gradient there is $-\frac{b^2x}{a^2y} = -\frac{4\cdot 9/5}{9\cdot 8/5} = -\frac{1}{2}$.
- Substituting PQ into the ellipse equation should give a quadratic with a repeated

root. We checked this, and the formulae for Q and PQ, numerically for a thousand random a, b, k.

- As $k \to 0^+$, $T \to (a, 0)$, and as $k \to 1^-$, $T \to (0, b)$. This agrees with the two end cases.

Common slips

- Getting the sign wrong in the gradient of PQ: the denominator $\frac{a(1-k)}{1+k} - a$ is negative.
- Saying the result fails at $k = 0$ because $\operatorname{cosec} 0$ is undefined. The *formula* breaks down, but the geometry doesn't: PQ is still the tangent at $\alpha = 0$.

Beyond STEP

The substitution $t = \tan(\alpha/2)$ is a rational parametrisation of the ellipse: $\left(a\frac{1-t^2}{1+t^2}, b\frac{2t}{1+t^2}\right)$. It is the same idea that turns trigonometric integrals into rational ones and finds all Pythagorean triples. Every conic can be parametrised rationally like this, and constructions like the one here, built only from parallels and intersections, are the subject of projective geometry. For a first look at how this connects with number theory, try Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP II 2006, Question 8

Pure • Vector Geometry of Lines

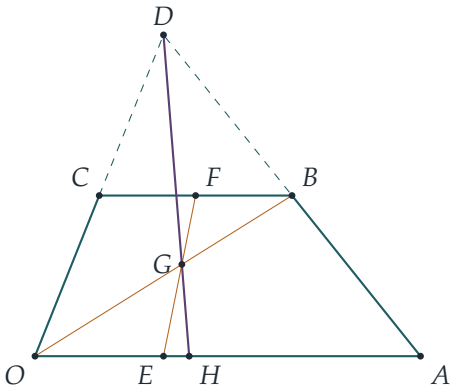
STEP II 2006, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q8

The key idea

Every point in the question is defined as the intersection of two lines, so the whole question runs on one routine: write each line in the form from the first part, set the two expressions equal, and compare coefficients of $\mathbf{a}$ and $\mathbf{c}$. That comparison is legitimate because $\mathbf{a}$ and $\mathbf{c}$ are not parallel, so a vector can be written as $p\mathbf{a} + q\mathbf{c}$ in only one way.



Solution

The equation of a line

The line through the points with position vectors $\mathbf{x}$ and $\mathbf{y}$ passes through $\mathbf{x}$ and has direction $\mathbf{y} - \mathbf{x}$, so its points are

$$\mathbf{r} = \mathbf{x} + \alpha(\mathbf{y} - \mathbf{x}) = (1 - \alpha)\mathbf{x} + \alpha\mathbf{y},$$

as α runs through all real numbers. Note that $\alpha = 0$ gives $\mathbf{x}$, $\alpha = 1$ gives $\mathbf{y}$, and the coefficients always add up to 1. That is the required form.

Throughout, $\mathbf{a}$ and $\mathbf{c}$ are not parallel (otherwise O, A, C would be collinear and there

would be no trapezium), so if $p\mathbf{a}+q\mathbf{c}=p'\mathbf{a}+q'\mathbf{c}$ then $p=p'$ and $q=q'$.

Part (i)

Since CB is parallel to OA, the vector $\overrightarrow{CB}$ is a scalar multiple of $\overrightarrow{OA}=\mathbf{a}$; call it $\lambda\mathbf{a}$. Then

$$\mathbf{b}=\overrightarrow{OC}+\overrightarrow{CB}=\mathbf{c}+\lambda\mathbf{a}.$$

Going round $OABC$ in order, $\overrightarrow{CB}$ points the same way as $\overrightarrow{OA}$, so $\lambda>0$; and $CB<OA$ gives $\lambda<1$. So $0<\lambda<1$, and in particular $\lambda\neq 1$, which we shall need.

Part (ii)

E **and** F**.** These come straight from the ratios:

$$\mathbf{e}=\tfrac{1}{3}\mathbf{a},\qquad \mathbf{f}=\tfrac{1}{2}(\mathbf{c}+\mathbf{b})=\tfrac{1}{2}\lambda\mathbf{a}+\mathbf{c}.$$

D**.** Points of line OC are $\mu\mathbf{c}$. Points of line AB are

$$(1-\beta)\mathbf{a}+\beta(\lambda\mathbf{a}+\mathbf{c})=(1-\beta+\beta\lambda)\mathbf{a}+\beta\mathbf{c}.$$

At D these agree. Comparing coefficients gives $\beta=\mu$ and $1-\beta(1-\lambda)=0$, so $\beta=\dfrac{1}{1-\lambda}$ (fine, since $\lambda\neq 1$). Hence

$$\mathbf{d}=\frac{1}{1-\lambda}\,\mathbf{c}.$$

Since $\frac{1}{1-\lambda}>1$, D does lie on OC produced beyond C, as it should.

G**.** Points of OB are $t(\lambda\mathbf{a}+\mathbf{c})$. Points of EF are

$$(1-s)\tfrac{1}{3}\mathbf{a}+s\big(\tfrac{1}{2}\lambda\mathbf{a}+\mathbf{c}\big)=\big(\tfrac{1}{3}-\tfrac{1}{3}s+\tfrac{1}{2}\lambda s\big)\mathbf{a}+s\mathbf{c}.$$

Comparing $\mathbf{c}$-coefficients gives $s=t$; then the $\mathbf{a}$-coefficients give

$$\lambda t=\tfrac{1}{3}-\tfrac{1}{3}t+\tfrac{1}{2}\lambda t\quad\Longrightarrow\quad t\Big(\frac{3\lambda+2}{6}\Big)=\frac{1}{3}\quad\Longrightarrow\quad t=\frac{2}{3\lambda+2}.$$

So

$$\mathbf{g}=\frac{2}{3\lambda+2}\,(\lambda\mathbf{a}+\mathbf{c}).$$

H**.** Points of line DG are

$$(1-k)\mathbf{d}+k\mathbf{g}=\frac{2k\lambda}{3\lambda+2}\,\mathbf{a}+\Big(\frac{1-k}{1-\lambda}+\frac{2k}{3\lambda+2}\Big)\mathbf{c}.$$

Ask yourself How do we pick out the point of this line that lies on OA?

Points of line OA are multiples of $\mathbf{a}$ alone, so we need the $\mathbf{c}$-coefficient to vanish. Multiplying by $(1-\lambda)(3\lambda+2)$,

$$\begin{aligned}(1-k)(3\lambda+2)+2k(1-\lambda)&=0\\(3\lambda+2)-k\big(3\lambda+2-2+2\lambda\big)&=0\\k&=\frac{3\lambda+2}{5\lambda}.\end{aligned}$$

The $\mathbf{a}$-coefficient is then

$$\frac{2\lambda}{3\lambda+2}\cdot\frac{3\lambda+2}{5\lambda}=\frac{2}{5},\qquad\text{so}\qquad\mathbf{h}=\tfrac{2}{5}\,\mathbf{a}.$$

Collecting the answers:

$$\mathbf{d}=\frac{\mathbf{c}}{1-\lambda},\quad\mathbf{e}=\tfrac{1}{3}\mathbf{a},\quad\mathbf{f}=\tfrac{1}{2}\lambda\mathbf{a}+\mathbf{c},\quad\mathbf{g}=\frac{2(\lambda\mathbf{a}+\mathbf{c})}{3\lambda+2},\quad\mathbf{h}=\tfrac{2}{5}\mathbf{a}.$$

Since $OH=\frac{2}{5}OA$ and $HA=\frac{3}{5}OA$,

$$OH:HA=2:3,$$

whatever the shape of the trapezium.

Checks

- Take $\mathbf{a}=(6,0)$, $\mathbf{c}=(1,2.5)$ and $\lambda=\frac{1}{2}$ (the diagram). The formulas give $D=(2,5)$, $G\approx(2.286,1.429)$ and $H=(2.4,0)$; the slope from D to G is $-3.571/0.286=-12.5$, and so is the slope from D to H, namely $-5/0.4$. So D, G, H really are collinear.
- The answer for H doesn't depend on λ, $\mathbf{a}$ or $\mathbf{c}$. A computer repeated the whole construction in exact rational arithmetic for 2000 random trapezia and found $OH:HA=2:3$ every time.
- As $\lambda\to1$ the sides OC and AB become parallel, and $\mathbf{d}=\mathbf{c}/(1-\lambda)$ runs off to infinity, as it should.

Common slips

- Using the same parameter letter for two different lines. Each line needs its own parameter before you set them equal.
- Comparing coefficients without saying why it's allowed: it works because $\mathbf{a}$ and $\mathbf{c}$ are not parallel.
- Writing $\mathbf{b}=\mathbf{a}+\lambda\mathbf{c}$. It is $\overrightarrow{CB}$, not $\overrightarrow{AB}$, that is parallel to $\mathbf{a}$.

Beyond STEP

The answer $2:3$ doesn't depend on the shape of the trapezium because every step uses only lines, intersections and ratios along lines, and an affine map (a linear map

followed by a translation) preserves all of these. So you could send the trapezium to a convenient one, say with $OABC$ having C directly above O, and compute there. This is the idea of *affine geometry*; the coefficient form $(1-\alpha)\mathbf{x}+\alpha\mathbf{y}$ is the start of barycentric coordinates. Strang's *Introduction to Linear Algebra* covers linear maps and coordinates, and Siklos's *Advanced Problems in Mathematics* has more vector geometry questions in this style.

WORKED SOLUTION

STEP II 2006, Question 9

Mechanics • Friction and Toppling

STEP II 2006, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

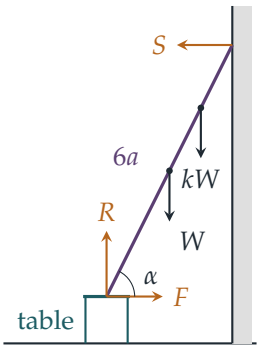
step.maths.org/questions/06-s2-q9

The key idea

Treat the ladder and the table as two separate bodies, linked by the contact at the foot of the ladder. The ladder alone tells us the reaction and the friction at its foot, and these depend on how high the painter is. Then Newton's third law hands the same forces, reversed, to the table. For the table we compare two thresholds: the push that makes it slide, and the push whose moment about the far legs beats the moment of the weight on it.

Solution

We model the ladder as a uniform rod, the painter as a particle, the wall as smooth and the table as a rigid body with its top at height a (we ignore its thickness). Write $\alpha = \arctan 2$, so $\tan\alpha = 2$, $\sin\alpha = \frac{2}{\sqrt{5}}$ and $\cos\alpha = \frac{1}{\sqrt{5}}$. Let the painter stand a distance x along the ladder from its foot, where $0 \le x \le 6a$.



Part (i)

The forces on the ladder are: its weight W at its midpoint, the painter's weight kW, the horizontal reaction S from the smooth wall, and at the foot a normal reaction R

upwards and a friction force F towards the wall. (Friction must point towards the wall, since it is the only horizontal force that can balance S.)

Resolving vertically and horizontally,

$$R = (k+1)W, \qquad F = S.$$

Taking moments about the foot of the ladder, the top is at height $6a\sin\alpha$ above the foot and the weights act at horizontal distances $3a\cos\alpha$ and $x\cos\alpha$, so

$$S \cdot 6a\sin\alpha = W \cdot 3a\cos\alpha + kW \cdot x\cos\alpha.$$

Dividing by $6a\sin\alpha$ and using $\cot\alpha = \frac{1}{2}$,

$$F = S = \frac{W}{12}\Big(3 + \frac{kx}{a}\Big).$$

This increases with x, so the largest friction is needed when the painter is at the very top, $x = 6a$:

$$F_{\max} = \frac{(3+6k)W}{12} = \frac{(2k+1)W}{4}.$$

The greatest friction available is

$$\tfrac{1}{2}R = \frac{(k+1)W}{2} = \frac{(2k+2)W}{4}.$$

Since $2k+1 < 2k+2$, we have $F \le F_{\max} < \frac{1}{2}R$ for every position of the painter, and for every k. With the table fixed, the ladder therefore never slips, however high the painter stands, as required.

Part (ii)

Now $k = 9$, so $R = 10W$ and

$$F = \frac{W}{4}\Big(1 + \frac{3x}{a}\Big),$$

which rises from $\frac{1}{4}W$ at $x = 0$ to $\frac{19}{4}W$ at $x = 6a$. Part (i) still applies, so the ladder does not slip on the table top.

By Newton's third law the ladder pushes on the table top, at its centre, with a force $R = 10W$ downwards and a horizontal force F *away from* the wall. The table also has its own weight $2W$ acting at its centre, and the ground acts on its legs with normal reactions and friction.

Ask yourself How can a horizontal push at the centre of the table top make the table tilt?

The push F acts at height a, so it has a moment about the bottom of the two legs furthest from the wall. If the table tilts, it rotates about the line joining the feet of those legs. The legs are at the corners of a square of side $\frac{1}{2}a$, so that line is a horizontal distance $\frac{1}{4}a$ from the centre of the table.

Tilting. About the line of the far legs' feet, the ground's forces on the far legs have no moment, and nor does ground friction on any leg (it acts at ground level). The downward forces $10W + 2W = 12W$ at the centre have a restoring moment $12W \cdot \frac{1}{4}a = 3Wa$. The push F has an overturning moment Fa. Taking moments about this line, the normal reaction N_1 (in total) on the two legs nearest the wall satisfies

$$N_1 \cdot \tfrac{1}{2}a = 3Wa - Fa,$$

so $N_1 \geq 0$ exactly when $F \leq 3W$. The table is about to tilt when $N_1 = 0$, that is when

$$\frac{W}{4}\left(1 + \frac{3x}{a}\right) = 3W \quad \Longleftrightarrow \quad x = \frac{11a}{3}.$$

Slipping. Resolving vertically for the table, the total normal reaction from the ground is $12W$, so the greatest total friction available is $\frac{1}{3} \cdot 12W = 4W$. Resolving horizontally, the friction needed is F. So the table is about to slip when

$$\frac{W}{4}\left(1 + \frac{3x}{a}\right) = 4W \quad \Longleftrightarrow \quad x = 5a.$$

Both values lie within the ladder's length $6a$, and F increases steadily as the painter climbs. It reaches $3W$ (at $x = \frac{11}{3}a$) before it reaches $4W$ (at $x = 5a$). So **the table tilts first**. This happens when the painter is $\frac{11}{3}a$ along the ladder, a height $\frac{11}{3}a \sin\alpha = \frac{22a}{3\sqrt{5}} \approx 3.28a$ above the table top. At that moment the friction needed at the ground is only $3W$, less than the $4W$ available, so the table has not slipped.

Checks

- With no painter on the ladder ($x = 0$), $F = \frac{1}{4}W$, and the near legs carry $N_1 = \frac{11}{2}W$ while the far legs carry $\frac{13}{2}W$. The far legs take slightly more, as they should with the push directed away from the wall.
- A computer computes the forces for many values of k and x. In (i) the largest ratio of friction needed to friction available is just under 1 (it is $\frac{2k+1}{2k+2}$ at the top). In (ii) it finds the near-leg reaction vanishing at $x \approx 3.667a$ and the ground friction reaching its limit at $x = 5a$.

Common slips

- Taking the tilting axis at the table's edge a distance $\frac{1}{2}a$ from the centre: the table top has side $\frac{1}{2}a$, so the half-width is $\frac{1}{4}a$.
- Forgetting that the push on the table acts at height a, the top of the legs, not at ground level. It is this height that gives the push its overturning moment.
- Leaving out the ladder and painter's $10W$ when finding the ground's normal reaction on the table. Only the table's own weight $2W$ is not enough.

Beyond STEP

This question is about how a system of rigid bodies can fail in competing ways: by sliding, or by toppling. The method is to find the threshold for each and see which the load reaches first. Engineers apply the same idea to retaining walls, cranes and stacked loads, and in statics courses it leads on to the principle of virtual work. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body equilibrium and friction with care.

WORKED SOLUTION

STEP II 2006, Question 10

Mechanics • Collisions and Restitution

STEP II 2006, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q10

The key idea

Each collision is a two-equation problem: conservation of momentum along the line, and Newton's law of restitution (speed of separation equals e times speed of approach). Solve the A–B collision, then feed B's new velocity into the B–C collision. A second A–B collision happens exactly when A, which is behind B, ends up moving faster in the forward direction than B does after B has hit C. Part (ii) is then a pursuit problem with constant velocities.

Solution

We model A, B and C as particles moving along a straight line on a smooth surface, so no horizontal forces act except during collisions. Take the direction in which A is projected as positive. All velocities below are signed.

Part (i)

Collision of A and B. Let A and B have velocities v_A and v_B just afterwards. Momentum and restitution (with $e = \frac{1}{2}$) give

$$mv_A + km\,v_B = mu, \qquad v_B - v_A = \tfrac{1}{2}u.$$

Substituting $v_A = v_B - \frac{1}{2}u$ into the first equation gives $(1+k)v_B = \frac{3}{2}u$, so

$$v_B = \frac{3u}{2(1+k)}, \qquad v_A = v_B - \frac{u}{2} = \frac{(2-k)\,u}{2(1+k)}.$$

Since $v_B > 0$, B does move on towards C, and since $v_A < v_B$, A cannot catch B before B reaches C.

Collision of B and C. Let B and C have velocities w_B and w_C just afterwards. With $e = \frac{1}{4}$,

$$km\,w_B + 3m\,w_C = km\,v_B, \qquad w_C - w_B = \tfrac{1}{4}v_B.$$

Substituting $w_C = w_B + \frac{1}{4}v_B$ gives $(k+3)w_B = \left(k - \frac{3}{4}\right)v_B$, so

$$w_B = \frac{(4k-3)\,v_B}{4(k+3)} = \frac{3(4k-3)\,u}{8(1+k)(k+3)}.$$

Also $w_C > w_B$, so B and C separate and never meet again. After this collision, nothing else acts on A or B until they meet, if they ever do.

Ask yourself What exactly is the condition for A and B to collide again?

Once B has hit C, both A and B move with constant velocities, and A is behind B (it is on the negative side of B). The gap between them shrinks exactly when $v_A > w_B$, and then it shrinks to zero in finite time. If $v_A \le w_B$ the gap never decreases. This one inequality covers every sign case: for instance, if A is moving backwards but B is moving backwards even faster, B still catches A, and that is also $v_A > w_B$.

So we need

$$\frac{(2-k)u}{2(1+k)} > \frac{3(4k-3)u}{8(1+k)(k+3)}.$$

Multiplying by $8(1+k)(k+3)/u$, which is positive since $k > 0$, gives the equivalent inequality

$$\begin{aligned} 4(2-k)(k+3) &> 3(4k-3) \\ 24 - 4k - 4k^2 &> 12k - 9 \\ 4k^2 + 16k - 33 &< 0 \\ (2k-3)(2k+11) &< 0. \end{aligned}$$

Since $2k + 11 > 0$, this holds if and only if $2k - 3 < 0$. Because k is a mass ratio, $k > 0$. Hence A and B collide for a second time exactly when

$$0 < k < \tfrac{3}{2}.$$

At $k = \frac{3}{2}$ we get $v_A = w_B = \frac{u}{10}$: the two move on together at the same speed and never actually meet again, so the endpoint is excluded.

Part (ii)

With $k = 1$ the formulas above give

$$v_A = \frac{u}{4}, \qquad v_B = \frac{3u}{4}, \qquad w_B = \frac{3u}{64}, \qquad w_C = \frac{15u}{64}.$$

Put the origin where A and B first collide (that is, at B's starting point), and measure time t from that collision.

Up to the B–C collision. B covers the distance d to C at speed $\frac{3}{4}u$, which takes

$$t_1 = \frac{d}{3u/4} = \frac{4d}{3u}.$$

In this time A has moved $\frac{u}{4}\cdot\frac{4d}{3u}=\frac{d}{3}$. So A is at $\frac{d}{3}$ and B is at d: the gap is $\frac{2d}{3}$.

The chase. Now A moves at $\frac{u}{4}=\frac{16u}{64}$ and B at $\frac{3u}{64}$, so A closes the gap at the relative speed $\frac{13u}{64}$. That takes

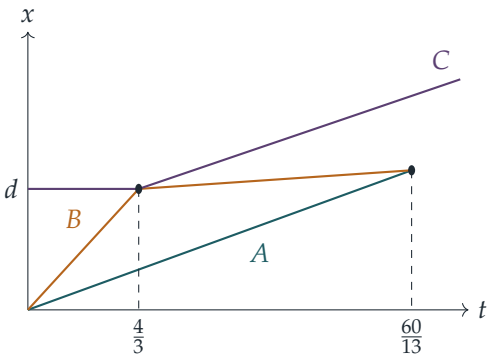
$$t_2=\frac{2d/3}{13u/64}=\frac{128d}{39u}.$$

Total. The time between the two A–B collisions is

$$t_1+t_2=\frac{4d}{3u}+\frac{128d}{39u}=\frac{52d+128d}{39u}=\frac{180d}{39u}=\frac{60d}{13u},$$

as required.

The distance–time graph below shows the motion, with $u=d=1$. The lines for A and B meet again at $t=\frac{60}{13}$.



Checks

- Momentum: with $k=1$, after the first collision $m\cdot\frac{u}{4}+m\cdot\frac{3u}{4}=mu$, and after the second $m\cdot\frac{3u}{64}+3m\cdot\frac{15u}{64}=\frac{48}{64}mu=m\cdot\frac{3u}{4}$. Both balance.
- The boundary $k=\frac{3}{2}$: both v_A and w_B come out as $\frac{u}{10}$, which confirms the critical value.
- Sanity check: for large k, A bounces backwards ($v_A<0$) while B continues forwards, so there's clearly no second collision. This agrees with the range.
- We checked numerically that $v_A>w_B$ exactly when $k<\frac{3}{2}$ for 10^5 random rational k, and a time-stepped simulation with $u=d=1$ gave the second collision at $t\approx 4.6154$, matching $\frac{60}{13}\approx 4.6154$.

Common slips

- Using speeds instead of signed velocities in the restitution law. With signed velocities, separation speed is "front minus back", and the sign of v_A takes care of itself.

- Requiring $v_A > 0$ as part of the condition. A could in principle move backwards and still be caught by B (if $w_B < v_A$). The single condition $v_A > w_B$ is the right one.
- In (ii), timing the chase from the first collision instead of from the moment B's velocity changes, so missing the $\frac{d}{3}$ head start that A has already covered.

Beyond STEP

Momentum and restitution together make every collision a linear map from velocities before to velocities after. Chains of collisions, such as Newton's cradle or the famous problem of counting collisions between two blocks and a wall, are then studied by composing these maps. In first-year mechanics this leads to the centre-of-mass frame, where the restitution law becomes a simple scaling of relative velocity, and on to collisions in two and three dimensions. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions and the centre-of-mass frame clearly.

WORKED SOLUTION

STEP II 2006, Question 14

Probability and Statistics • Continuous Random Variables

STEP II 2006, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q14

The key idea

The whole question rests on one antiderivative. The denominator $x \ln x$ is $\ln x$ multiplied by x, and $1/x$ is the derivative of $\ln x$, so the integrand has the form $g'(x)/g(x)$ with $g(x) = \ln x$. Hence

$$\int \frac{1}{x \ln x}\,\mathrm{d}x = \ln|\ln x| + C.$$

After that, each part is a matter of applying "total probability is 1" and keeping careful track of signs and of where the density is actually non-zero.

Solution

The sketch

Write $y = \dfrac{1}{x \ln x}$ for $x > 0$, $x \neq 1$.

Sign. Since $x > 0$, y has the sign of $\ln x$: $y < 0$ on $0 < x < 1$ and $y > 0$ on $x > 1$. The curve never meets the axes.

Behaviour at the ends.

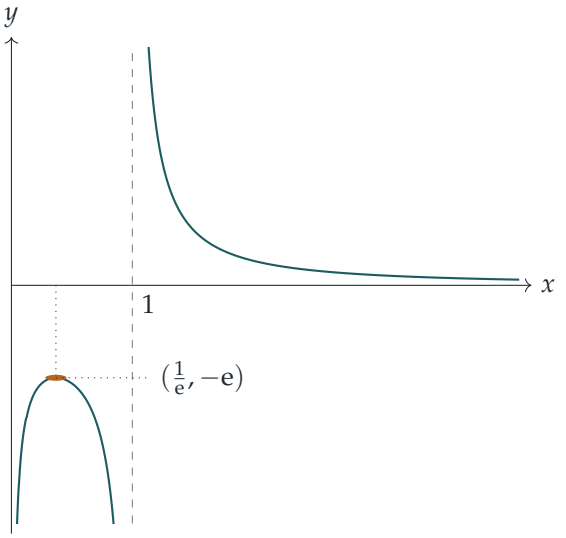
- As $x \to 0^+$: we are told $x \ln x \to 0$, and $x \ln x < 0$ there, so $y \to -\infty$.
- As $x \to 1$: $x \ln x \to 0$, negative from the left and positive from the right, so $y \to -\infty$ as $x \to 1^-$ and $y \to +\infty$ as $x \to 1^+$. The line $x = 1$ is a vertical asymptote.
- As $x \to \infty$: $x \ln x \to \infty$, so $y \to 0^+$, and the x-axis is an asymptote.

Turning points. By the quotient (or chain) rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{\ln x + 1}{(x \ln x)^2},$$

which is zero only when $\ln x = -1$, that is $x = 1/\mathrm{e}$, where $y = \dfrac{1}{(1/\mathrm{e})(-1)} = -\mathrm{e}$. For

$0 < x < 1/\mathrm{e}$ we have $\ln x + 1 < 0$, so y is increasing; for $x > 1/\mathrm{e}$ (with $x \neq 1$) it is decreasing. So $(1/\mathrm{e}, -\mathrm{e})$ is a local maximum of the left-hand branch, and the right-hand branch decreases steadily from $+\infty$ to 0.



Part (i)

Here $a = \frac{1}{4}$ and $b = \frac{1}{2}$, both in $(0, 1)$, where $x \ln x < 0$. Using the antiderivative above,

$$\int_{1/4}^{1/2} \frac{\mathrm{d}x}{x \ln x} = \Big[\ln|\ln x|\Big]_{1/4}^{1/2} = \ln|\ln \tfrac{1}{2}| - \ln|\ln \tfrac{1}{4}|$$
$$= \ln(\ln 2) - \ln(2 \ln 2) = -\ln 2.$$

The total probability must be 1, so $\lambda \cdot (-\ln 2) = 1$, giving

$$\lambda = -\frac{1}{\ln 2}.$$

The negative sign is exactly what we need: $x \ln x < 0$ on $[\frac{1}{4}, \frac{1}{2}]$, so λ must be negative for f to be non-negative there.

Part (ii)

With $\lambda = 1$ the density must be non-negative, so the interval $[a, b]$ must lie where $x \ln x > 0$, which it does since $b > a > 1$. Then $\ln a, \ln b > 0$ and

$$1 = \int_a^b \frac{\mathrm{d}x}{x \ln x} = \ln(\ln b) - \ln(\ln a) = \ln \frac{\ln b}{\ln a}.$$

So $\dfrac{\ln b}{\ln a} = \mathrm{e}$, that is $\ln b = \mathrm{e} \ln a = \ln\left(a^{\mathrm{e}}\right)$, and since ln is one-to-one, $b = a^{\mathrm{e}}$, as required.

Part (iii)

With $a = \mathrm{e}$, part (ii) gives $b = \mathrm{e}^{\mathrm{e}}$. Since $1 < \frac{3}{2} < 2 < \mathrm{e}$, the whole interval $[\mathrm{e}^{3/2}, \mathrm{e}^2]$ lies inside $[a, b] = [\mathrm{e}, \mathrm{e}^{\mathrm{e}}]$, so

$$\mathrm{P}(\mathrm{e}^{3/2} \le X \le \mathrm{e}^2) = \Big[\ln(\ln x)\Big]_{\mathrm{e}^{3/2}}^{\mathrm{e}^2} = \ln 2 - \ln \tfrac{3}{2} = \ln \tfrac{4}{3}.$$

To turn this into a fraction, use the Maclaurin series $\ln(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \cdots$ with $t = \frac{1}{3}$, keeping four terms:

$$\ln \tfrac{4}{3} \approx \frac{1}{3} - \frac{1}{18} + \frac{1}{81} - \frac{1}{324} = \frac{108 - 18 + 4 - 1}{324} = \frac{93}{324} = \frac{31}{108}.$$

Ask yourself How do we know four terms is a sensible place to stop?

The series alternates with terms decreasing in size, so the error after four terms is less than the fifth term, $\frac{1}{5 \cdot 3^5} = \frac{1}{1215} < 0.001$. So $\mathrm{P} = \ln \frac{4}{3} \approx \frac{31}{108}$, as required.

Part (iv)

Now $a = \mathrm{e}^{1/2}$, so by part (ii)

$$b = a^{\mathrm{e}} = \mathrm{e}^{\mathrm{e}/2}.$$

Ask yourself Before integrating, where is the density actually non-zero?

Since $\mathrm{e} < 3$, we have $\mathrm{e}/2 < \frac{3}{2}$, so $b = \mathrm{e}^{\mathrm{e}/2} < \mathrm{e}^{3/2}$. The interval $[\mathrm{e}^{3/2}, \mathrm{e}^2]$ lies entirely to the right of b, where $\mathrm{f}(x) = 0$. Hence

$$\mathrm{P}(\mathrm{e}^{3/2} \le X \le \mathrm{e}^2) = 0.$$

(Blindly integrating $1/(x \ln x)$ from $\mathrm{e}^{3/2}$ to e^2 would give $\ln \frac{4}{3}$ again, which is wrong.)

Checks

- Numerically, $\mathrm{e}^{-1} \approx 0.368$ and a scan of y on $(0, 1)$ gives its largest value $\approx -2.718 = -\mathrm{e}$ there.
- Simpson's rule gives $\int_{1/4}^{1/2} \frac{\mathrm{d}x}{x \ln x} \approx -0.6931 = -\ln 2$, so $\lambda \approx -1.4427$.
- For several values of $a > 1$ (e.g. $a = 2$, $a = 7.3$), $\int_a^{a^{\mathrm{e}}} \frac{\mathrm{d}x}{x \ln x}$ comes out as 1.0000.
- $\ln \frac{4}{3} \approx 0.28768$ and $\frac{31}{108} \approx 0.28704$, agreeing to within 0.0007.
- In (iv), $b = \mathrm{e}^{\mathrm{e}/2} \approx 3.89$ while $\mathrm{e}^{3/2} \approx 4.48$.

Common slips

- Writing $\ln(\ln x)$ in part (i), where $\ln x < 0$: the antiderivative needs the modulus, $\ln|\ln x|$.
- Giving $\lambda = \frac{1}{\ln 2}$: a positive λ would make the "density" negative.
- In part (iv), integrating over $[e^{3/2}, e^2]$ without checking that it lies inside the support $[a, b]$.

Beyond STEP

The substitution $u = \ln x$ turns this density into $1/u$ on $[\ln a, \ln b]$, and the condition $b = a^e$ is just $\int_{\ln a}^{\ln b} \frac{du}{u} = 1$. Transforming random variables like this (here $U = \ln X$) is a central tool in first-year probability, where the density of a function of a random variable is found from the chain rule. The same function $1/(x \ln x)$ is also famous in analysis: $\sum 1/(n \ln n)$ diverges, by comparison with this integral, even though its terms tend to zero. See Grimmett and Welsh, *Probability: An Introduction*, for transformations of random variables.

WORKED SOLUTION

STEP III 2006, Question 1

Pure • Curve Sketching

STEP III 2006, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q1

The key idea

Divide out first: writing the curve as a straight line plus a small correction gives the oblique asymptote, tells you which side of it the curve lies, and makes the derivative easy. The derivative turns out to exceed 2 everywhere, and that single fact does most of the work in the root-counting parts: each equation rearranges to "curve = something simple", and we compare slopes.

Solution

The curve

Since $2x^3 - 10x = 2x(x^2 - 4) - 2x$,

$$y = \frac{2x(x^2-5)}{x^2-4} = 2x - \frac{2x}{x^2-4}.$$

Some features we can read off straight away:

- y is an *odd* function, so the curve has rotational symmetry of order 2 about the origin.
- It crosses the x-axis at $x = 0$ and $x = \pm\sqrt{5}$.
- **Asymptotes.** The denominator vanishes at $x = \pm 2$ (where the numerator is $\pm 4 \cdot (-1) \cdot 2 \neq 0$), so $x = 2$ and $x = -2$ are vertical asymptotes. As $|x| \to \infty$ the correction $2x/(x^2 - 4) \to 0$, so $y = 2x$ is an oblique asymptote.
- **Side of the asymptote.** $y - 2x = -2x/(x^2 - 4)$. For $x > 2$ this is negative (curve just below $y = 2x$); for $x < -2$ it is positive (just above). For $0 < x < 2$ it is positive and for $-2 < x < 0$ negative.
- **Behaviour near $x = \pm 2$.** As $x \to 2^+$, $-2x/(x^2 - 4) \to -\infty$, and as $x \to 2^-$ it tends to $+\infty$; by oddness, $y \to +\infty$ as $x \to -2^-$ and $y \to -\infty$ as $x \to -2^+$.

Differentiating the divided form,

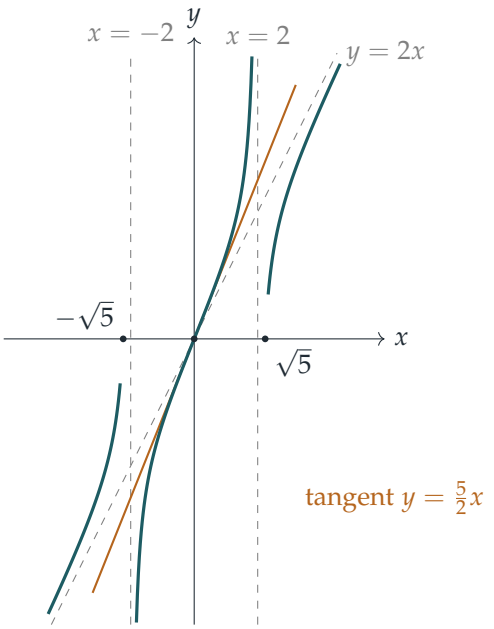
$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2 - \frac{2(x^2-4) - 2x\cdot 2x}{(x^2-4)^2} = 2 + \frac{2(x^2+4)}{(x^2-4)^2}.$$

So $\mathrm{d}y/\mathrm{d}x > 2$ for every $x \neq \pm 2$: there are no stationary points, and on each of the three intervals $x < -2$, $-2 < x < 2$, $x > 2$ the curve is strictly increasing, rising from $-\infty$ to $+\infty$.

Tangent at the origin. At $x = 0$ the gradient is $2 + 8/16 = \frac{5}{2}$, so the tangent is $y = \frac{5}{2}x$. Differentiating once more,

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{4x(x^2+12)}{(x^2-4)^3},$$

which vanishes only at $x = 0$ and changes sign there: the origin is a point of inflection. On $0 < x < 2$ the second derivative is positive (numerator and denominator both negative), so the middle branch is convex there and lies *above* its tangent $y = \frac{5}{2}x$; on $-2 < x < 0$ it is concave and lies below it.



Summary. Asymptotes $x = 2$, $x = -2$ and $y = 2x$; tangent at the origin $y = \frac{5}{2}x$.

In each part below, $x = \pm 2$ is never a root (the left-hand side is non-zero there while the right-hand side is zero), so we may divide by $x^2 - 4$ and count intersections of the curve with another graph.

Part (i)

Dividing by $\frac{3}{2}(x^2-4)$, the equation becomes

$$\frac{2x(x^2-5)}{x^2-4} = \tfrac{2}{3}(x+3),$$

so we want the intersections of the curve with the line $y = \frac{2}{3}x + 2$, of gradient $\frac{2}{3}$.

Ask yourself Why does a line of gradient less than 2 meet each branch exactly once?

Let $h(x)$ be the curve's y minus the line's. On each branch $h'(x) > 2 - \frac{2}{3} > 0$, so h is strictly increasing and has at most one zero. At the vertical asymptote ends $h \to \pm\infty$ just as y does (the line stays finite). At the outer ends, $h \approx 2x - \frac{2}{3}x = \frac{4}{3}x$, which tends to $-\infty$ as $x \to -\infty$ and to $+\infty$ as $x \to +\infty$. So on each of the three branches h runs from $-\infty$ to $+\infty$ and, by the intermediate value theorem, vanishes exactly once.

Hence equation (i) has **three** real roots. (Indeed it is the cubic $2x^3 - 3x^2 - 11x + 12 = 0$, with $x = 1$ as one root.)

Part (ii)

Dividing by $2(x^2 - 4)$, we need the intersections with $y = \frac{5}{2}x - 1$: the tangent at the origin moved down by 1. Its gradient now exceeds 2, so the argument of (i) no longer applies, and we look branch by branch.

- $x > 2$: the curve lies below $y = 2x$, while the line satisfies $\frac{5}{2}x - 1 - 2x = \frac{1}{2}x - 1 > 0$, so it lies above $y = 2x$. No intersection.
- $x < -2$: the curve lies above $y = 2x$, and the line lies below it since $\frac{1}{2}x - 1 < 0$. No intersection.
- $0 \le x < 2$: the curve lies on or above its tangent $y = \frac{5}{2}x$ (convexity), which is above the line. No intersection.
- $-2 < x < 0$: as $x \to -2^+$ the curve tends to $-\infty$ while the line tends to -6; at $x = 0$ the curve is at 0, above the line's -1. So there is at least one crossing. Because $y'' < 0$ here, y' is decreasing on $(-2, 0)$, so $y' > y'(0) = \frac{5}{2}$; the difference "curve minus line" is strictly increasing, and the crossing is unique.

Hence equation (ii) has exactly **one** real root, lying between -2 and 0. (As a cubic it is $x^3 - 2x^2 + 8 = 0$, whose stationary values 8 and $\frac{184}{27}$ are both positive, confirming a single real root, near $x \approx -1.51$.)

Part (iii)

Dividing by $(x^2 - 4)^2$ gives

$$\left(\frac{2x(x^2 - 5)}{x^2 - 4}\right)^2 = x^2 + 1,$$

so we need the intersections of the curve with the rectangular hyperbola $y^2 - x^2 = 1$, that is, with $y = \sqrt{x^2 + 1}$ (upper branch) or $y = -\sqrt{x^2 + 1}$ (lower branch).

Ask yourself Can we use the symmetry to halve the work?

Yes. The map $(x, y) \mapsto (-x, -y)$ sends the curve to itself (it is odd) and swaps the two branches of the hyperbola. So intersections with the lower branch correspond one-to-one with intersections with the upper branch, and we need only count the latter.

Let $u(x) = \sqrt{x^2+1}$. Then $|u'(x)| = |x|/\sqrt{x^2+1} < 1$, so the difference $y - u$ has derivative greater than $2 - 1 = 1 > 0$ on each branch of the curve, and each branch gives at most one intersection. Now check the ends:

- $-2 < x < 2$: $y - u$ runs from $-\infty$ to $+\infty$. One root.
- $x > 2$: $y - u \to -\infty$ as $x \to 2^+$, and $y - u \approx 2x - x \to +\infty$ as $x \to \infty$. One root.
- $x < -2$: $y - u \approx 2x - |x| = 3x \to -\infty$ as $x \to -\infty$, and $y - u \to +\infty$ as $x \to -2^-$. One root.

So the curve meets the upper branch three times and, by symmetry, the lower branch three times. The two branches have no points in common ($u \geq 1$), so these six points have six distinct x-values, and equation (iii) has **six** real roots. Since it is a polynomial equation of degree 6 (namely $3x^6 - 33x^4 + 92x^2 - 16 = 0$), all its roots are real.

Checks

- Near $x = 0$, $y \approx 2x(-5)/(-4) = \frac{5}{2}x$, agreeing with the tangent.
- At $x = 1000$, $y - 2x \approx -0.002$: small and negative, as predicted.
- A numerical sign-change search on the original equations finds roots $x \approx -2.212,\ 1,\ 2.713$ for (i); only $x \approx -1.510$ for (ii); and $x \approx \pm 0.431,\ \pm 2.155,\ \pm 2.484$ for (iii).

Common slips

- Forgetting to check that $x = \pm 2$ are not roots before dividing by $x^2 - 4$.
- In (iii), counting only the intersections with $y = \sqrt{x^2+1}$ and missing the negative square root.
- In (ii), assuming "steep line, so one root per branch" as in (i): with gradient above 2 the outer branches need a separate argument.

Beyond STEP

The root counting here is the intermediate value theorem combined with monotonicity: a continuous, strictly increasing function that goes from $-\infty$ to $+\infty$ has exactly one zero. In a first-year analysis course both facts are proved carefully from the completeness of the real numbers, and the same reasoning underpins results such as "every real polynomial of odd degree has a real root". Abbott's *Understanding Analysis* develops this beautifully.

WORKED SOLUTION

STEP III 2006, Question 2

Pure • Definite Integrals by Symmetry

STEP III 2006, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q2

The key idea

The interval $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$ is symmetric, so replacing θ by $-\theta$ gives a second expression for I; adding the two clears the awkward $\sin\theta$ from the denominator. Once everything is written in terms of $\tan\theta$, the integrals I and J have the same denominator, and a well-chosen combination of them has integrand exactly 1. Throughout, keep an eye on the sign of $\cos 2\alpha$: it is what changes in the final part.

Solution

Write $s = \sin 2\alpha$ and $c = \cos 2\alpha$, so $s^2 + c^2 = 1$. In both ranges of α considered ($\alpha \neq \frac{1}{4}\pi$) we have $0 < s < 1$, so $1 \mp s\sin\theta \geq 1 - s > 0$: the integrand of I is continuous on the whole interval.

Part (i)

Substitute $\theta = -\phi$, so $\mathrm{d}\theta = -\mathrm{d}\phi$ and the limits swap. Since $\cos^2(-\phi) = \cos^2\phi$ and $\sin(-\phi) = -\sin\phi$,

$$I = \int_{\frac{1}{2}\pi}^{-\frac{1}{2}\pi} \frac{\cos^2\phi}{1 + \sin\phi\, s}\,(-\mathrm{d}\phi) = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{\cos^2\phi}{1 + s\sin\phi}\,\mathrm{d}\phi,$$

which is the first result (renaming ϕ as θ).

Adding the two expressions for I,

$$2I = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \cos^2\theta \left(\frac{1}{1 - s\sin\theta} + \frac{1}{1 + s\sin\theta}\right)\mathrm{d}\theta = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{2\cos^2\theta}{1 - s^2\sin^2\theta}\,\mathrm{d}\theta.$$

Now divide top and bottom by $\cos^2\theta$ (non-zero inside the interval):

$$\frac{2\cos^2\theta}{1 - s^2\sin^2\theta} = \frac{2}{\sec^2\theta - s^2\tan^2\theta} = \frac{2}{1 + \tan^2\theta\,(1 - s^2)} = \frac{2}{1 + c^2\tan^2\theta},$$

using $\sec^2\theta = 1 + \tan^2\theta$ and $1 - \sin^2 2\alpha = \cos^2 2\alpha$. Hence

$$2I = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{2}{1 + \tan^2\theta \cos^2 2\alpha}\, \mathrm{d}\theta,$$

as required. (At $\theta = \pm\frac{1}{2}\pi$ this integrand tends to 0, so nothing goes wrong at the ends.)

Part (ii)

The $\sec^2\theta$ in the numerator invites $t = \tan\theta$, $\mathrm{d}t = \sec^2\theta\,\mathrm{d}\theta$. As θ runs from $-\frac{1}{2}\pi$ to $\frac{1}{2}\pi$, t runs over the whole real line:

$$J = \int_{-\infty}^{\infty} \frac{\mathrm{d}t}{1 + c^2t^2}.$$

Since $0 < \alpha < \frac{1}{4}\pi$, we have $0 < 2\alpha < \frac{1}{2}\pi$ and so $c > 0$. Putting $u = ct$,

$$J = \frac{1}{c}\int_{-\infty}^{\infty} \frac{\mathrm{d}u}{1 + u^2} = \frac{1}{c}\Big[\arctan u\Big]_{-\infty}^{\infty} = \frac{\pi}{c} = \pi\sec 2\alpha.$$

Part (iii)

By part (i), $I = \displaystyle\int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{\mathrm{d}\theta}{1 + c^2\tan^2\theta}$, which has the same denominator as J. So

$$s^2I + c^2J = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{s^2 + c^2\sec^2\theta}{1 + c^2\tan^2\theta}\, \mathrm{d}\theta.$$

? **Ask yourself** Why should this particular combination be any easier?

Look at the numerator: $s^2 + c^2\sec^2\theta = s^2 + c^2 + c^2\tan^2\theta = 1 + c^2\tan^2\theta$. It is exactly the denominator, so the integrand is 1 and

$$s^2I + c^2J = \pi.$$

Using $J = \pi/c$ from part (ii),

$$s^2I = \pi - \pi c = \pi(1 - c) \quad\Longrightarrow\quad I = \frac{\pi(1-c)}{1 - c^2} = \frac{\pi}{1 + c},$$

where we used $s^2 = 1 - c^2 = (1 - c)(1 + c)$ and cancelled $1 - c \neq 0$ (as $c < 1$). Finally $1 + \cos 2\alpha = 2\cos^2\alpha$, so

$$I = \frac{\pi}{2\cos^2\alpha} = \tfrac{1}{2}\pi\sec^2\alpha. \qquad \checkmark$$

Part (iv)

Now $\frac{1}{4}\pi < \alpha < \frac{1}{2}\pi$, so $\frac{1}{2}\pi < 2\alpha < \pi$: still $0 < s < 1$, but now $c < 0$. Part (i) used only $s^2 + c^2 = 1$, so it still holds, and so does the identity $s^2I + c^2J = \pi$ from part (iii). The one thing that changes is part (ii).

Ask yourself Where exactly did part (ii) use $\alpha < \frac{1}{4}\pi$?

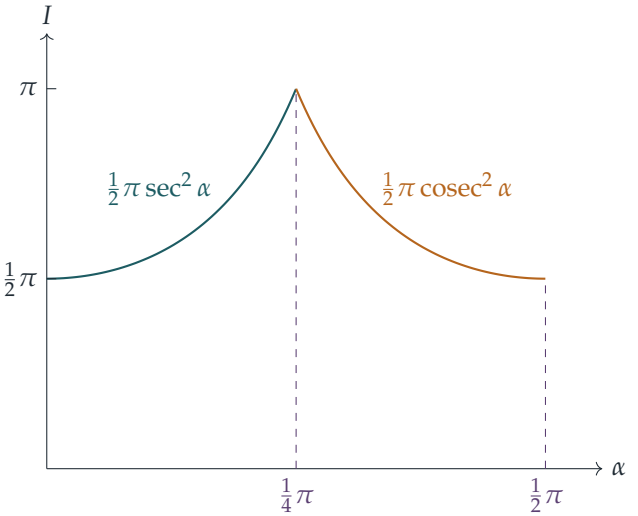
In the substitution $u = ct$: if $c < 0$ the limits of u come out as $+\infty$ to $-\infty$, and swapping them back gives a minus sign. In other words, $J = \pi/|c| = -\pi/c$. (This must be so: J is the integral of a positive function, so it cannot equal the negative number $\pi \sec 2\alpha$.) Then

$$s^2 I = \pi - c^2\Big(-\frac{\pi}{c}\Big) = \pi(1+c) \quad \Longrightarrow \quad I = \frac{\pi(1+c)}{(1-c)(1+c)} = \frac{\pi}{1-c},$$

cancelling $1 + c \neq 0$ (as $c > -1$). Since $1 - \cos 2\alpha = 2\sin^2\alpha$,

$$I = \frac{\pi}{2\sin^2\alpha} = \tfrac{1}{2}\pi \operatorname{cosec}^2\alpha \qquad \big(\tfrac{1}{4}\pi < \alpha < \tfrac{1}{2}\pi\big).$$

There is a quick confirmation: I depends on α only through $\sin 2\alpha$, and $\sin 2(\frac{1}{2}\pi - \alpha) = \sin 2\alpha$. So $I(\alpha) = I(\frac{1}{2}\pi - \alpha) = \frac{1}{2}\pi \sec^2(\frac{1}{2}\pi - \alpha) = \frac{1}{2}\pi \operatorname{cosec}^2\alpha$. The graph of I is symmetric about $\alpha = \frac{1}{4}\pi$, where both formulas give π:



Checks

- At $\alpha = \frac{1}{4}\pi$ ($c = 0$) the 2I-form of part (i) has integrand 2, giving $I = \pi$, matching both formulas.
- As $\alpha \to 0$, $I \to \int \cos^2\theta \, d\theta = \frac{1}{2}\pi$, matching $\frac{1}{2}\pi \sec^2 0$.
- Numerically (Simpson's rule on the original integral), $\alpha = 0.5$ gives $I \approx 2.039595$, equal to $\frac{1}{2}\pi \sec^2 0.5$; and $\alpha = 1.1$ gives $I \approx 1.977709 = \frac{1}{2}\pi \operatorname{cosec}^2 1.1$. In every case tested, $s^2 I + c^2 J$ came out as 3.141593.

Common slips

- Writing $J = \pi \sec 2\alpha$ in part (iv) as well: this is negative there, which is impossible for the integral of a positive function.
- Forgetting that the substitution $t = \tan\theta$ turns the finite interval into the whole real line, so the limits become $\pm\infty$.

Beyond STEP

The trick in part (i), averaging $f(\theta)$ and $f(-\theta)$, is the start of splitting a function into even and odd parts: only the even part survives integration over a symmetric interval. Integrals of rational functions of $\sin\theta$ and $\cos\theta$ over a full period are a classic application of contour integration (the residue theorem) in a first complex-analysis course, where results like this one drop out in a few lines; see Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP III 2006, Question 3

Pure • Power Series Coefficients

STEP III 2006, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q3

The key idea

We never have to differentiate $\tan x$ over and over. Each trigonometric identity links two of these functions, so it turns into a link between their coefficients. Once we compare coefficients of x^n, all three series are fixed by the single sequence b_n. The Pythagorean identity $\cot^2 x + 1 = \operatorname{cosec}^2 x$ then gives equations that the b_n must satisfy, and we can solve them one at a time.

Solution

Part (i)

A power series that converges to a function near 0 is that function's Maclaurin series, so $a_n = \tan^{(n)}(0)/n!$. Now tan is an odd function: $\tan(-x) = -\tan x$. Differentiating an odd function gives an even function, and differentiating an even function gives an odd one. So every even-order derivative of tan is odd, and an odd function vanishes at 0. Hence $\tan^{(n)}(0) = 0$ for even n, and so $a_n = 0$ for even n. Equivalently, the series contains only odd powers of x. The same argument applies to $\cot x - \frac{1}{x} = \sum b_n x^n$, which is also odd, so $b_n = 0$ for even n. We will also get this from the relation below.

For the identity, put everything over $\sin x \cos x$:

$$\cot x - \tan x = \frac{\cos^2 x - \sin^2 x}{\sin x \cos x} = \frac{\cos 2x}{\frac{1}{2}\sin 2x} = 2\cot 2x.$$

Now take $0 < x < \frac{1}{4}\pi$, so that $2x$ also lies in $(0, \frac{1}{2}\pi)$ and we may use the series for $\cot 2x$, with $2x$ in place of x:

$$2\cot 2x = 2\left(\frac{1}{2x} + \sum_{n=0}^{\infty} b_n 2^n x^n\right) = \frac{1}{x} + \sum_{n=0}^{\infty} 2^{n+1} b_n x^n.$$

The left-hand side of the identity is

$$\cot x - \tan x = \frac{1}{x} + \sum_{n=0}^{\infty} (b_n - a_n) x^n.$$

The $\frac{1}{x}$ terms cancel. We are left with two power series that are equal for every x in $(0, \frac{1}{4}\pi)$.

Ask yourself Why are we allowed to compare coefficients of two infinite series?

If two power series have the same sum throughout an interval, their coefficients agree. Subtract them: the difference is a power series that is zero on the whole interval. Then all its derivatives are zero there, and letting $x \to 0^+$ shows that every coefficient is zero. So

$$b_n - a_n = 2^{n+1} b_n, \qquad \text{that is,} \qquad a_n = (1 - 2^{n+1})\, b_n,$$

as required. Since $1 - 2^{n+1} \neq 0$, we have $b_n = a_n/(1 - 2^{n+1})$, so $b_n = 0$ whenever $a_n = 0$, which is for even n.

Part (ii)

In the same way,

$$\cot x + \tan x = \frac{\cos^2 x + \sin^2 x}{\sin x \cos x} = \frac{1}{\frac{1}{2}\sin 2x} = 2 \operatorname{cosec} 2x.$$

For $0 < x < \frac{1}{4}\pi$, the series for cosec with $2x$ in place of x gives

$$2 \operatorname{cosec} 2x = \frac{1}{x} + \sum_{n=0}^{\infty} 2^{n+1} c_n x^n, \qquad \cot x + \tan x = \frac{1}{x} + \sum_{n=0}^{\infty} (b_n + a_n) x^n.$$

Comparing coefficients gives $b_n + a_n = 2^{n+1} c_n$. Substituting $a_n = (1 - 2^{n+1}) b_n$ from part (i),

$$2^{n+1} c_n = (2 - 2^{n+1})\, b_n \quad \Longrightarrow \quad c_n = \frac{2 - 2^{n+1}}{2^{n+1}}\, b_n = (2^{-n} - 1)\, b_n,$$

which is the required result. In particular $c_n = 0$ for even n as well.

Part (iii)

Multiplying the series for $\cot x$ and $\operatorname{cosec} x$ by x,

$$1 + x \sum_{n=0}^{\infty} b_n x^n = x \cot x, \qquad 1 + x \sum_{n=0}^{\infty} c_n x^n = x \operatorname{cosec} x.$$

So the claimed equation says $x^2 \cot^2 x + x^2 = x^2 \operatorname{cosec}^2 x$. This is x^2 times the identity $\cot^2 x + 1 = \operatorname{cosec}^2 x$, which comes from dividing $\cos^2 x + \sin^2 x = 1$ by $\sin^2 x$. That proves it.

Now we expand. The even-indexed b_n and c_n are zero, so

$$x\cot x = 1 + b_1x^2 + b_3x^4 + \cdots, \qquad x\operatorname{cosec} x = 1 + c_1x^2 + c_3x^4 + \cdots,$$

and the equation becomes

$$\left(1 + b_1x^2 + b_3x^4 + \cdots\right)^2 + x^2 = \left(1 + c_1x^2 + c_3x^4 + \cdots\right)^2.$$

Squaring,

$$1 + (2b_1 + 1)x^2 + (2b_3 + b_1^2)x^4 + \cdots = 1 + 2c_1x^2 + (2c_3 + c_1^2)x^4 + \cdots.$$

From part (ii), $c_1 = (\frac{1}{2} - 1)b_1 = -\frac{1}{2}b_1$ and $c_3 = (\frac{1}{8} - 1)b_3 = -\frac{7}{8}b_3$.

Coefficient of x^2:

$$2b_1 + 1 = 2c_1 = -b_1 \quad\Longrightarrow\quad b_1 = -\tfrac{1}{3}.$$

Then part (i) gives $a_1 = (1 - 2^2)b_1 = (-3)(-\frac{1}{3}) = 1$, as required.

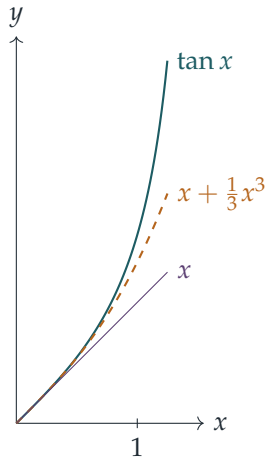
Coefficient of x^4: with $b_1 = -\frac{1}{3}$ and $c_1 = \frac{1}{6}$,

$$2b_3 + \tfrac{1}{9} = -\tfrac{7}{4}b_3 + \tfrac{1}{36} \quad\Longrightarrow\quad \tfrac{15}{4}\,b_3 = -\tfrac{3}{36} = -\tfrac{1}{12} \quad\Longrightarrow\quad b_3 = -\tfrac{1}{45}.$$

Hence

$$a_3 = (1 - 2^4)\,b_3 = (-15)\left(-\tfrac{1}{45}\right) = \tfrac{1}{3}.$$

So $\tan x = x + \frac{1}{3}x^3 + \cdots$.



Checks

- At $x = 0.3$, $\tan 0.3 \approx 0.309336$, and $0.3 + \frac{1}{3}(0.3)^3 = 0.309$. The difference is about 3×10^{-4}, which is the size of the next term, $\frac{2}{15}x^5$.
- The value $b_1 = -\frac{1}{3}$ matches the familiar $\cot x \approx \frac{1}{x} - \frac{x}{3}$. Also $c_1 = \frac{1}{6}$ matches $\operatorname{cosec} x \approx \frac{1}{x} + \frac{x}{6}$, which you can get from $\sin x \approx x - \frac{x^3}{6}$.

- We checked all three relations exactly (in rational arithmetic) for every n up to 48 by dividing the series for sin and cos. We also checked that the partial sums agree with tan, cot and cosec at random points in $(0, 1)$.

Common slips

- Comparing coefficients while forgetting the $\frac{1}{x}$ terms. They cancel, but only because of the factor 2 in $2\cot 2x$ and $2\operatorname{cosec} 2x$.
- Forgetting that $b_0 = b_2 = 0$ when expanding the squares in part (iii). Keeping b_0 adds spurious odd powers of x, and you would have to show separately that their coefficients vanish.
- Getting $c_3 = -\frac{7}{8}b_3$ with the wrong sign or power of 2. Note that $2^{-3} - 1 = -\frac{7}{8}$.

Beyond STEP

Work through this method to higher order and the coefficients turn out to be the Bernoulli numbers B_{2k}: $x\cot x = \sum_k (-1)^k 2^{2k} B_{2k}\, x^{2k}/(2k)!$. The factor $1 - 2^{n+1}$ from part (i) then gives the general formula for the coefficients of $\tan x$. Why does the series for $\tan x$ stop converging at $|x| = \frac{1}{2}\pi$? Complex analysis answers this: a power series converges out to the nearest point where the function blows up, which for tan is the pole at $\pm\frac{1}{2}\pi$. Needham's *Visual Complex Analysis* explains this beautifully, and Abbott's *Understanding Analysis* makes the term-by-term comparison of power series rigorous.

WORKED SOLUTION

STEP III 2006, Question 4

Pure • Functional Equations

STEP III 2006, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q4

The key idea

The opening identity is Cauchy's functional equation. Putting $y = x$ gives a relation between f at x and at $2x$, and differentiating that relation n times and setting $x = 0$ kills almost every Maclaurin coefficient. Each later part is the same equation in disguise: a change of variable (logarithm, exponential, tangent) turns the new identity back into $(*)$, so we can use the answer we already have.

Solution

The basic equation

Put $y = x$ in $(*)$: $\mathrm{f}(x) + \mathrm{f}(x) = \mathrm{f}(2x)$, that is,

$$2\mathrm{f}(x) \equiv \mathrm{f}(2x).$$

This is an identity in x, so we may differentiate both sides. By the chain rule, each differentiation of $\mathrm{f}(2x)$ brings out a factor of 2, so after n differentiations

$$2\mathrm{f}^{(n)}(x) \equiv 2^n \mathrm{f}^{(n)}(2x).$$

Setting $x = 0$ gives $(2 - 2^n)\,\mathrm{f}^{(n)}(0) = 0$. With $n = 2$ this reads $-2\mathrm{f}''(0) = 0$, so $\mathrm{f}''(0) = 0$, as required.

The same argument works for every $n \neq 1$: the factor $2 - 2^n$ is zero only when $n = 1$, so

$$\mathrm{f}(0) = 0 \quad \text{and} \quad \mathrm{f}^{(n)}(0) = 0 \ \text{ for all } n \geq 2.$$

(The case $n = 0$ is just $2\mathrm{f}(0) = \mathrm{f}(0)$.) Only the linear term of the Maclaurin series survives:

$$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)\,x + \tfrac{1}{2!}\mathrm{f}''(0)\,x^2 + \cdots = kx, \qquad k = \mathrm{f}'(0).$$

Conversely $kx + ky = k(x + y)$, so every such function works. The most general solution of $(*)$ is $\mathrm{f}(x) = kx$, where k is any constant.

Part (i)

To take a logarithm we need $\mathrm{g} > 0$, so first we deal with the sign. Putting $x = y = \frac{1}{2}s$ gives $\mathrm{g}(s) = \big(\mathrm{g}(\frac{1}{2}s)\big)^2 \geq 0$. If $\mathrm{g}(a) = 0$ for some a, then for every x,

$$\mathrm{g}(x) = \mathrm{g}(a)\,\mathrm{g}(x-a) = 0,$$

so $\mathrm{g} \equiv 0$, which does satisfy the identity. Otherwise $\mathrm{g}(x) > 0$ for all x, and $\mathrm{G} = \ln \mathrm{g}$ is defined everywhere. Taking logarithms of $\mathrm{g}(x)\mathrm{g}(y) = \mathrm{g}(x+y)$ gives

$$\mathrm{G}(x) + \mathrm{G}(y) = \mathrm{G}(x+y),$$

which is exactly $(*)$. So $\mathrm{G}(x) = kx$, and

$$\mathrm{g}(x) = \mathrm{e}^{kx}$$

for a constant k, which clearly works since $\mathrm{e}^{kx}\mathrm{e}^{ky} = \mathrm{e}^{k(x+y)}$. Apart from the trivial solution $\mathrm{g} \equiv 0$, these are all the solutions.

Part (ii)

Every positive x is e^u for exactly one real u, namely $u = \ln x$. So we can write $x = \mathrm{e}^u$ and $y = \mathrm{e}^v$, where u and v range over *all* real numbers. Then $xy = \mathrm{e}^{u+v}$, and the identity becomes

$$\mathrm{H}(u) + \mathrm{H}(v) = \mathrm{h}(\mathrm{e}^u) + \mathrm{h}(\mathrm{e}^v) = \mathrm{h}(\mathrm{e}^{u+v}) = \mathrm{H}(u+v)$$

for all real u, v. This is $(*)$ again, so $\mathrm{H}(u) = ku$. Since $u = \ln x$,

$$\mathrm{h}(x) = k \ln x \qquad (x > 0),$$

and this works because $\ln x + \ln y = \ln(xy)$.

Part (iii)

The expression $\dfrac{x+y}{1-xy}$ is the tangent addition formula in disguise. Write $x = \tan u$ and $y = \tan v$ with $u, v \in (-\frac{\pi}{2}, \frac{\pi}{2})$, so that $u = \arctan x$ and $v = \arctan y$. Then

$$z = \frac{\tan u + \tan v}{1 - \tan u \tan v} = \tan(u+v).$$

Define $\mathrm{T}(u) = \mathrm{t}(\tan u)$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$. Whenever $u+v$ also lies in $(-\frac{\pi}{2}, \frac{\pi}{2})$, the identity says

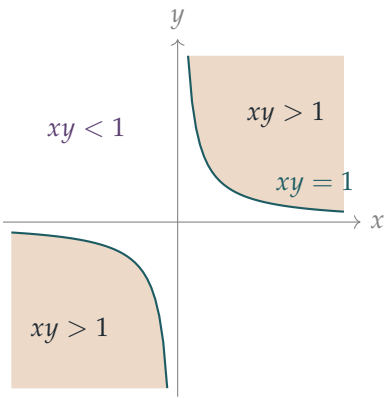
$$\mathrm{T}(u) + \mathrm{T}(v) = \mathrm{t}(x) + \mathrm{t}(y) = \mathrm{t}\big(\tan(u+v)\big) = \mathrm{T}(u+v).$$

Ask yourself The argument for $(*)$ used the identity for *all* x and y. Does it still work when we only know it for u, v and $u+v$ in $(-\frac{\pi}{2}, \frac{\pi}{2})$?

Yes. The Maclaurin argument only looks near 0. For $|u| < \frac{\pi}{4}$ we may put $v = u$ (then $|2u| < \frac{\pi}{2}$), which gives $2\mathrm{T}(u) = \mathrm{T}(2u)$ near 0. Differentiating n times at $u = 0$ gives $(2 - 2^n)\mathrm{T}^{(n)}(0) = 0$ exactly as before, so $\mathrm{T}(u) = ku$. Since $u = \arctan x$,

$$\mathrm{t}(x) = k \arctan x.$$

For which x and y does this really work? If $xy < 1$ then $u + v \in (-\frac{\pi}{2}, \frac{\pi}{2})$. When u and v have opposite signs this is automatic. When both are positive, $\tan u \tan v < 1$ means $\tan v < \cot u = \tan(\frac{\pi}{2} - u)$, so $u + v < \frac{\pi}{2}$, and the negative case is the same by symmetry. So $\arctan x + \arctan y = \arctan z$, and the identity holds. If $xy > 1$, however, $u + v$ lies outside $(-\frac{\pi}{2}, \frac{\pi}{2})$, and then $\arctan z = u + v \mp \pi$. For example, $x = y = 2$ gives $z = -\frac{4}{3}$, and $2\arctan 2 - \arctan(-\frac{4}{3}) = \pi$.



So $\mathrm{t}(x) = k \arctan x$ satisfies the identity for every pair with $xy < 1$, the region containing the origin in the figure. That is the answer the question intends. In the shaded regions $xy > 1$, the two sides differ by $\pm k\pi$. If we insist on the identity for literally every pair with $xy \neq 1$, only $k = 0$ survives, and the only solution is $\mathrm{t} \equiv 0$.

Checks

- For part (iii), differentiate the identity with respect to y and then put $y = 0$. Since $\partial z/\partial y = (1 + x^2)/(1 - xy)^2$, which is $1 + x^2$ at $y = 0$, we get $\mathrm{t}'(0) = \mathrm{t}'(x)(1 + x^2)$. So $\mathrm{t}'(x) = k/(1 + x^2)$, and with $\mathrm{t}(0) = 0$ this integrates to $k \arctan x$. The same trick on $(*)$ gives $\mathrm{f}'(x) = \mathrm{f}'(0)$ at once.
- We tested all four answers numerically on 10^4 random pairs each. In part (iii), $k \arctan x$ satisfied the identity on every one of about 5×10^4 random pairs with $xy < 1$, and missed by exactly $k\pi$ in size on every pair with $xy > 1$.

Common slips

- Forgetting the constant term: $\mathrm{f}(0) = 0$ comes from the case $n = 0$, so the answer is kx, not $kx + c$.

- Taking $\ln g$ without checking that $g > 0$. The zero function is a genuine solution of part (i).
- Assuming $\arctan x + \arctan y = \arctan \dfrac{x+y}{1-xy}$ always holds. It needs $xy < 1$.

Beyond STEP

Without the assumption of a Maclaurin series, Cauchy's equation $(*)$ needs much less: if f is merely continuous (or even just bounded on some interval), then $f(x) = kx$ still follows, starting from $f(q) = q\,f(1)$ for rational q. Using the axiom of choice, though, one can build wild solutions that are not linear, whose graphs are dense in the plane. Part (i) also shows that the exponential function is the essentially unique continuous homomorphism from addition to multiplication, an idea that reappears throughout group theory. Abbott's *Understanding Analysis* is a good place to see continuity and density arguments of this kind done carefully.

WORKED SOLUTION

STEP III 2006, Question 5

Pure • Complex Numbers and Geometry

STEP III 2006, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q5

The key idea

A triangle is equilateral exactly when one side, viewed from a vertex, is another side turned through $60°$. In complex numbers that turn is multiplication by $e^{\pm i\pi/3}$, and these two numbers are precisely the roots of $t^2 - t + 1 = 0$. Shifting the origin to one vertex turns the symmetric expression into exactly that quadratic. After that, the cubic part is Vieta's formulae, and the last part is the observation that $z \mapsto pw + q$ only rotates, scales and shifts the picture.

Solution

The equilateral condition

Write $u = \beta - \alpha$ and $v = \gamma - \alpha$, the two sides leaving the vertex α. Since the points are distinct, $u \neq 0$, $v \neq 0$ and $u \neq v$. First,

$$\alpha^2 + \beta^2 + \gamma^2 - \beta\gamma - \gamma\alpha - \alpha\beta = \tfrac{1}{2}\left[(\alpha - \beta)^2 + (\beta - \gamma)^2 + (\gamma - \alpha)^2\right],$$

as expanding the right-hand side shows. Now $\alpha - \beta = -u$, $\gamma - \alpha = v$ and $\beta - \gamma = u - v$, so the bracket is $u^2 + (u - v)^2 + v^2 = 2(u^2 - uv + v^2)$. Hence

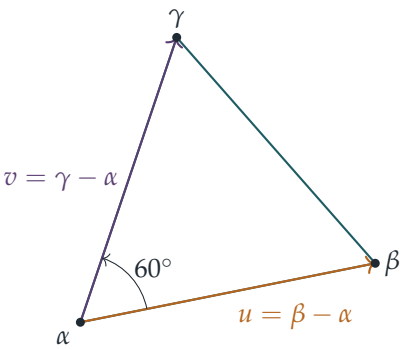
$$\alpha^2 + \beta^2 + \gamma^2 - \beta\gamma - \gamma\alpha - \alpha\beta = u^2 - uv + v^2 = u^2\left(t^2 - t + 1\right), \qquad t = \frac{v}{u}.$$

Since $u \neq 0$, the expression vanishes if and only if $t^2 - t + 1 = 0$, that is,

$$t = \frac{1 \pm i\sqrt{3}}{2} = e^{\pm i\pi/3}.$$

Ask yourself Why does $\dfrac{\gamma - \alpha}{\beta - \alpha} = e^{\pm i\pi/3}$ say exactly that the triangle is equilateral?

The modulus of v/u is $|\gamma-\alpha|/|\beta-\alpha|$ and its argument is the angle at α from side $\alpha\beta$ to side $\alpha\gamma$. So $v/u = e^{\pm i\pi/3}$ says: two sides from α are equal and the angle between them is $60°$. Such an isosceles triangle has base angles $\frac{1}{2}(180° - 60°) = 60°$, so it is equilateral. Conversely, if the triangle is equilateral then $|v| = |u|$ and the angle at α is $60°$, turned one way or the other according to the orientation, so $v/u = e^{i\pi/3}$ or $e^{-i\pi/3}$.



So the distinct points form an equilateral triangle (in either order) if and only if the expression is zero, as required.

The cubic

Let the roots of $(*)$ be α, β, γ. By Vieta's formulae, $\alpha+\beta+\gamma = -a$ and $\alpha\beta+\beta\gamma+\gamma\alpha = b$. So

$$\begin{aligned}\alpha^2+\beta^2+\gamma^2-\beta\gamma-\gamma\alpha-\alpha\beta &= (\alpha+\beta+\gamma)^2 - 3(\alpha\beta+\beta\gamma+\gamma\alpha)\\ &= a^2 - 3b.\end{aligned}$$

By the first result, the (distinct) roots form an equilateral triangle if and only if $a^2 - 3b = 0$, i.e. $a^2 = 3b$.

One caveat: the first result needs distinct points. If two roots coincide, say $\alpha = \beta$, the expression reduces to $(\gamma-\alpha)^2$, so it vanishes only when all three roots are equal. Thus $a^2 = 3b$ with repeated roots happens only for a triple root, $z^3 + az^2 + bz + c = (z + \frac{a}{3})^3$, i.e. $c = a^3/27$ as well. Apart from this degenerate "triangle" of one point, $a^2 = 3b$ is exactly the condition.

The transformation

Substitute $z = pw + q$ into $(*)$ and divide by $p^3 \neq 0$:

$$w^3 + \frac{3q+a}{p}\,w^2 + \frac{3q^2+2aq+b}{p^2}\,w + \frac{q^3+aq^2+bq+c}{p^3} = 0.$$

So

$$A = \frac{3q+a}{p}, \qquad B = \frac{3q^2+2aq+b}{p^2},$$

and therefore

$$A^2 - 3B = \frac{(9q^2 + 6aq + a^2) - (9q^2 + 6aq + 3b)}{p^2} = \frac{a^2 - 3b}{p^2}.$$

If the roots of $(*)$ form an equilateral triangle then they are distinct and $a^2 = 3b$, so $A^2 = 3B$. The roots of $(**)$ are $w = (z - q)/p$ for the roots z of $(*)$; as $p \neq 0$ this map is one-to-one, so they are distinct too. By the previous result, the roots of $(**)$ form an equilateral triangle.

Geometrically, this is no surprise: $w = (z - q)/p$ translates by $-q$, rotates by $-\arg p$ and scales by $1/|p|$, and none of these spoils an equilateral triangle. In the language of the first part, each side is divided by p, so the ratio v/u is unchanged.

Checks

- The cube roots of unity $1, \omega, \omega^2$ form an equilateral triangle and are the roots of $z^3 - 1 = 0$, with $a = b = 0$, so $a^2 = 3b$ holds.
- With $\alpha = 0$, $\beta = 1$ and $\gamma = \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$, the expression is $1 + \gamma^2 - \gamma$, which is zero because $\gamma = \mathrm{e}^{\mathrm{i}\pi/3}$ is a root of $t^2 - t + 1$.
- A numerical check over 1000 random triples confirmed $\sum \alpha^2 - \sum \alpha\beta = a^2 - 3b$ for the cubic with those roots. The same check confirmed $A^2 - 3B = (a^2 - 3b)/p^2$ and that equilateral roots stay equilateral under random p and q.

Common slips

- Only checking one orientation: $\mathrm{e}^{\mathrm{i}\pi/3}$ and $\mathrm{e}^{-\mathrm{i}\pi/3}$ both appear, and the "if and only if" needs both.
- Forgetting that $u \neq 0$ (distinct points) is what lets us divide by u^2.
- Getting B wrong in the substitution: the w coefficient collects $3q^2$ from the cube, $2aq$ from the square and b, all over p^2.

Beyond STEP

The maps $z \mapsto pz + q$ are the direct similarities of the plane. Quantities they leave unchanged, like the ratio $(\gamma - \alpha)/(\beta - \alpha)$ or the vanishing of $a^2 - 3b$, are *invariants*, a central idea in geometry and algebra. The expression $a^2 - 3b$ is (up to a factor) the discriminant of f', where $f(z) = z^3 + az^2 + bz + c$. This links to a pretty fact, Marden's theorem: the roots of f' are the foci of the ellipse that touches the sides of the triangle of roots of f at their midpoints. For an equilateral triangle that ellipse is a circle, and its two foci merge into the double root of f'. Needham's *Visual Complex Analysis* develops complex geometry in exactly this spirit.

WORKED SOLUTION

STEP III 2006, Question 6

Pure • Polar Curves and Reflection

STEP III 2006, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q6

The key idea

The gradient formula is just the chain rule applied to $x = r\cos\theta$, $y = r\sin\theta$. For the mirror, put the pole at O with the initial line along L: the ray reflected at a point P then runs straight back along the radius, so its direction is given by θ alone. The law of reflection says the normal bisects the angle between the two rays, which pins the normal (and so the tangent) down in terms of θ. That turns the geometry into a separable differential equation for $r(\theta)$.

Solution

The gradient in polar coordinates

A curve given by $r = r(\theta)$ has Cartesian coordinates $x = r\cos\theta$, $y = r\sin\theta$, with r a function of θ. Differentiating with respect to θ by the product rule,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta, \qquad \frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta.$$

By the chain rule, wherever $\mathrm{d}x/\mathrm{d}\theta \neq 0$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}\theta}{\mathrm{d}x/\mathrm{d}\theta} = \frac{\dfrac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta}{\dfrac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta} = \frac{\dfrac{\mathrm{d}r}{\mathrm{d}\theta}\tan\theta + r}{\dfrac{\mathrm{d}r}{\mathrm{d}\theta} - r\tan\theta},$$

dividing top and bottom by $\cos\theta$ (allowed when $\cos\theta \neq 0$). This is the required gradient.

It is worth keeping the undivided version too: the vector

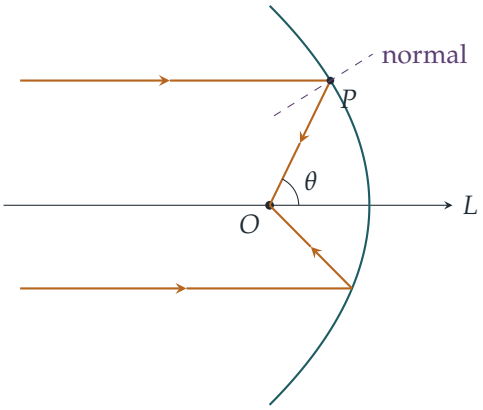
$$\mathbf{t} = \left(\tfrac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta,\ \tfrac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta\right)$$

points along the tangent at every point of the curve, including where $\tan\theta$ or the gradient is undefined.

The mirror

Take any plane Π containing L, and let P be a point where the mirror meets Π. The incident ray at P is parallel to L, so it lies in Π; the reflected ray passes through O, which is on L, so it lies in Π too. The law of reflection also puts the normal at P in the plane of the two rays, so the normal lies in Π, and the tangent to the curve of intersection at P is the line in Π perpendicular to it. So everything happens in the plane Π, and we can work in two dimensions.

In Π use polar coordinates with pole O and initial line along L, oriented so that the incident light travels in the direction $\theta = 0$ (left to right in the figure). Let P be (r, θ) with $-\pi < \theta < \pi$. (The point $\theta = \pi$ would be on L behind O, where a ray would pass straight through O without needing a mirror.)



The direction of the normal. At P take the two unit vectors pointing away from P along the rays:

$$\mathbf{a} = (-1, 0) \quad \text{(back along the incident ray)}, \qquad \mathbf{b} = (-\cos\theta, -\sin\theta) \quad \text{(towards } O\text{)}.$$

The normal makes equal angles with $\mathbf{a}$ and $\mathbf{b}$, so it lies along the bisector of the angle between them, which is the direction of $\mathbf{a} + \mathbf{b}$ (the diagonal of the rhombus formed by two unit vectors bisects their angle). Now

$$\mathbf{a} + \mathbf{b} = -(1 + \cos\theta,\ \sin\theta) = -2\cos\tfrac{\theta}{2}\left(\cos\tfrac{\theta}{2},\ \sin\tfrac{\theta}{2}\right),$$

using $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$. Since $-\pi < \theta < \pi$, $\cos\frac{\theta}{2} > 0$, so this is non-zero and the normal at P is along

$$\mathbf{n} = \left(\cos\tfrac{\theta}{2},\ \sin\tfrac{\theta}{2}\right) :$$

the normal makes angle $\frac{\theta}{2}$ with L.

Ask yourself Why not just set the gradient formula equal to the gradient of the tangent?

We can: the tangent is perpendicular to $\mathbf{n}$, so its gradient is $-\cot\frac{\theta}{2}$, and equating this to the polar formula and clearing fractions leads to the same equation as below. But that route divides by $\tan\theta$, $\cot\frac{\theta}{2}$ and the denominator of the formula, so it breaks at $\theta = 0$ and $\theta = \pm\frac{\pi}{2}$. The tangent vector $\mathbf{t}$ avoids all of that.

The differential equation. The tangent is perpendicular to the normal, so $\mathbf{t}\cdot\mathbf{n} = 0$:

$$\frac{\mathrm{d}r}{\mathrm{d}\theta}\left(\cos\theta\cos\tfrac{\theta}{2} + \sin\theta\sin\tfrac{\theta}{2}\right) + r\left(\cos\theta\sin\tfrac{\theta}{2} - \sin\theta\cos\tfrac{\theta}{2}\right) = 0.$$

By the compound-angle formulae the brackets are $\cos\left(\theta - \frac{\theta}{2}\right) = \cos\frac{\theta}{2}$ and $\sin\left(\frac{\theta}{2} - \theta\right) = -\sin\frac{\theta}{2}$. So

$$\frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\tfrac{\theta}{2} = r\sin\tfrac{\theta}{2}, \qquad \text{that is,} \qquad \frac{1}{r}\frac{\mathrm{d}r}{\mathrm{d}\theta} = \tan\tfrac{\theta}{2},$$

dividing by $r\cos\frac{\theta}{2}$, which is positive because $P \neq O$ and $-\pi < \theta < \pi$.

Solving it. The equation is separable. Integrating with respect to θ, and noting $\int \tan\frac{\theta}{2}\,\mathrm{d}\theta = -2\ln\cos\frac{\theta}{2}$ (valid as $\cos\frac{\theta}{2} > 0$),

$$\ln r = -2\ln\cos\tfrac{\theta}{2} + C \quad\Longrightarrow\quad r\cos^2\tfrac{\theta}{2} = k,$$

where $k = \mathrm{e}^C > 0$ is constant along the curve (the curve is a single connected piece $r = r(\theta)$ on the interval $-\pi < \theta < \pi$). Using $2\cos^2\frac{\theta}{2} = 1 + \cos\theta$,

$$r(1 + \cos\theta) = 2k.$$

Converting to Cartesians. With $x = r\cos\theta$ and $r = \sqrt{x^2 + y^2}$ this reads $\sqrt{x^2 + y^2} = 2k - x$. Squaring (both sides are non-negative, since $2k - x = r \geq 0$),

$$x^2 + y^2 = 4k^2 - 4kx + x^2 \qquad\Longrightarrow\qquad y^2 = 4k(k - x).$$

This is a parabola with axis L, vertex at $(k, 0)$ and focus at the origin O, opening back towards the incoming light. Since Π was any plane containing L, the mirror meets every such plane in a parabola, as required. ■

Checks

- Run it backwards: on $y^2 = 4k(k - x)$ the normal is along $(4k, 2y)$; reflecting the direction $(1, 0)$ in this normal gives a ray through O. We checked this numerically at 10^4 random points with $k = 1.3$, and also checked that $r = 2k/(1 + \cos\theta)$ satisfies $r'/r = \tan\frac{\theta}{2}$ by finite differences.
- The gradient formula agreed with a finite-difference gradient of random curves $r(\theta)$ to about 10^{-8}.
- At $\theta = 0$ the normal lies along L and the tangent is perpendicular to L: that is the vertex $(k, 0)$, where a ray along L reflects straight back through O.

Common slips

- Using the external bisector $\mathbf{a} - \mathbf{b}$ for the normal. That direction makes supplementary angles with the two rays, not equal ones, and leads to $r'/r = -\cot\frac{\theta}{2}$, whose solution $r(1-\cos\theta) = 2k$ is a parabola facing the other way, focusing light that arrives from the opposite direction.
- Forgetting to explain why the problem is two-dimensional: the normal must lie in the plane containing L and P, or the curve of intersection need not have the tangent we used.
- Squaring $\sqrt{x^2+y^2} = 2k - x$ without noting both sides are non-negative.

Beyond STEP

The mirror is a paraboloid of revolution, and the property proved here is the focus–directrix property in disguise: $r(1+\cos\theta) = 2k$ says that the distance from O equals the distance from the line $x = 2k$. This is why satellite dishes, car headlamps and reflecting telescopes are paraboloids. More generally, conics all have the polar form $r(1+e\cos\theta) = \ell$, which is exactly what appears in Newton's solution of planetary orbits; Kleppner and Kolenkow, *An Introduction to Mechanics*, derives it from the inverse-square law.

WORKED SOLUTION

STEP III 2006, Question 7

Pure • First-Order Differential Equations

STEP III 2006, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q7

The key idea

Both differential equations are quadratics in $\mathrm{d}y/\mathrm{d}x$. Rather than attacking them as differential equations straight away, we solve each quadratic for the derivative, which splits the equation into two ordinary first-order equations. Hyperbolic identities make the roots collapse to something simple, and the conditions at $x = 0$ tell us which root to follow.

Solution

Part (i)

By the quadratic formula,

$$u = -\sinh x \pm \sqrt{\sinh^2 x + 1} = -\sinh x \pm \cosh x,$$

since $\sinh^2 x + 1 = \cosh^2 x$ and $\cosh x > 0$. Using $\cosh x + \sinh x = \mathrm{e}^x$ and $\cosh x - \sinh x = \mathrm{e}^{-x}$, the two roots are

$$u = \mathrm{e}^{-x} \qquad \text{or} \qquad u = -\mathrm{e}^x.$$

The differential equation is this quadratic with $u = \mathrm{d}y/\mathrm{d}x$, so at every x we must have

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{e}^{-x} \qquad \text{or} \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = -\mathrm{e}^x.$$

The first root is always positive and the second always negative, so they never coincide. A solution with a continuous derivative therefore cannot jump from one root to the other, and the condition $\mathrm{d}y/\mathrm{d}x > 0$ at $x = 0$ picks out the first one throughout. Integrating, $y = -\mathrm{e}^{-x} + c$, and $y = 0$ at $x = 0$ gives $c = 1$:

$$y = 1 - \mathrm{e}^{-x}.$$

Part (ii)

Write $p = \mathrm{d}y/\mathrm{d}x$. If $\sinh y = 0$, that is $y = 0$, then the equation reduces to $2p = 0$. This is how $y \equiv 0$ comes to be a solution, and it is the one we are told to set aside. Where $y \neq 0$, the quadratic formula gives

$$p = \frac{-1 \pm \sqrt{1 + \sinh^2 y}}{\sinh y} = \frac{-1 \pm \cosh y}{\sinh y}.$$

With the half-angle forms $\cosh y - 1 = 2\sinh^2 \frac{y}{2}$, $\cosh y + 1 = 2\cosh^2 \frac{y}{2}$ and $\sinh y = 2\sinh \frac{y}{2} \cosh \frac{y}{2}$, the two roots are

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \tanh \frac{y}{2} \qquad \text{or} \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = -\coth \frac{y}{2}.$$

These never coincide, because their product is -1, whereas equal roots would have product $\tanh^2 \frac{y}{2} \geq 0$.

Ask yourself Which of the two equations can give a solution through the origin that is not identically zero?

The first equation, $y' = \tanh \frac{y}{2}$, has the solution $y \equiv 0$. Its right-hand side is a smooth function of y, so through $(0, 0)$ that is its only solution: this is the trivial solution again. The second equation is singular at $y = 0$, where $|\mathrm{d}y/\mathrm{d}x| \to \infty$, so its solution curve can meet $y = 0$ with a vertical tangent. That is the one we want. Separating the variables,

$$\int \tanh \frac{y}{2} \, \mathrm{d}y = -\int \mathrm{d}x \quad \Longrightarrow \quad 2\ln\cosh \frac{y}{2} = -x + C.$$

Putting $y = 0$ at $x = 0$ gives $C = 0$, so $\cosh^2 \frac{y}{2} = \mathrm{e}^{-x}$. Since $\cosh y = 2\cosh^2 \frac{y}{2} - 1$,

$$\cosh y = 2\mathrm{e}^{-x} - 1.$$

We can check this directly. Differentiating gives $\sinh y \, y' = -2\mathrm{e}^{-x} = -(\cosh y + 1)$, which is exactly $y' = -\coth \frac{y}{2}$. This works for both signs of y, so the whole curve, upper and lower halves, is a solution. As $\cosh y \geq 1$, the curve exists only for $x \leq 0$. It is symmetric in the x-axis. Near the origin $1 + \frac{1}{2}y^2 \approx 1 - 2x$, so the curve looks like the parabola $y^2 = -4x$ there.

Asymptotes. As $x \to -\infty$ we have $\cosh y \to \infty$, so $|y| \to \infty$. Write $w = 2\mathrm{e}^{-x} - 1$. Then

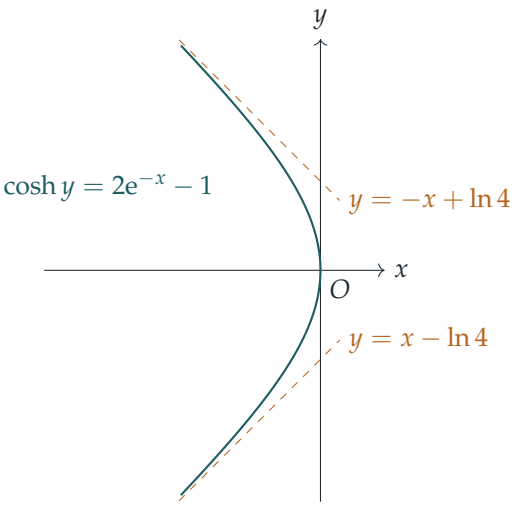
$$|y| = \operatorname{arcosh} w = \ln\bigl(w + \sqrt{w^2 - 1}\bigr),$$

so

$$|y| - (-x + \ln 4) = \ln \frac{w + \sqrt{w^2 - 1}}{4\mathrm{e}^{-x}} = \ln\Bigl(\frac{w}{4\mathrm{e}^{-x}}\Bigl(1 + \sqrt{1 - w^{-2}}\Bigr)\Bigr).$$

As $x \to -\infty$, $w/(4\mathrm{e}^{-x}) = \frac{1}{2} - \frac{1}{4}\mathrm{e}^{x} \to \frac{1}{2}$ and $1 + \sqrt{1 - w^{-2}} \to 2$, so the argument of the logarithm tends to 1 and the difference tends to 0. Hence the upper half of

the curve approaches $y = -x + \ln 4$ and, by symmetry, the lower half approaches $y = -(-x + \ln 4)$. The asymptotes are $y = \pm(-x + \ln 4)$, as required. (For $x \to 0^-$ the curve stays bounded, so these are the only asymptotes.)



Checks

- In (i), $y = 1 - e^{-x}$ has $y' = e^{-x}$, and $e^{-2x} + 2e^{-x}\sinh x - 1 = e^{-2x} + (1 - e^{-2x}) - 1 = 0$.
- In (ii), stepping $y' = -\coth\frac{y}{2}$ numerically from near the origin out to $x = -6$ gives $y \approx 7.38505$, matching $\operatorname{arcosh}(2e^6 - 1)$. At $x = -10$ the gap between the curve and $-x + \ln 4$ is about 2×10^{-5}.

Common slips

- Taking the root $-\sinh x - \cosh x = -e^x$ in (i): its derivative is negative at $x = 0$.
- Picking $y' = \tanh\frac{y}{2}$ in (ii) because it "behaves well" at $y = 0$. Through the origin it gives only $y \equiv 0$.
- Forgetting the lower half of the curve, which has $y < 0$ and is needed for the second asymptote.

Beyond STEP

An equation that is not linear in y', like these, is a *first-order implicit* equation. Solving for y' splits it into several branches. Part (ii) also shows that uniqueness of solutions can fail: at $y = 0$ the right-hand side $-\coth\frac{y}{2}$ is not Lipschitz, which is why a non-trivial solution can start from the same point as $y \equiv 0$. Boyce and DiPrima, *Elementary*

Differential Equations, covers existence and uniqueness theorems and when they apply.

WORKED SOLUTION

STEP III 2006, Question 8

Pure • Axiomatic Differentiation

STEP III 2006, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q8

The key idea

The four rules are exactly the properties of differentiation you use every day: linearity and the product rule, plus the single fact that x "differentiates to" 1. Every polynomial is built from the number 1 and the letter x by multiplying and adding, so if we know what $\triangle$ does to 1 and to x, the rules leave it no freedom anywhere else. The product rule applied to $x \cdot x^n$ gives an induction on powers, and linearity then finishes the job.

Solution

Throughout, "rule (i)" to "rule (iv)" refer to the four defining rules, in the order given.

Constants

First take the constant polynomial 1. Since $1 = 1 \cdot 1$, rule (iv) with $f(x) = g(x) = 1$ gives

$$\triangle 1 = \triangle(1 \cdot 1) = 1 \cdot \triangle 1 + 1 \cdot \triangle 1 = 2\triangle 1.$$

Subtracting $\triangle 1$ from both sides gives $\triangle 1 = 0$ (the zero polynomial).

Ask yourself Why not just say "$\triangle$ of a constant is zero, like a derivative"?

Because we don't yet know that $\triangle$ *is* a derivative; that is what the last part asks us to prove. All we may use are the four rules, and none of them mentions constants directly. The trick is that 1 is its own square, so the product rule forces $\triangle 1$ to equal twice itself.

Now let $f(x) = \lambda$ be any constant. Then $\lambda = \lambda \cdot 1$, so by rule (iii)

$$\triangle\lambda = \triangle(\lambda \cdot 1) = \lambda\triangle 1 = \lambda \cdot 0 = 0.$$

So $\triangle f(x) = 0$ for every polynomial of degree zero, as required.

The square and the cube

Writing $x^2 = x \cdot x$ and using rule (iv), then rule (i),

$$\triangle x^2 = x\,\triangle x + x\,\triangle x = x \cdot 1 + x \cdot 1 = 2x.$$

Writing $x^3 = x \cdot x^2$ and using rule (iv), then the result just found and rule (i),

$$\triangle x^3 = x\,\triangle x^2 + x^2\,\triangle x = x \cdot 2x + x^2 \cdot 1 = 3x^2.$$

The general polynomial

Step 1: powers of x. We prove by induction that

$$\triangle x^n = n\,x^{n-1} \qquad \text{for every integer } n \geq 1.$$

For $n = 1$ this says $\triangle x = 1$, which is rule (i). Suppose it holds for some $n = k \geq 1$. Since $x^{k+1} = x \cdot x^k$, rule (iv) gives

$$\begin{aligned} \triangle x^{k+1} &= x\,\triangle x^k + x^k\,\triangle x \\ &= x \cdot k\,x^{k-1} + x^k \cdot 1 \\ &= (k+1)\,x^k, \end{aligned}$$

using the induction hypothesis and rule (i) in the second line. So the result holds for $n = k+1$, and by induction for all $n \geq 1$. For $n = 0$, $x^0 = 1$ and we showed $\triangle 1 = 0$, which agrees with $\dfrac{\mathrm{d}}{\mathrm{d}x}1 = 0$. So for every $n \geq 0$,

$$\triangle x^n = \frac{\mathrm{d}}{\mathrm{d}x}\,x^n.$$

Step 2: sums of many terms. Rule (ii) is stated for two polynomials, but it extends to any finite number. If $\triangle(f_1 + \cdots + f_m) = \triangle f_1 + \cdots + \triangle f_m$, then applying rule (ii) to the two polynomials $f_1 + \cdots + f_m$ and f_{m+1} gives

$$\triangle(f_1 + \cdots + f_{m+1}) = \triangle(f_1 + \cdots + f_m) + \triangle f_{m+1} = \triangle f_1 + \cdots + \triangle f_{m+1},$$

so by induction on m it holds for any number of terms.

Step 3: putting it together. Any polynomial can be written $h(x) = \sum_{k=0}^{n} a_k x^k$ with constants a_k. Then

$$\begin{aligned} \triangle h(x) &= \sum_{k=0}^{n} \triangle\left(a_k x^k\right) && \text{by rule (ii), extended in Step 2} \\ &= \sum_{k=0}^{n} a_k\,\triangle x^k && \text{by rule (iii)} \\ &= \sum_{k=1}^{n} k\,a_k\,x^{k-1} && \text{by Step 1 (the } k = 0 \text{ term is 0)} \\ &= \frac{\mathrm{d}h(x)}{\mathrm{d}x}, \end{aligned}$$

the last line being the usual derivative of a polynomial. This holds for every polynomial $h(x)$, so $\triangle h(x) \equiv \dfrac{\mathrm{d}h(x)}{\mathrm{d}x}$, as required.

Checks

It is reassuring that ordinary differentiation obeys all four rules, so the rules aren't contradictory and $\mathrm{d}/\mathrm{d}x$ really is an operation of the kind described. You can also test the product rule against the formula: for $f = x^2$ and $g = x^3$, rule (iv) gives $x^2 \cdot 3x^2 + x^3 \cdot 2x = 5x^4$, which is $\triangle x^5$. In a computer we built $\triangle$ using only the rules (as in Steps 1 to 3) and compared it with the derivative, exactly, for 2000 random polynomials of degree up to 9; for instance $\triangle\left(\frac{1}{2}x^4 - 3x + 7\right) = 2x^3 - 3$, which also matches a numerical difference quotient.

Common slips

- Asserting $\triangle(\text{constant}) = 0$ "because it's a derivative". That assumes the final answer; it has to be derived from the rules.
- Using rule (ii) for a long sum without comment. It's stated for two polynomials, so a sentence of induction is needed for more terms.
- Not saying which rule is used where: the question explicitly asks for this.

Beyond STEP

An operation obeying linearity and the product rule is called a *derivation*, and the argument here shows that on polynomials a derivation is completely fixed by its value on x. If instead $\triangle x = p(x)$ for some polynomial p, the same proof gives $\triangle h = p(x)\,h'(x)$. The idea that differentiation can be defined algebraically, with no limits at all, is central in abstract algebra and in differential geometry, where vector fields are exactly the derivations on functions. Allenby's *Rings, Fields and Groups* is a good place to meet polynomial rings treated algebraically.

WORKED SOLUTION

STEP III 2006, Question 9

Mechanics • Energy Conservation

STEP III 2006, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

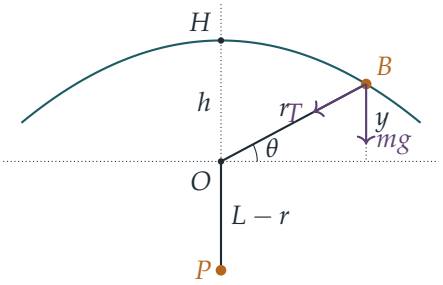
step.maths.org/questions/06-s3-q9

The key idea

The string has fixed length, so whatever the bead gains in distance from O, the particle P rises by the same amount. That ties the two potential energies together: the total depends only on $y + r$. "Equilibrium in every position" means the potential energy has no preferred position, so it must be constant along the wire. Once it is constant, energy conservation says the total kinetic energy never changes, and the rest is writing the bead's speed in polar coordinates.

Solution

Model. P and B are particles of mass m; the string is light and inextensible; the ring and the wire are smooth, so the string tension does no net work and the wire's normal reaction does no work. Take O as origin, with x horizontal and y vertically up, and let the string have length L.



The shape of the wire

The length of string from O to B is r, so the length OP is $L - r$ and P is a distance $L - r$ below O. Taking the horizontal plane through O as the zero of potential energy, the

total potential energy is

$$U = mgy - mg(L - r) = mg(y + r) - mgL.$$

The system has one degree of freedom (the position of the bead on the wire), and an equilibrium position is a stationary point of U along the wire. If every position is an equilibrium, then $\mathrm{d}U/\mathrm{d}s = 0$ everywhere along the wire (s being arc length), so U is constant along the wire. Hence $y + r$ is constant.

At H the bead is directly above O at height h, so $y = h$ and $r = h$ there, and the constant is $2h$. So the wire is

$$y + r = 2h,$$

as required. (In Cartesian form, $x^2 + y^2 = (2h - y)^2$ gives $y = h - x^2/(4h)$: a parabola with focus O, as in the picture. You can also confirm the equilibrium directly: the tension is mg, since P is in equilibrium, and the resultant of mg along BO and mg downwards bisects the angle between those directions, which is exactly the normal to a parabola at a point, by its reflection property.)

The rate of change of θ

On the wire $y = 2h - r$, so

$$\sin\theta = \frac{y}{r} = \frac{2h}{r} - 1.$$

Differentiating with respect to time,

$$\cos\theta\,\dot{\theta} = -\frac{2h\dot{r}}{r^2}.$$

Since $\theta = \arcsin(y/r)$ lies in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, we have $\cos\theta \geq 0$, so

$$\cos\theta = \sqrt{1 - \left(\frac{2h - r}{r}\right)^2} = \frac{\sqrt{r^2 - (2h - r)^2}}{r} = \frac{\sqrt{4rh - 4h^2}}{r} = \frac{2\sqrt{rh - h^2}}{r}.$$

After the bead leaves H we have $r > h$ (because $r = 2h - y$ and $y < h$ away from H), so $\cos\theta \neq 0$ and we may divide:

$$\dot{\theta} = -\frac{2h\dot{r}}{r^2} \cdot \frac{r}{2\sqrt{rh - h^2}} = -\frac{h\dot{r}}{r\sqrt{rh - h^2}},$$

which is the required result.

The speed of P

Ask yourself Why must the total kinetic energy stay equal to $\frac{1}{2}mV^2$?

The only forces that do work on the system are gravity and the (internal) string tension. The tension does no net work: the string is inextensible and the ring smooth, so the work done by the tension on B cancels that done on P. The wire is smooth, so its

reaction does no work. Hence kinetic plus potential energy is conserved. But we have just shown that the potential energy U is *constant* along this wire, so the kinetic energy is constant too.

At the start the bead is at H, where r has its minimum value h (since y is greatest there). So $\dot r = 0$ initially, and P, whose speed is $|\dot r|$, starts at rest. The initial kinetic energy is therefore $\frac{1}{2}mV^2$.

In polar coordinates about O, the bead's speed squared is $\dot r^2 + r^2\dot\theta^2$. Using the result for $\dot\theta$,

$$r^2\dot\theta^2 = \frac{h^2\dot r^2}{rh - h^2} = \frac{h\dot r^2}{r-h}, \qquad \text{so} \qquad v_B^2 = \dot r^2\Big(1 + \frac{h}{r-h}\Big) = \frac{r\dot r^2}{r-h}.$$

Conservation of kinetic energy, after cancelling $\frac{1}{2}m$, gives

$$\frac{r\dot r^2}{r-h} + \dot r^2 = V^2 \quad\Longrightarrow\quad \dot r^2\,\frac{2r-h}{r-h} = V^2.$$

Since $r > h$, both $r-h$ and $2r-h$ are positive, and the speed of P is

$$|\dot r| = V\Big(\frac{r-h}{2r-h}\Big)^{\frac{1}{2}},$$

as required.

Checks

- At H, $r = h$ and the formula gives 0: correct, since P starts at rest.
- As $r \to \infty$ the speed of P tends to $V/\sqrt{2}$. Far out the bead moves almost radially, so B and P have nearly equal speeds and share the kinetic energy equally.
- Numerically, with $h = 1$ and $V = 2$, we stepped the bead's equation of motion along $y = 1 - \frac{1}{4}x^2$ (with RK4) and compared finite-difference values of $\dot\theta$ and $\dot r$ with the formulae. At $r \approx 4.043$ the speed of P was 1.31064, matching $2\sqrt{3.043/7.087}$ to six figures.

Common slips

- Taking the potential energy of P as $+mg(L-r)$: P is *below* O, so its height is $-(L-r)$.
- Forgetting that P moves too, and writing $\frac{1}{2}mv_B^2 = \frac{1}{2}mV^2$.
- Dividing by $\cos\theta$ without noting that it vanishes only at H.

Beyond STEP

Finding a curve on which a system is in "neutral" equilibrium everywhere is a small instance of the principle that equilibria are stationary points of the potential energy,

which leads on to Lagrangian mechanics, where constraints like the string and wire are built in by choosing good coordinates. Here the single coordinate r does the job, and energy conservation is the first integral. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers energy methods and polar coordinates in this spirit.

WORKED SOLUTION

STEP III 2006, Question 10

Mechanics • Conservation of Angular Momentum

STEP III 2006, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q10

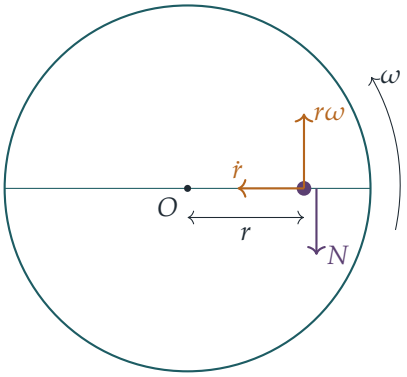
The key idea

Nothing external acts on the disc-and-particle system to turn it about the axis, and the smooth groove does no net work, so there are two conservation laws: angular momentum about the axis and kinetic energy. These give ω and $\dot{r}$ as functions of r alone, and the strange-looking condition linking V, Ω, a and k is there to make the energy expression collapse. Dividing $\dot{r}$ by $\dot{\theta} = \omega$ removes time, which leaves a separable equation for the path.

Solution

The model

We treat the particle as a point mass m, and the groove as smooth, so the only horizontal force on the particle is a reaction N perpendicular to the groove. The disc turns freely about a fixed vertical axis, so the bearing exerts no torque about it. Gravity and the vertical reactions are vertical and play no part. Let θ be the angle the disc has turned through, so $\omega = \dot{\theta}$, and let r be the particle's distance from the axis. Because the particle stays in the groove, its velocity has a component $\dot{r}$ along the groove and a component $r\omega$ perpendicular to it.



Angular speed

The reaction N acts on the particle and $-N$ acts on the disc at the same point, so their moments about the axis cancel. With no external torque about the axis, the total angular momentum about it is constant. The disc contributes $mk^2\omega$, and the particle contributes $mr \cdot r\omega = mr^2\omega$ (only the transverse component of its velocity has a moment). At $t = 0$ the total is $m(k^2 + a^2)\Omega$, so

$$(k^2 + r^2)\,\omega = (k^2 + a^2)\,\Omega, \qquad \text{that is,} \qquad \omega = \frac{\Omega(k^2 + a^2)}{k^2 + r^2},$$

which is the first result.

Radial speed

Ask yourself The groove pushes the particle sideways while the particle moves. Why is energy conserved?

The reaction N is perpendicular to the groove. The particle and the point of the disc in contact with it have the same velocity component perpendicular to the groove, namely $r\omega$; they differ only by the sliding velocity $\dot{r}$ along the groove, which is perpendicular to N. So the work N does on the particle is cancelled exactly by the work $-N$ does on the disc. The bearing at the axis does no work, so the total kinetic energy is constant.

Cancelling the common factor $\frac{1}{2}m$, conservation of energy gives

$$k^2\omega^2 + \dot{r}^2 + r^2\omega^2 = k^2\Omega^2 + V^2 + a^2\Omega^2.$$

Now we use the given condition $V^2 = \Omega^2 a^2(k^2 + a^2)/k^2$ on the right:

$$(k^2 + a^2)\Omega^2 + \frac{\Omega^2 a^2(k^2 + a^2)}{k^2} = \frac{\Omega^2(k^2 + a^2)(k^2 + a^2)}{k^2} = \frac{\Omega^2(k^2 + a^2)^2}{k^2}.$$

On the left, $(k^2+r^2)\omega^2 = \Omega^2(k^2+a^2)^2/(k^2+r^2)$ by the first result. Hence

$$\begin{aligned}\dot{r}^2 &= \Omega^2(k^2+a^2)^2\left(\frac{1}{k^2}-\frac{1}{k^2+r^2}\right)\\ &= \frac{\Omega^2 r^2(k^2+a^2)^2}{k^2(k^2+r^2)},\end{aligned}$$

as required.

The equation of the path

While the particle moves towards the axis, $\dot{r} < 0$, so taking the negative square root,

$$\dot{r} = -\frac{\Omega r(k^2+a^2)}{k\sqrt{k^2+r^2}}.$$

Since $\omega = \dot{\theta} > 0$, we can use the chain rule, $\dfrac{\mathrm{d}r}{\mathrm{d}\theta} = \dot{r}\Big/\dot{\theta} = \dot{r}/\omega$:

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = -\frac{\Omega r(k^2+a^2)}{k\sqrt{k^2+r^2}}\cdot\frac{k^2+r^2}{\Omega(k^2+a^2)} = -\frac{r\sqrt{k^2+r^2}}{k},$$

so $k\dfrac{\mathrm{d}r}{\mathrm{d}\theta} = -r(k^2+r^2)^{1/2}$, as required.

Solving for r

Put $u = k/r$, so $u > 0$. Then

$$\frac{\mathrm{d}u}{\mathrm{d}\theta} = -\frac{k}{r^2}\frac{\mathrm{d}r}{\mathrm{d}\theta} = \frac{k}{r^2}\cdot\frac{r\sqrt{k^2+r^2}}{k} = \sqrt{\frac{k^2}{r^2}+1} = \sqrt{1+u^2}.$$

This separates:

$$\int\frac{\mathrm{d}u}{\sqrt{1+u^2}} = \int\mathrm{d}\theta \quad\Longrightarrow\quad \operatorname{arsinh} u = \theta+\alpha$$

for some constant α. So $k/r = \sinh(\theta+\alpha)$, that is,

$$r\sinh(\theta+\alpha) = k.$$

At $t = 0$ we have $\theta = 0$ and $r = a$, so $\sinh\alpha = k/a$, as required.

The particle never reaches the axis

> **Ask yourself** $r = k/\sinh(\theta+\alpha)$ is never zero, but could θ become infinite in a finite time?

No. Since $r^2 \geq 0$, the first result gives $\omega \leq \Omega(k^2+a^2)/k^2$, so $\theta \leq \Omega(k^2+a^2)t/k^2$ is finite at every finite time t. Then $r = k/\sinh(\theta+\alpha) > 0$ at every time.

We should also check that the formulae stay valid, that is, that the particle keeps moving towards the axis. By the formula for $\dot{r}^2$, $\dot{r}$ can only vanish when $r = 0$. So while $r > 0$, $\dot{r}$ keeps its initial negative sign, and the argument above applies for all $t > 0$. The particle spirals ever closer to the axis, with $r \to 0$ only as $\theta \to \infty$, but it never reaches it.

Checks

- At $r = a$ the formula for $\dot{r}^2$ gives $\Omega^2 a^2 (k^2 + a^2)/k^2 = V^2$, matching the initial condition.
- With $k = a = \Omega = 1$ (so $V = \sqrt{2}$), we integrated the equations of motion $\ddot{r} = r\omega^2$ and $(k^2 + r^2)\omega =$ constant numerically. The particle reached $r = 0.01$ after the disc had turned through $\theta \approx 4.4170$, and $\operatorname{arsinh} 100 - \operatorname{arsinh} 1 \approx 5.2983 - 0.8814 = 4.4169$ agrees. Energy, $\dot{r}^2$ and $r \sinh(\theta + \alpha) = k$ all held along the run to about 10^{-11}.

Common slips

- Forgetting the particle's own angular momentum $mr^2\omega$, or its transverse kinetic energy $\frac{1}{2}mr^2\omega^2$.
- Taking the positive square root for $\dot{r}$, which gives the wrong sign in $\mathrm{d}r/\mathrm{d}\theta$.
- Concluding "never reaches the axis" from $r \neq 0$ alone, without noting that θ stays finite at finite times.

Beyond STEP

The particle feels no radial force, yet in the rotating frame of the disc it seems to be pushed outwards. That is the centrifugal term in $\ddot{r} = r\omega^2$, and the sideways reaction N balances the Coriolis force. Undergraduate mechanics handles problems like this with Lagrangian methods: the angle θ does not appear in the energy, so its conjugate momentum, here the angular momentum, is conserved automatically. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rotating frames and angular momentum at the right level.

WORKED SOLUTION

STEP III 2006, Question 11

Mechanics • Collisions in Connected Systems

STEP III 2006, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q11

The key idea

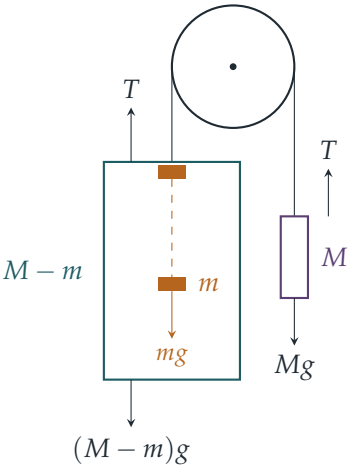
Once the tile lets go, the lift is lighter than the counterweight, so the lift starts to rise while the tile falls: the tile has to cover the height h *relative to the lift*. For the impact, the lift and counterweight act as a single body of mass $2M - m$ joined to the tile through the floor, and the given condition on the cable's impulses is exactly what lets us write a momentum-type conservation law for that body and the tile. That "momentum" turns out to be zero just before impact, which makes the energy bookkeeping very clean.

Solution

We model the lift, counterweight and tile as particles (for vertical motion), the cable as light and inextensible, and the pulley as smooth. Before the tile detaches the system is at rest, so the lift *including* the tile balances the counterweight: the lift has mass M with the tile, and $M - m$ without it.

The fall

Take upwards as positive for the lift and the tile. Since the cable is inextensible, when the lift moves up with speed u and acceleration a, the counterweight moves down with the same speed and acceleration. Let T be the tension.



Newton's second law for the lift (upwards) and the counterweight (downwards) gives

$$(M-m)a = T-(M-m)g, \qquad Ma = Mg - T.$$

Adding, $(2M-m)a = mg$, so

$$a = \frac{mg}{2M-m}.$$

The tile is in free fall with acceleration g downwards (the guides are smooth and nothing else touches it). Relative to the lift, the tile therefore accelerates downwards at

$$g + a = g + \frac{mg}{2M-m} = \frac{2Mg}{2M-m},$$

starting from rest relative to the lift. It meets the floor after a time t with $\frac{1}{2}(g+a)t^2 = h$, so

$$t^2 = \frac{(2M-m)h}{Mg}.$$

At that moment the tile's speed is

$$v = gt = \sqrt{g^2 \cdot \frac{(2M-m)h}{Mg}} = \sqrt{\frac{(2M-m)gh}{M}},$$

as required. The lift (and counterweight) then have speed

$$U = at = \frac{m}{2M-m}\,gt = \frac{mv}{2M-m}.$$

In this time the lift rises $\frac{1}{2}at^2 = \dfrac{mh}{2M} < h$, so the condition on the distance to the pulley guarantees the lift doesn't reach the pulley first.

The impact

Let J be the impulse of the floor on the tile (upwards), and let I be the impulse of the cable's tension, which by the given condition has the same magnitude on the lift and on the counterweight (it acts upwards on each). Gravity gives no impulse during the instantaneous impact. Let the tile's velocity just after be w (upwards), and let the lift's velocity just after be U' (upwards), so the counterweight's downward velocity is also U'. Then:

$$\begin{aligned} \text{tile:} \quad & m\big(w-(-v)\big) = J, \\ \text{lift:} \quad & (M-m)(U'-U) = I - J, \\ \text{counterweight (downwards):} \quad & M(U'-U) = -I. \end{aligned}$$

Adding all three eliminates J and I:

$$mw + (2M-m)U' = -mv + (2M-m)U.$$

Ask yourself Why is it legitimate to add the counterweight's equation, which uses "downwards" as positive, to the others?

Because of the cable, "counterweight down" and "lift up" are the same coordinate. What we have shown is that the combination $mw + (2M-m)U'$ (tile's upward momentum plus the momentum of lift and counterweight measured along the cable) is unchanged by the impact. This is where the equal-impulse condition is used: without it, I would not cancel.

Now the key observation: since $U = \dfrac{mv}{2M-m}$, the right-hand side is $-mv + mv = 0$. So

$$mw + (2M-m)U' = 0 = -mv + (2M-m)U.$$

Newton's experimental law at the floor says that the speed of separation is e times the speed of approach:

$$w - U' = e\big(U-(-v)\big) = e(U+v).$$

Both the before and after velocities satisfy the same "zero momentum" equation, so each of the pairs (w, U') and $(-v, U)$ is a multiple of $(2M-m,\ -m)$. The restitution equation then fixes the multiple: $w = ev$ and $U' = -eU$. (Check: $mev - (2M-m)eU = e(mv - mv) = 0$, and $w - U' = e(v+U)$.)

The energy loss

Just before impact, the kinetic energy is

$$\tfrac{1}{2}mv^2 + \tfrac{1}{2}(2M-m)U^2 = \tfrac{1}{2}mv^2\left(1 + \frac{m}{2M-m}\right) = \frac{Mmv^2}{2M-m} = mgh,$$

using $U = \frac{mv}{2M-m}$ and then $v^2 = \frac{(2M-m)gh}{M}$. Just after, every velocity has been multiplied by $\pm e$, so the kinetic energy is $e^2\,mgh$. The impact is instantaneous, so the potential energy doesn't change during it, and the loss of energy of the system is

$$mgh - e^2 mgh = mgh(1-e^2),$$

as required.

Comment

The answer does not depend on M at all: it is exactly the loss we would get if the tile fell a height h inside a *fixed* lift and bounced with coefficient e. That makes sense. Just before impact the system's kinetic energy is mgh, which is simply the potential energy released (the tile and the lift close up a gap of h, and the lift's rise is exactly balanced by the counterweight's fall). And because the "momentum" along the cable is zero, the impact behaves like a collision viewed in the zero-momentum frame, where a fraction $1 - e^2$ of the kinetic energy is always lost. In particular, if $e = 0$ all the kinetic energy is lost and the whole system comes to rest; if $e = 1$ nothing is lost.

Checks

- Take $M = 5$, $m = 2$, $h = 3$, $g = 10$, $e = \frac{1}{3}$. Then $v^2 = 48$ and $U^2 = 3$, so the kinetic energy before impact is $\frac{1}{2} \cdot 2 \cdot 48 + \frac{1}{2} \cdot 8 \cdot 3 = 60 = mgh$, and the loss is $60 \cdot \frac{8}{9} = \frac{160}{3}$.
- Stepping the motion numerically (tile in free fall, lift accelerating up at $\frac{mg}{2M-m}$) for 200 random choices of M, m, h, e reproduces the impact speed v, the lift speed U and the loss $mgh(1 - e^2)$.
- As $m \to 0$ the lift hardly moves and $v \to \sqrt{2gh}$, the free-fall speed, as it should.
- The tension impulse $I = M(U - U') = M(1 + e)U$ is positive, so the cable stays taut during the impact.

Common slips

- Treating the lift as having mass M after the tile has fallen. Then nothing would accelerate and the tile's speed would be $\sqrt{2gh}$.
- Forgetting that the floor rises to meet the tile, so the tile falls less than h relative to the ground.
- Applying conservation of momentum to the tile and lift alone, forgetting the counterweight, which is linked to the lift through the cable.

Beyond STEP

The quantity conserved in the impact is not ordinary momentum but the momentum conjugate to the single coordinate describing the cable's position: a first taste of generalised coordinates and Lagrangian mechanics. The fact that a collision with zero total momentum loses exactly the fraction $1 - e^2$ of the kinetic energy is the centre-of-mass frame idea that runs through all of collision theory. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both momentum in the centre-of-mass frame and systems of connected bodies carefully.

WORKED SOLUTION

STEP III 2006, Question 12

Probability and Statistics • Normal Approximation Expectation

STEP III 2006, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q12

The key idea

The extra bus is only worth what it earns on average, so the licence fee the company should consider is at most the *expected* extra profit per year. The extra bus earns nothing on a day when the existing fleet already copes, and it can never carry more than its 32 seats. So we need the expectation of a "clipped" random variable. The numbers are chosen so that, under the normal approximation, the old and new capacities land on very tidy points of the distribution.

Solution

Setting up the model

On a single arrival, let X be the number of the 1024 tourists directed to the bus station. Each tourist goes there with probability $\frac{1}{2}$, and we assume the tourists are directed independently of one another. Then

$$X \sim \mathrm{B}\left(1024, \tfrac{1}{2}\right), \qquad \mathrm{E}(X) = 512, \qquad \mathrm{Var}(X) = 1024 \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = 256.$$

Since n is large and $p = \frac{1}{2}$, we use the normal approximation $X \approx \mathrm{N}(512, 16^2)$, and write $X = 512 + 16Z$ with $Z \sim \mathrm{N}(0,1)$. As instructed, we make no continuity correction.

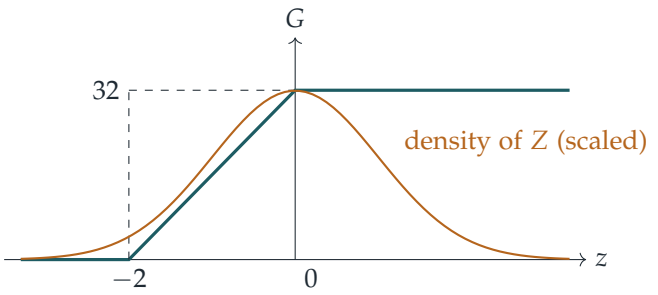
The 15 buses carry $15 \times 32 = 480$ passengers, so the current profit on one arrival is $£\min(X, 480)$. With a 16th bus the capacity is 512 and the profit is $£\min(X, 512)$. The extra profit on one arrival is therefore

$$G = \min(X,512) - \min(X,480) = \begin{cases} 0, & X \le 480, \\ X - 480, & 480 \le X \le 512, \\ 32, & X \ge 512. \end{cases}$$

In terms of Z, the two break points are $X = 480 \iff Z = -2$ and $X = 512 \iff Z =$

0, and in the middle range $X - 480 = 32 + 16Z$. So

$$G = \begin{cases} 0, & Z \le -2, \\ 32 + 16Z, & -2 \le Z \le 0, \\ 32, & Z \ge 0. \end{cases}$$



The expected extra profit

Let $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$ be the standard normal density. Splitting the expectation over the three ranges,

$$\mathrm{E}(G) = 32\,\mathrm{P}(Z \ge 0) + \int_{-2}^{0} (32 + 16z)\,\phi(z)\,\mathrm{d}z.$$

The first term is $32 \times \frac{1}{2} = 16$. For the integral, the constant part gives

$$32 \int_{-2}^{0} \phi(z)\,\mathrm{d}z = 32\big(\Phi(0) - \Phi(-2)\big) = 32\big(\tfrac{1}{2} - (1 - \Phi(2))\big) = 32\Phi(2) - 16,$$

using the symmetry $\Phi(-2) = 1 - \Phi(2)$.

? **Ask yourself** How do we integrate $z\,\phi(z)$ without a table?

Differentiate the density: $\phi'(z) = -z\,\phi(z)$. So $z\,\phi(z)$ has the exact antiderivative $-\phi(z)$, and

$$16 \int_{-2}^{0} z\,\phi(z)\,\mathrm{d}z = 16\big[-\phi(z)\big]_{-2}^{0} = -16\big(\phi(0) - \phi(2)\big) = -\frac{16}{\sqrt{2\pi}}\big(1 - e^{-2}\big).$$

(It is negative, as it should be: $z < 0$ throughout the range.)

Adding the three pieces, the 16s cancel:

$$\mathrm{E}(G) = 16 + \big(32\Phi(2) - 16\big) - \frac{16}{\sqrt{2\pi}}\big(1 - e^{-2}\big) = 32\Phi(2) - \frac{16}{\sqrt{2\pi}}\big(1 - e^{-2}\big).$$

The licence fee

There are 50 arrivals a year, so by linearity of expectation the expected extra annual profit from the 16th bus is

$$50\,\mathrm{E}(G) = 1600\,\Phi(2) - \frac{800}{\sqrt{2\pi}}\left(1 - \mathrm{e}^{-2}\right) \text{ pounds.}$$

Paying a licence fee larger than this would, on average, lose the company money, while any smaller fee leaves an expected gain. So this is (approximately, because of the normal approximation) the largest annual fee worth considering, which is the required result. Numerically it is about £1288.

Checks

- Sanity bounds: the extra bus earns between 0 and 32 per trip, so the annual figure must lie between 0 and 1600. Our value $1600 \times 0.97725 - 319.15 \times 0.86466 \approx 1287.6$ does.
- Exact computation: summing $\min(k, 512) - \min(k, 480)$ against the exact $\mathrm{B}(1024, \frac{1}{2})$ probabilities and multiplying by 50 gives 1287.69, agreeing with the normal approximation 1287.64 to within 6p.

Common slips

- Taking the extra profit as $32\,\mathrm{P}(X > 480)$, as if the new bus always fills whenever it is needed. On days with between 480 and 512 bus passengers it is only partly full.
- Sign errors in $\int_{-2}^{0} z\phi(z)\,\mathrm{d}z$: the answer must be negative, since the integrand is.

Beyond STEP

The quantity $\mathrm{E}\big(\min(X, c)\big)$ is the expected sales of a newsvendor with stock c, and the "newsvendor problem" of choosing c is a classic of operations research; the same clipped payoff $\min(\max(X - a, 0), b - a)$ is a "call spread" in financial mathematics. The trick $\phi'(z) = -z\phi(z)$ is the start of the theory of truncated normal moments. Grimmett and Welsh, *Probability: An Introduction*, covers the normal approximation to the binomial and expectations of functions of random variables.

WORKED SOLUTION

STEP III 2006, Question 13

Probability and Statistics • Geometric Probability

STEP III 2006, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q13

The key idea

The perimeter is made of two very different pieces, an arc of length π and a diameter of length 2, so we condition on which piece each point lands on. In each of the three cases the area of the triangle has a simple formula, $\frac{1}{2}ab\sin C$ with the vertex C at the centre, and the conditional expectations are short integrals. The law of total expectation then puts the pieces back together.

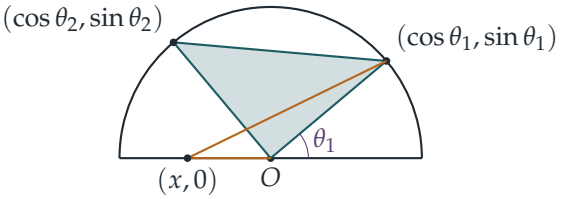
Solution

Setting up

Put the centre O of the diameter at the origin, with the diameter along the x-axis from $(-1,0)$ to $(1,0)$ and the arc in the upper half-plane. "At random on the perimeter" means uniformly distributed by length along the whole boundary, whose total length is $\pi+2$. So each point independently

- lies on the arc with probability $\dfrac{\pi}{\pi+2}$, and then it is $(\cos\theta,\sin\theta)$ with θ uniform on $[0,\pi]$;
- lies on the diameter with probability $\dfrac{2}{\pi+2}$, and then it is $(x,0)$ with x uniform on $[-1,1]$.

(Uniform by length on the arc is the same as uniform in angle, because arc length on a unit circle equals the angle.)



Let D be the number of the two points that lie on the diameter. Then $D = 2, 1, 0$ with probabilities

$$\frac{4}{(\pi+2)^2}, \qquad \frac{2\cdot 2\pi}{(\pi+2)^2} = \frac{4\pi}{(\pi+2)^2}, \qquad \frac{\pi^2}{(\pi+2)^2},$$

using independence of the two points (the factor 2 in the middle case counts which point is on the diameter). In every case the triangle has a vertex at O, so its area is $\frac{1}{2}\,|OP|\,|OQ|\sin\angle POQ$.

Both points on the diameter

All three vertices are on a line, so the triangle is degenerate and $A = 0$. Hence $\mathrm{E}(A \mid D = 2) = 0$.

One point on the arc, one on the diameter

Say the points are $P = (\cos\theta, \sin\theta)$ and $R = (x, 0)$. Then $|OP| = 1$, $|OR| = |x|$, and the base OR lies along the diameter while P is at height $\sin\theta \geq 0$ above it, so

$$A = \tfrac{1}{2}\,|x|\sin\theta.$$

Since x and θ are independent,

$$\mathrm{E}(A \mid D = 1) = \tfrac{1}{2}\,\mathrm{E}|x|\,\mathrm{E}(\sin\theta) = \tfrac{1}{2}\cdot\frac{1}{2}\int_{-1}^{1}|x|\,\mathrm{d}x\cdot\frac{1}{\pi}\int_0^{\pi}\sin\theta\,\mathrm{d}\theta = \tfrac{1}{2}\cdot\tfrac{1}{2}\cdot\frac{2}{\pi} = \frac{1}{2\pi}.$$

Both points on the arc

Now $P = (\cos\theta_1, \sin\theta_1)$ and $Q = (\cos\theta_2, \sin\theta_2)$ with θ_1, θ_2 independent and uniform on $[0, \pi]$. Both radii have length 1 and the angle POQ is $|\theta_1 - \theta_2|$, so

$$A = \tfrac{1}{2}\,|\sin(\theta_1 - \theta_2)|.$$

Ask yourself Do we need the distribution of $\theta_1 - \theta_2$ to find $\mathrm{E}|\sin(\theta_1 - \theta_2)|$?

No: condition on θ_2 first. For a fixed θ_2, as θ_1 runs over $[0, \pi]$ the angle $\theta_1 - \theta_2$ runs over an interval of length π. The function $|\sin u|$ has period π, so its integral over *any* interval of length π is the same as over $[0, \pi]$:

$$\int_0^{\pi}|\sin(\theta_1 - \theta_2)|\,\mathrm{d}\theta_1 = \int_0^{\pi}\sin u\,\mathrm{d}u = 2.$$

So $\mathrm{E}\big(|\sin(\theta_1-\theta_2)| \,\big|\, \theta_2\big) = 2/\pi$ whatever θ_2 is, and therefore

$$\mathrm{E}(A \mid D=0) = \tfrac{1}{2}\cdot\frac{2}{\pi} = \frac{1}{\pi}.$$

Putting it together

By the law of total expectation,

$$\begin{aligned}\mathrm{E}(A) &= \frac{4}{(\pi+2)^2}\cdot 0 + \frac{4\pi}{(\pi+2)^2}\cdot\frac{1}{2\pi} + \frac{\pi^2}{(\pi+2)^2}\cdot\frac{1}{\pi}\\ &= \frac{2+\pi}{(\pi+2)^2} = \frac{1}{2+\pi},\end{aligned}$$

which is $(2+\pi)^{-1}$, as required.

Checks

- The three case probabilities $\dfrac{4+4\pi+\pi^2}{(\pi+2)^2}$ add to 1.
- Each conditional expectation is at most the largest possible area in that case: $\frac{1}{\pi} \approx 0.318 \le \frac{1}{2}$ (two radii at right angles) and $\frac{1}{2\pi} \approx 0.159 \le \frac{1}{2}$.
- A Monte Carlo simulation with 10^6 pairs of points gave $\mathrm{E}(A) \approx 0.1943$, against $1/(2+\pi) \approx 0.1945$; quadrature confirmed the conditional values $1/\pi$ and $1/(2\pi)$.

Common slips

- Choosing each point on the arc or the diameter with probability $\frac{1}{2}$ each. "At random on the perimeter" weights each piece by its length.
- Forgetting the factor 2 in $\mathrm{P}(D=1)$: either point may be the one on the diameter.
- Writing $A = \frac{1}{2}\sin(\theta_1-\theta_2)$ without the modulus, which gives expectation 0 by symmetry.

Beyond STEP

This is a small piece of *geometric probability*, whose classic examples include Buffon's needle and Bertrand's paradox. Bertrand's paradox shows why the phrase "at random" must be pinned down, exactly as we did by taking length along the perimeter as the measure. The tool doing the work here, conditioning on a partition and averaging (the law of total expectation), is central to first-year probability. Grimmett and Welsh, *Probability: An Introduction*, covers it well.

WORKED SOLUTION

STEP III 2006, Question 14

Probability and Statistics • Covariance and Independence

STEP III 2006, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q14

The key idea

Everything here runs on two facts: expectation is linear, and for independent variables the expectation of a product is the product of the expectations. Variances come from $\mathrm{Var}(Y) = \mathrm{E}(Y^2) - [\mathrm{E}(Y)]^2$, so we only ever need expectations. To show two variables are *not* independent, it is enough to find one consequence of independence that fails, and the easiest one to test is $\mathrm{E}(UV) = \mathrm{E}(U)\,\mathrm{E}(V)$.

Solution

The two facts

For any random variables and constants a, b,

$$\mathrm{E}(aX_1 + bX_2) = a\,\mathrm{E}(X_1) + b\,\mathrm{E}(X_2),$$

and if X_1 and X_2 are independent then

$$\mathrm{E}(X_1X_2) = \mathrm{E}(X_1)\,\mathrm{E}(X_2).$$

The second fact also applies to functions of the variables: if X_1, X_2 are independent then so are X_1^2 and X_2, X_1 and X_2^2, and X_1^2 and X_2^2. We shall also use $\mathrm{E}(X_i^2) = \sigma_i^2 + \mu_i^2$, which is just $\mathrm{Var}(X_i) = \mathrm{E}(X_i^2) - \mu_i^2$ rearranged.

Perimeter and area

The perimeter is $P = 2X_1 + 2X_2$ and the area is $A = X_1X_2$. By linearity, and by independence,

$$\mathrm{E}(P) = 2(\mu_1 + \mu_2), \qquad \mathrm{E}(A) = \mu_1\mu_2.$$

For the variance of P we work from $\mathrm{E}(P^2)$:

$$\begin{aligned}\mathrm{E}(P^2) &= 4\,\mathrm{E}(X_1^2 + 2X_1X_2 + X_2^2)\\ &= 4\big(\sigma_1^2 + \mu_1^2 + 2\mu_1\mu_2 + \sigma_2^2 + \mu_2^2\big)\\ &= 4(\sigma_1^2 + \sigma_2^2) + 4(\mu_1 + \mu_2)^2,\end{aligned}$$

using linearity and $\mathrm{E}(X_1X_2) = \mu_1\mu_2$. Subtracting $[\mathrm{E}(P)]^2 = 4(\mu_1 + \mu_2)^2$,

$$\mathrm{Var}(P) = 4(\sigma_1^2 + \sigma_2^2), \qquad \text{so the standard deviation of } P \text{ is } 2\sqrt{\sigma_1^2 + \sigma_2^2}.$$

For A, since X_1^2 and X_2^2 are independent,

$$\mathrm{E}(A^2) = \mathrm{E}(X_1^2)\,\mathrm{E}(X_2^2) = (\sigma_1^2 + \mu_1^2)(\sigma_2^2 + \mu_2^2),$$

and subtracting $\mu_1^2\mu_2^2$ leaves

$$\mathrm{Var}(A) = \sigma_1^2\sigma_2^2 + \mu_1^2\sigma_2^2 + \mu_2^2\sigma_1^2,$$

so the standard deviation of A is

$$\sqrt{\sigma_1^2\sigma_2^2 + \mu_1^2\sigma_2^2 + \mu_2^2\sigma_1^2}.$$

P and A are not independent

If P and A were independent we would have $\mathrm{E}(PA) = \mathrm{E}(P)\,\mathrm{E}(A)$. Now $PA = 2X_1^2X_2 + 2X_1X_2^2$, so by linearity and independence of X_1 and X_2,

$$\begin{aligned}\mathrm{E}(PA) &= 2(\sigma_1^2 + \mu_1^2)\mu_2 + 2\mu_1(\sigma_2^2 + \mu_2^2)\\ &= 2(\mu_1 + \mu_2)\mu_1\mu_2 + 2(\mu_2\sigma_1^2 + \mu_1\sigma_2^2)\\ &= \mathrm{E}(P)\,\mathrm{E}(A) + 2(\mu_2\sigma_1^2 + \mu_1\sigma_2^2).\end{aligned}$$

A length and a breadth are positive, so $\mu_1, \mu_2 > 0$, and $\sigma_1, \sigma_2 \neq 0$, so the extra term $2(\mu_2\sigma_1^2 + \mu_1\sigma_2^2)$ is strictly positive. Hence $\mathrm{E}(PA) \neq \mathrm{E}(P)\,\mathrm{E}(A)$, and P and A are not independent.

The variable $Z = P - \alpha A$

Again, independence of Z and A would force $\mathrm{E}(ZA) = \mathrm{E}(Z)\,\mathrm{E}(A)$. By linearity,

$$\begin{aligned}\mathrm{E}(ZA) - \mathrm{E}(Z)\,\mathrm{E}(A) &= \big[\mathrm{E}(PA) - \mathrm{E}(P)\,\mathrm{E}(A)\big] - \alpha\big[\mathrm{E}(A^2) - [\mathrm{E}(A)]^2\big]\\ &= 2(\mu_2\sigma_1^2 + \mu_1\sigma_2^2) - \alpha\big(\mu_1^2\sigma_2^2 + \mu_2^2\sigma_1^2 + \sigma_1^2\sigma_2^2\big),\end{aligned}$$

using the two results above. The bracket multiplying α is $\mathrm{Var}(A)$, which is positive (it contains $\sigma_1^2\sigma_2^2 > 0$), so this expression vanishes for exactly one value of α:

$$\alpha_0 = \frac{2(\mu_1\sigma_2^2 + \mu_2\sigma_1^2)}{\mu_1^2\sigma_2^2 + \mu_2^2\sigma_1^2 + \sigma_1^2\sigma_2^2}.$$

If $\alpha \neq \alpha_0$, then $\mathrm{E}(ZA) \neq \mathrm{E}(Z)\,\mathrm{E}(A)$, so Z and A are not independent, as required.

Ask yourself Does $\alpha = \alpha_0$ mean that Z and A *are* independent?

No. Zero covariance is a consequence of independence, not a guarantee of it. At $\alpha = \alpha_0$ our test simply gives no information, and we need a different argument. The final part shows that in one concrete case independence fails there too.

The final part

Here $\mu_1 = \mu_2 = 2$ and $\sigma_1^2 = \sigma_2^2 = 1$, so

$$\alpha_0 = \frac{2(2+2)}{4+4+1} = \frac{8}{9}.$$

By the previous part, Z and A are not independent for any $\alpha \neq \frac{8}{9}$. For a single argument that covers every α, list the four equally likely outcomes:

(X_1, X_2)	A	P	$Z = P - \alpha A$
$(1,1)$	1	4	$4 - \alpha$
$(1,3)$, $(3,1)$	3	8	$8 - 3\alpha$
$(3,3)$	9	12	$12 - 9\alpha$

The key observation is that P, and hence Z, is completely determined by A. Let $z = 4 - \alpha$. Knowing $A = 1$ forces $Z = z$, so

$$\mathrm{P}(Z = z,\ A = 1) = \mathrm{P}(A = 1) = \tfrac{1}{4}.$$

Independence would need this to equal $\mathrm{P}(Z = z)\,\mathrm{P}(A = 1)$, that is, $\mathrm{P}(Z = z) = 1$, so Z would have to take the value $4 - \alpha$ on every outcome. That needs $8 - 3\alpha = 4 - \alpha$, so $\alpha = 2$, and also $12 - 9\alpha = 4 - \alpha$, so $\alpha = 1$. These can't both hold, so $\mathrm{P}(Z = z) < 1$ and independence fails.

Hence Z and A are not independent for any value of α. For instance, at $\alpha = \frac{8}{9}$ the values of Z are $\frac{28}{9}$, $\frac{16}{3}$ and 4, and $\mathrm{P}(Z = \frac{28}{9}, A = 1) = \frac{1}{4}$, whereas $\mathrm{P}(Z = \frac{28}{9})\,\mathrm{P}(A = 1) = \frac{1}{16}$.

Checks

- With no variation ($\sigma_1 = \sigma_2 = 0$) both standard deviations reduce to 0, as they should.
- In the final part, $\mathrm{Var}(A) = 1 + 4 + 4 = 9$ by the formula; directly, A takes values $1, 3, 9$ with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$, giving $\mathrm{E}(A) = 4$, $\mathrm{E}(A^2) = 25$ and variance 9.
- We checked the general mean, variance and covariance formulas exactly, using fractions, on 200 random discrete distributions. We also confirmed that $\mathrm{E}(ZA) = \mathrm{E}(Z)\mathrm{E}(A)$ at α_0, that $\alpha_0 = \frac{8}{9}$ in the final part, and that Z and A fail independence there for $\alpha = \frac{8}{9}$ and 500 other values of α.

Common slips

- Writing $\mathrm{E}(X_1^2) = \mu_1^2$. The variance term σ_1^2 is exactly what makes $\mathrm{Var}(A)$ and the covariance non-zero.
- Concluding that Z and A are independent when $\alpha = \alpha_0$. Zero covariance does not imply independence.
- Forgetting to say why $\mu_2\sigma_1^2 + \mu_1\sigma_2^2 \neq 0$. It is the positivity of lengths that rules out cancellation.

Beyond STEP

The quantity $\mathrm{E}(UV) - \mathrm{E}(U)\mathrm{E}(V)$ is the covariance, and the special value α_0 is $\mathrm{Cov}(P, A)/\mathrm{Var}(A)$, the slope of the least-squares regression of P on A. Choosing $\alpha = \alpha_0$ makes the residual Z uncorrelated with A, but not independent of it. For jointly normal variables the two notions coincide, which is one reason the normal distribution is so convenient. Grimmett and Welsh, *Probability: An Introduction*, treats covariance, correlation and independence carefully.

Beyond STEP

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